A minimalist intro to topology

Anton Petrunin

Here are incomplete lecture notes from my introductory course in topology at Penn State. They aim to be minimalist, conservative, and rigorous. The emphasis is on topology as a working tool. I include the basic notions and examples that are useful elsewhere. The main prerequisite is mathematical maturity; students should be comfortable with proofs. No topology is assumed; we use elementary set theory, calculus, and a small amount of group theory.

Exercises that are used later in the text are marked with an exclamation point: **Exercise!**

I used several sources, including Allen Hatcher's texts [13, 14], lectures of Sergei Ivanov [18], the textbook of Czes Kosniowski [19], and the textbook of Oleg Viro, Oleg Ivanov, Nikita Netsvetaev, and Viatcheslav Kharlamov [27]. I want to thank Rostislav Matveev for his help.

Anton Petrunin

Contents

Chapter 0

Puzzles and anecdotes

You have probably heard that for topologists, *two shapes are the same if one can be turned into the other by stretching and squeezing* (without ripping, piercing, gluing, and so on).

This statement is neither precise nor correct, but it helps build intuition. So, imagine an object made from endlessly stretchy material that can be twisted and pulled like chewing gum; if you can get one shape from another, then they have the same topology. Note that the size plays no role.

The picture above suggests that a square and a disc have the same topology. It might be more impressive to see that a donut is topologically equivalent to a coffee mug. On the other hand, a disc and an annulus are topologically different. It might be obvious, but this is a nontrivial statement (it will follow from 16.1). So, at least not all shapes are topologically equivalent.

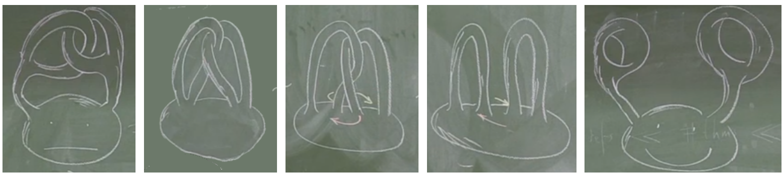
0.1. Exercise. *In the morning, a topologist looks at his clothes; there should be a T-shirt, a pair of pants, and socks. Help him to identify each piece of clothing. Which piece has a hole in it?*

0.2. Exercise. *A topologist dropped his mug and noticed that the handle had snapped off. "Its topology has changed," he thaut, "but it can still be used for its intended purpose."*

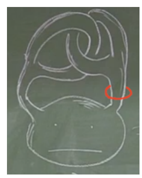
He picked the mug up, turned it over, and said: "Oh, I was wrong—the topology has not changed, but it is now impossible to use."

What did he see?

0.3. Exercise. *The topologist decided to fix the broken mug with superglue. Being a rather clumsy person, he somehow managed to glue his index finger to his thumb—in fact, on both hands. Worse still, the*



finger rings he'd made were linked together. But after some thinking and stretching he managed to separate the rings.



What would happen if he had a hair tie on his hand?

At this point, you might think that topology has no practical value, and you are almost right, but it helps in solving the following puzzles; these puzzles can also help you to start thinking topologically.

0.4. Exercise.

(a) *Put a loop of cord thru the handles of a pair of scissors as shown. Have someone hold the ends. Try to get the scissors off without cutting the cord or releasing the ends.*

(b) *Tie your hands together with a long cord; then tie the hands of your friend in the same way, linking your cords, as shown. Now try to get them apart without cutting the string or untying the knots.*

(c) *If there is no friend nearby, try undoing the knot in the middle of the cord shown in the last picture. Again, no cutting the string or untying the little knots.*

Chapter 1

Metric spaces

In this chapter we discuss *metric spaces* — a motivating example that will guide us toward the definition of our main object of study — *topological spaces.*

Examples of metric spaces were considered for thousands of years, but the first general definition was given only in 1906 by Maurice Fréchet [10].

A Definition

The following definition grabs together the most important properties of the intuitive notion of *distance.*

1.1. Definition. *Let $\mathcal{X}$ be a set with a function that returns a real number, denoted as $|x-y|$, for any pair $x, y \in \mathcal{X}$. Assume that the following conditions are satisfied for any $x, y, z \in \mathcal{X}$:*

(a) $|x-y| \geqslant 0$.

(b) $x = y$ *if and only if* $|x-y| = 0$.

(c) $|x-y| = |y-x|$.

(d) $|x-y| + |y-z| \geqslant |x-z|$; *this property is called the triangle inequality.*

In this case, we say that $\mathcal{X}$ is a metric space and the function

$$(x, y) \mapsto |x-y|$$

is called a metric. The elements of $\mathcal{X}$ are called points of the metric space. Given two points $x, y \in \mathcal{X}$, the value $|x-y|$ is called the distance from x to y.

For two points x and y in a metric space the difference $x-y$ may have no meaning, but $|x-y|$ means the distance.

Typically, we consider only one metric on a set, but if a few metrics are needed, we can distinguish them by an index, say $|x-y|_{\bullet}$ or $|x-y|_{239}$. If we need to emphasize that the distance is taken in the metric space $\mathcal{X}$ we write $|x-y|_{\mathcal{X}}$ instead of $|x-y|$.

1.2. Exercise. *Show that*

$$|x-y|_{\natural} = (x-y)^2$$

is not a metric on the real line $\mathbb{R}$.

1.3. Exercise. *Show that if* $(x,y) \mapsto |x-y|$ *is a metric, then so is*

$$(x,y) \mapsto |x-y|_{\min} = \min\{1, |x-y|\}.$$

B Examples

Discrete space. Let $\mathcal{X}$ be an arbitrary set. For any $x, y \in \mathcal{X}$, set $|x-y| = 0$ if $x = y$ and $|x-y| = 1$ otherwise. This metric is called the discrete metric on $\mathcal{X}$ and the obtained metric space is called discrete.

Real line is the set $\mathbb{R}$ of all real numbers with the metric defined by $|x-y|$. (Unless it is stated otherwise, the real line $\mathbb{R}$ will be considered with this metric.)

Metrics on the plane. Let us denote by $\mathbb{R}^2$ the set of all pairs (x,y) of real numbers. Consider two points $p = (x_p, y_p)$ and $q = (x_q, y_q)$ in $\mathbb{R}^2$. One can equip $\mathbb{R}^2$ with the following metrics:

- Euclidean metric,

$$|p-q|_2 = \sqrt{(x_p - x_q)^2 + (y_p - y_q)^2}.$$

  (Unless it is stated otherwise, the plane $\mathbb{R}^2$ will be considered with the Euclidean metric.)
- Manhattan metric,

$$|p-q|_1 = |x_p - x_q| + |y_p - y_q|.$$

- Maximum metric,

$$|p-q|_\infty = \max\{|x_p - x_q|, |y_p - y_q|\}.$$

1.4. Exercise. *Prove that (a)* $|{*}-{*}|_1$*; (b)* $|{*}-{*}|_2$ *and (c)* $|{*}-{*}|_\infty$ *are metrics on* $\mathbb{R}^2$.

C Subspaces

Let us recall the set-builder notation: Given a set $\mathcal{X}$ and a property $P(x)$ that depends on $x \in \mathcal{X}$, we denote by

$$\{ x \in \mathcal{X} : P(x) \}$$

the subset of all elements $x \in \mathcal{X}$ for which the property $P(x)$ holds. For example, $\{ x \in \mathbb{R} : x > 0 \}$ denotes the set of positive reals.

Any subset $\mathcal{A}$ of a metric space $\mathcal{X}$ forms a metric space on its own; it is called a subspace of $\mathcal{X}$. This construction produces many more examples of metric spaces. For example, the disc

$$\mathbb{D}^2 = \{ (x, y) \in \mathbb{R}^2 : x^2 + y^2 \leqslant 1 \}$$

and the circle

$$\mathbb{S}^1 = \{ (x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1 \}$$

are metric spaces with the metrics taken from the Euclidean plane. Similarly, the interval $[0, 1)$ is a metric space with metric taken from $\mathbb{R}$.

D Continuous maps

Recall that a real-to-real function f is called *continuous* if *for any $x_0 \in \mathbb{R}$ and any $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$|x_0 - x| < \delta \quad \Rightarrow \quad |f(x_0) - f(x)| < \varepsilon.$$

This definition admits the following straightforward generalization to metric spaces:

1.5. Definition. *A map $f : \mathcal{X} \to \mathcal{Y}$ between metric spaces is called continuous if for any $x_0 \in \mathcal{X}$ and any $\varepsilon > 0$ there exists $\delta > 0$ such that $|f(x_0) - f(x)|_{\mathcal{Y}} < \varepsilon$, for any $x \in \mathcal{X}$ such that $|x_0 - x|_{\mathcal{X}} < \delta$.*

1.6. Exercise. *Let $\mathcal{X}$ be a metric space and $z \in \mathcal{X}$ be a fixed point. Show that the function*

$$f(x) := |x - z|_{\mathcal{X}}$$

is continuous.

1.7. Claim. *Let $f : \mathcal{X} \to \mathcal{Y}$ and $g : \mathcal{Y} \to \mathcal{Z}$ be continuous maps between metric spaces. Then the composition $h = g \circ f : \mathcal{X} \to \mathcal{Z}$ is also continuous.*

Proof. Fix $\varepsilon > 0$, $x \in \mathcal{X}$, and $y \in \mathcal{Y}$ such that $f(x) = y$.

Since g is continuous, there is $\delta_1 > 0$ such that

$$|y' - y|_{\mathcal{Y}} < \delta_1 \quad \Rightarrow \quad |g(y') - g(y)|_{\mathcal{Z}} < \varepsilon.$$

Since f is continuous, there is $\delta_2 > 0$ such that

$$|x' - x|_{\mathcal{X}} < \delta_2 \quad \Rightarrow \quad |f(x' - f(x)|_{\mathcal{Y}} < \delta_1.$$

Since $f(x) = y$, we get that

$$|x' - x|_{\mathcal{X}} < \delta_2 \quad \Rightarrow \quad |h(x') - h(x)|_{\mathcal{Z}} < \varepsilon.$$

Hence the result. □

1.8. Exercise. *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a distance-preserving map between metric spaces; that is,*

$$|x_0 - x_1|_{\mathcal{X}} = |f(x_0) - f(x_1)|_{\mathcal{Y}}$$

for any $x_0, x_1 \in \mathcal{X}$.

(a) Show that f is continuous.

(b) Show that f is injective; that is, if $x_0 \neq x_1$, then $f(x_0) \neq$ $\neq f(x_1)$.

1.9. Exercise. *Let $\mathcal{X}$ be a discrete metric space (defined in 1B) and $\mathcal{Y}$ be an arbitrary metric space. Show that any map $f\colon \mathcal{X} \to \mathcal{Y}$ is continuous.*

1.10. Advanced exercise. *Construct a continuous function*

$$f\colon [0,1] \to [0,1]$$

that takes every value in $[0,1]$ an infinite number of times.

E Balls

Let x be a point in a metric space $\mathcal{X}$, and $r > 0$. The set of points in $\mathcal{X}$ that lie at distance less than r is called the open ball of radius r centered at x. It is denoted as $\mathrm{B}(x, r)$ or $\mathrm{B}(x, r)_{\mathcal{X}}$; the latter notation is used if we need to emphasize that it is taken in the space $\mathcal{X}$.[1]

The ball $\mathrm{B}(x, r)$ is also called an r-neighborhood of x.

[1]Many authors use the notations $\mathrm{B}_r(x)$ and $\mathrm{B}_r(x)_{\mathcal{X}}$ as well.

Analogously we may define *closed balls*

$$\bar{\mathrm{B}}[x,r] = \bar{\mathrm{B}}[x,r]_{\mathcal{X}} = \{\, y \in \mathcal{X} \,:\, |x-y| \leqslant r \,\}.$$

1.11. Exercise. *Sketch the closed unit balls for the metrics $|{*}-{*}|_1$, $|{*}-{*}|_2$ and $|{*}-{*}|_\infty$ defined in 1B.*

1.12. Exercise. *Consider two balls $\mathrm{B}(x,r)$ and $\mathrm{B}(y,R)$ in a metric space such that $\mathrm{B}(x,r) \subsetneq \mathrm{B}(y,R)$. Show that $r < 2{\cdot}R$.*

Give an example of a metric space and a pair of balls as above with $r > R$.

Let us reformulate the definition of continuous map (1.5) using the introduced notion of ball.

1.13. Definition. *A map $f\colon \mathcal{X} \to \mathcal{Y}$ between metric spaces is called continuous if for every $x \in \mathcal{X}$ and every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$f(\mathrm{B}(x,\delta)_{\mathcal{X}}) \subset \mathrm{B}(f(x),\varepsilon)_{\mathcal{Y}};$$

that is, the image of the δ-ball centered at $x \in \mathcal{X}$ lies in the ε-ball centered at $f(x) \in \mathcal{Y}$.

1.14. Exercise. *Prove the equivalence of the definitions 1.5 and 1.13.*

F Open sets

1.15. Definition. *A subset V in a metric space $\mathcal{X}$ is called open if for any $x \in V$ there is $\varepsilon > 0$ such that $\mathrm{B}(x,\varepsilon) \subset V$.*

In other words, V is open if, together with each point, V contains its ε-neighborhood for some $\varepsilon > 0$.

For example, any set in a discrete metric space is open. Indeed a 1-neighborhood of any point p is the one-point set $\{p\}$, and it lies in any set containing p.

Further, the set of positive real numbers

$$(0,\infty) = \{\, x \in \mathbb{R} \,:\, x > 0 \,\}$$

is an open subset of $\mathbb{R}$; indeed, for any $x > 0$ its x-neighborhood lies in $(0,\infty)$. On the other hand, the set of nonnegative reals

$$[0,\infty) = \{\, x \in \mathbb{R} \,:\, x \geqslant 0 \,\}$$

is not open since there are negative numbers in any neighborhood of 0.

1.16. Claim. *Show that any open ball in a metric space is an open set.*

Proof. Choose an open ball $\mathrm{B}(x, R)$.

If $y \in \mathrm{B}(x, R)$, then $r = R - |x - y| > 0$. The triangle inequality implies that if $|y - z| < r$, then $|x - z| \leqslant |x - y| + |y - z| < R$.

In other words, if $z \in \mathrm{B}(y, r)$, then $z \in \mathrm{B}(x, R)$. Equivalently, $\mathrm{B}(y, r) \subset \mathrm{B}(x, R)$. Hence the result. $\square$

1.17. Exercise! *Prove the following:*

(a) The union of an arbitrary family of open sets is open.
(b) The intersection of two open sets is open.
(c) A set is open if and only if it is a union of open balls.

1.18. Exercise. *Give an example of a metric space $\mathcal{X}$ and an infinite sequence of open sets $V_1, V_2, \ldots$ such that their intersection $V_1 \cap V_2 \cap \ldots$ is not open.*

1.19. Exercise. *Show that the metrics $|{*}-{*}|_1$, $|{*}-{*}|_2$ and $|{*}-{*}|_\infty$ (defined in 1B) give rise to the same open sets in $\mathbb{R}^2$. That is, if $V \subset \mathbb{R}^2$ is open for one of these metrics, then it is open for the others.*

G Gateway to topology

The following result is the main gateway to topology. It says that continuous maps can be defined entirely in terms of open sets.

1.20. Proposition. *A map $f \colon \mathcal{X} \to \mathcal{Y}$ between two metric spaces is continuous if and only if the inverse image of any open set is open; that is, for any open set $W \subset \mathcal{Y}$ its inverse image*

$$f^{-1}(W) = \{\, x \in \mathcal{X} \;:\; f(x) \in W \,\}$$

is open.

The following exercise emphasizes that the proposition says nothing about the images of open sets; it is instructive to solve it before going into the proof (see also 5B).

1.21. Exercise. *Give an example of a continuous function $f \colon \mathbb{R} \to \mathbb{R}$ and an open set $V \subset \mathbb{R}$ such that the image $f(V) \subset \mathbb{R}$ is not open.*

Proof; only-if part. Let $W \subset \mathcal{Y}$ be an open set and $V = f^{-1}(W)$. Choose $x \in V$; note that $f(x) \in W$.

Since W is open,

❶ $$\mathrm{B}(f(x),\varepsilon)_{\mathcal{Y}} \subset W$$

for some $\varepsilon > 0$.

Since f is continuous, by Definition 1.13, there is $\delta > 0$ such that

$$f(\mathrm{B}(x,\delta)_{\mathcal{X}}) \subset \mathrm{B}(f(x),\varepsilon)_{\mathcal{Y}}.$$

It follows that together with any point $x \in V$, the set V contains $\mathrm{B}(x,\delta)$; so, V is open.

If part. Fix $x \in \mathcal{X}$ and $\varepsilon > 0$. According to 1.16,

$$W = \mathrm{B}(f(x),\varepsilon)_{\mathcal{Y}}$$

is an open set in $\mathcal{Y}$. Therefore its inverse image $f^{-1}(W)$ is open.

Clearly $x \in f^{-1}(W)$. By the definition of open set (1.15)

$$\mathrm{B}(x,\delta)_{\mathcal{X}} \subset f^{-1}(W)$$

for some $\delta > 0$. Or equivalently

$$f(\mathrm{B}(x,\delta)_{\mathcal{X}}) \subset W = \mathrm{B}(f(x),\varepsilon)_{\mathcal{Y}}.$$

Hence the if part follows. □

H Limits

1.22. Definition. *Let $x_1, x_2, \ldots$ be a sequence of points in a metric space $\mathcal{X}$. We say that it converges to a point $x_\infty \in \mathcal{X}$ if*

$$|x_\infty - x_n|_{\mathcal{X}} \to 0 \quad \textit{as} \quad n \to \infty.$$

In this case, we say that the sequence $x_1, x_2, \ldots$ is converging and x_∞ is its limit; it can be expressed by $x_n \to x_\infty$ as $n \to \infty$ or

$$x_\infty = \lim_{n\to\infty} x_n.$$

Note that we defined the convergence of points in a metric space using the convergence of real numbers $d_n = |x_\infty - x_n|_{\mathcal{X}}$, which we assume to be known.

1.23. Exercise. *Show that any sequence of points in a metric space has at most one limit.*

1.24. Exercise. *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a map between metric spaces. Show that f is continuous if and only if the following condition holds:*

- *If $x_n \to x_\infty$ as $n \to \infty$ in $\mathcal{X}$, then the sequence $y_n = f(x_n)$ converges to $y_\infty = f(x_\infty)$ as $n \to \infty$ in $\mathcal{Y}$.*

I Closed sets

Let A be a set in a metric space $\mathcal{X}$. A point $x \in \mathcal{X}$ is a limit point of A if there is a sequence $x_n \in A$ such that $x_n \to x$ as $n \to \infty$.[2]

The set of all limit points of A is called the closure of A and denoted as $\bar{A}$. Note that $\bar{A} \supset A$; indeed, any point $x \in A$ is a limit point of the constant sequence $x_n = x$.

If $\bar{A} = A$, then the set A is called closed.

1.25. Exercise. *Give an example of a subset $A \subset \mathbb{R}$ that is neither closed nor open.*

1.26. Exercise. *Show that the closure of any set in a metric space is a closed set; that is, $\bar{\bar{A}} = \bar{A}$.*

1.27. Exercise. *Show that a subset Q in a metric space $\mathcal{X}$ is closed if and only if its complement $V = \mathcal{X} \setminus Q$ is open.*

[2] Sometimes limit points are defined, assuming in addition that $x_n \neq x$ for any n — we do *not* follow this convention.

Chapter 2

Topological spaces

Topology starts here; the previous chapter provides motivation and examples.

In the previous chapter we defined open sets in metric spaces and showed that continuity could be defined using only the notion of open sets. In this chapter we collect key properties of open sets and state them as axioms. It will give us a definition of a *topological space* as a set with a distinguished class of subsets called *open sets*.

The first definition of topological spaces was given by Felix Hausdorff in 1914 [15, VII § 1]. In 1922, the definition was generalized slightly by Kazimierz Kuratowski [20], and it is now standard.

A Definitions

We are about to define *abstract open sets* without referring to metric spaces; this definition is based on two properties in exercises 1.17a and 1.17b.

2.1. Definition. *Suppose $\mathcal{X}$ is a set with a distinguished class of subsets, called open sets, such that*

(a) The empty set $\varnothing$ and the whole $\mathcal{X}$ are open.

(b) The union of any family of open sets is an open set. That is, if V_α is open for any α in the index set $\mathcal{I}$, then the set

$$W = \bigcup_{\alpha\in\mathcal{I}} V_\alpha = \{\, x \in \mathcal{X} : x \in V_\alpha \quad \text{for some} \quad \alpha \in \mathcal{I} \,\}$$

is open.

(c) The intersection of two open sets is an open set. That is, if V_1 and V_2 are open, then the intersection $W = V_1 \cap V_2$ is open.

In this case, $\mathcal{X}$ is called a topological space. The family of all open sets in $\mathcal{X}$ is called a topology on $\mathcal{X}$.

Usually we consider a set with just one topology; therefore, it is acceptable to use the same notation for the set and the corresponding topological space. Several times we will need to consider different topologies, say $\mathcal{T}_1$ and $\mathcal{T}_2$, on the same set $\mathcal{X}$; in this case, the corresponding topological spaces will be denoted by $(\mathcal{X}, \mathcal{T}_1)$ and $(\mathcal{X}, \mathcal{T}_2)$.

From 2.1c, it follows that the intersection of a finite family of open sets is open. That is, if V_1, V_2, ..., V_n are open, then the intersection

$$W = V_1 \cap \ldots \cap V_n$$

is open. The latter is proved by induction on n using the identity

$$V_1 \cap \cdots \cap V_n = (V_1 \cap \cdots \cap V_{n-1}) \cap V_n.$$

B Examples

For any set $\mathcal{X}$, we can define the following topologies:

- The discrete topology — the topology consisting of all subsets of a set $\mathcal{X}$.
- The concrete topology (also known as trivial topology) — the topology consisting of just the whole set $\mathcal{X}$ and the empty set, $\varnothing$.
- The cofinite topology — the topology consisting of the empty set, $\varnothing$, and the complements of finite sets.

By 1.17a and 1.17b any metric space is a topological space if one defines open sets as in the definition 1.15. For example, the real line $\mathbb{R}$ comes with a natural metric which defines a topology on $\mathbb{R}$; if not stated otherwise, the real line $\mathbb{R}$ will be considered with this topology. As follows from Exercise 1.19, different metrics on one set might define the same topology.

A topological space is called metrizable if its topology can be defined by a metric — these examples are most important.

2.2. Exercise. *Assume an infinite set $\mathcal{X}$ equipped with the cofinite topology. Show that $\mathcal{X}$ is not metrizable.*

The so-called connected two-point space is a simple but nontrivial example of a topological space. This space consists of two points

$$\mathcal{X} = \{a, b\}$$

and it has three open sets:

$$\varnothing, \quad \{a\} \quad \text{and} \quad \{a, b\}.$$

It is instructive to check that this is indeed a topology.

2.3. Exercise. *Show that a finite topological space (that is, a finite set equipped with a topology) is metrizable if and only if it is discrete. In particular, the connected two-point space is not metrizable.*

Let us also mention the Zariski topology: it is a topology on $\mathbb{R}^n$ in which the open sets are complements of solution sets of systems of algebraic equations. (For $n = 1$, it is the same as the cofinite topology.)

This topology plays an essential role in algebraic geometry. Unfortunately, proving that this indeed defines a topology requires some commutative algebra, which is beyond the scope of this text.

C Comparison of topologies

Let $\mathscr{W}$ and $\mathscr{S}$ be two topologies on the same set. Suppose $\mathscr{W} \subset \mathscr{S}$; that is, any open set in the $\mathscr{W}$-topology is open in the $\mathscr{S}$-topology. In this case, we say that $\mathscr{W}$ is weaker than $\mathscr{S}$, or, equivalently, $\mathscr{S}$ is stronger than $\mathscr{W}$.[1]

Note that on any set, the concrete topology is the weakest and the discrete topology is the strongest.

2.4. Exercise. *Let $\mathscr{W}$ and $\mathscr{S}$ be two topologies on one set. Suppose that for any point x and any $W \in \mathscr{W}$ such that $W \ni x$, there is $S \in \mathscr{S}$ such that $W \supset S \ni x$. Show that $\mathscr{W}$ is weaker than $\mathscr{S}$.*

D Continuous maps

Our challenge is to reformulate definitions from the previous chapter using only open sets. Continuous maps are first in line. The following definition is motivated by Proposition 1.20.

2.5. Definition. *A map between topological spaces $f\colon \mathcal{X} \to \mathcal{Y}$ is called continuous if the inverse image of any open set is open. That is, if W is an open subset of $\mathcal{Y}$, then its inverse image*

$$V = f^{-1}(W) = \{\, x \in \mathcal{X} \,:\, f(x) \in W \,\}$$

is an open subset of $\mathcal{X}$.

[1] Some authors use terms smaller or coarser for weaker topology and finer or larger for stronger topology.

2.6. Exercise. *Let $\mathbb{R}$ be the real line with the standard topology, and let $\mathcal{X} = \{a, b\}$ be the connected two-point space described in 2B — it has only three open sets: $\varnothing$, $\{a\}$, and $\{a, b\}$.*
(a) Construct a nonconstant continuous map $\mathbb{R} \to \mathcal{X}$.
(b) Show that any continuous map $\mathcal{X} \to \mathbb{R}$ is constant.

2.7. Claim. *The composition of continuous maps is continuous.*

Proof. Consider two continuous maps $\mathcal{X} \xrightarrow{f} \mathcal{Y} \xrightarrow{g} \mathcal{Z}$, and let

$$h = g \circ f\colon \mathcal{X} \to \mathcal{Z}.$$

Choose an open set $W \subset \mathcal{Z}$. Since g is continuous, $V = g^{-1}(W)$ is open. Furthermore, since f is continuous, $U = f^{-1}(V)$ is open. Observe that $U = h^{-1}(W)$ — hence the result. □

2.8. Exercise. *Let $\mathscr{T}$ be a family of subsets of $\mathbb{R}$ that consists of $\varnothing$, $\mathbb{R}$ and the intervals $[a, \infty)$, (a, ∞) for all $a \in \mathbb{R}$.*
(a) Show that $\mathscr{T}$ is a topology on $\mathbb{R}$.
(b) Show that the topological space $(\mathbb{R}, \mathscr{T})$ is not metrizable.
(c) Show that a function $f\colon \mathbb{R} \to \mathbb{R}$ is nondecreasing if and only if it defines a continuous map $(\mathbb{R}, \mathscr{T}) \to (\mathbb{R}, \mathscr{T})$.

In practice, continuity of maps between topological spaces is often verified indirectly.

In particular, one can use 2.7, the results from 1D, and the familiar fact from calculus that *any differentiable map defined on an open subset of $\mathbb{R}^n$ is continuous.* A bit later, we will be more flexible in constructing continuous maps; see 3.13 and 6.3.

2.9. Exercise. *Let f be a continuous real-valued function defined on a topological space. Show that the function $x \mapsto |f(x)|$ is continuous.*

E Neighborhoods and limits

Let x be a point in a topological space $\mathcal{X}$. A n e i g h b o r h o o d of x is any open set N containing x.

In topology, neighborhoods often replace balls, which make sense only in metric spaces.

2.10. Definition. *Suppose x_n is a sequence of points in a topological space $\mathcal{X}$. We say that x_n c o n v e r g e s to a point $x_\infty \in \mathcal{X}$ (briefly $x_n \to x_\infty$ as $n \to \infty$) if for any neighborhood N of x_∞, we have that $x_n \in N$ for all sufficiently large n.*

Observe that the above definition agrees with 1.22. In other words, a sequence of points $x_1, x_2, \dots$ in a metric space converges to a point x_∞ in the sense of the definition 1.22 if and only if it converges in the sense of the definition 2.10.

2.11. Exercise. *Show that a convergent sequence of points in a topological space is also convergent for every weaker topology.*

F A warning about sequences

Note that in a space with the concrete topology any sequence converges to any point. In particular, a sequence might have several different limits. Furthermore, if we equip $\mathbb{R}$ with the cofinite topology then any sequence of pairwise distinct numbers converges to every point in $\mathbb{R}$.

The following exercise shows that converging sequences do *not* adequately describe the topology of a space; namely, an analog of 1.24 does not hold. Unless you know what you are doing, avoid using limits in general topological spaces.[2]

Recall that a set is called countable if it admits a bijection to a subset of the set of natural numbers. In particular, all finite sets are countable.

2.12. Advanced exercise. *Let $\mathcal{X}$ be the set of real numbers with the so-called cocountable topology; its closed sets are either countable or the whole $\mathbb{R}$.*

(a) Show that a sequence $x_1, x_2, \dots \in \mathcal{X}$ converges to x_∞ if and only if $x_n = x_\infty$ for all sufficiently large n.

(b) Show that any set $A \subset \mathcal{X}$ is sequentially closed; that is, if a sequence $a_1, a_2, \dots \in A$ converges, then its limit lies in A. Observe that not all subsets of $\mathcal{X}$ are closed.

(c) Show that any map $f\colon \mathcal{X} \to \mathcal{X}$ is sequentially continuous; that is, if a sequence $x_1, x_2, \dots \in \mathcal{X}$ converges to x_∞ in $\mathcal{X}$, then $y_n = f(x_n)$ converges to $y_\infty = f(x_\infty)$. Observe that not all maps $f\colon \mathcal{X} \to \mathcal{X}$ are continuous.

[2]So-called nets [22] provide an appropriate generalization of sequences that works well in topological spaces, but we are not going to consider them.

Chapter 3

Subsets

This chapter starts with the definition of closed sets in a topological space. Further, we introduce constructions of interior, closure, and boundary.

A Closed sets

Let $\mathcal{X}$ be a topological space. A subset $K \subset \mathcal{X}$ is called closed if its complement $\mathcal{X} \setminus K$ is open.

Sometimes it is easier to use closed sets; for example, the cofinite topology can be defined by declaring that the whole space and all its finite sets are closed.[1]

3.1. Proposition. *Let $\mathcal{X}$ be a topological space.*

(a) *The empty set and $\mathcal{X}$ are closed.*

(b) *The intersection of any family of closed sets is a closed set. That is, if K_α is closed for any α in the index set $\mathcal{I}$, then the set*

$$Q = \bigcap_{\alpha \in \mathcal{I}} K_\alpha = \{\, x \in \mathcal{X} \,:\, x \in K_\alpha \quad \textit{for any} \quad \alpha \in \mathcal{I} \,\}$$

is closed.

(c) *The union of two closed sets (or any finite family of closed sets) is closed. That is, if K_1 and K_2 are closed, then the union $Q = K_1 \cup K_2$ is closed.*

[1] There is no particular reason why we define a topological space in terms of open sets — we could use closed sets instead. In fact, closed sets were considered before open sets — the former were introduced by Georg Cantor in 1884 [6], and the latter by René Baire in 1899 [2].

Proof. The proposition follows since union of complements is complement of intersection. More precisely, if $\{S_\alpha\}$ is a family of sets in $\mathcal{X}$, then

$$\bigcup_\alpha (\mathcal{X} \setminus S_\alpha) = \mathcal{X} \setminus \left(\bigcap_\alpha S_\alpha\right).$$

To prove this identity, notice that both sides can be written as

$$\{\, x \in \mathcal{X} \,:\, x \notin S_\alpha \text{ for some } \alpha \,\}.$$

It remains to apply the definition of closed sets and the definition of topological spaces; see 2.1. □

The following proposition is completely analogous to the original definition of continuous maps via open sets (2.5).

3.2. Proposition. *Let $\mathcal{X}$ and $\mathcal{Y}$ be topological spaces. A map $f\colon \mathcal{X} \to$ $\to \mathcal{Y}$ is continuous if and only if any closed set $Q \subset \mathcal{Y}$ has a closed inverse image $f^{-1}(Q) \subset \mathcal{X}$.*

Proof. In the proof, we will use the following set-theoretical identity. Suppose $A \subset \mathcal{Y}$ and $B = \mathcal{Y} \setminus A$ (or, equivalently, $A = \mathcal{Y} \setminus B$). Then

❶ $$f^{-1}(B) = \mathcal{X} \setminus f^{-1}(A)$$

for any map $f\colon \mathcal{X} \to \mathcal{Y}$. This identity is tautological; to prove it, observe that both sides can be spelled as

$$\{\, x \in \mathcal{X} \,:\, f(x) \notin A \,\}.$$

Only-if part. Let $B \subset \mathcal{Y}$ be a closed set. Then $A = \mathcal{Y} \setminus B$ is open. Since f is continuous, $f^{-1}(A)$ is open. By ❶, $f^{-1}(B)$ is the complement of $f^{-1}(A)$ in $\mathcal{X}$. Hence $f^{-1}(B)$ is closed.

The only-if part follows since B is an arbitrary closed set in $\mathcal{Y}$.

If part. Fix an open set B; its complement $A = \mathcal{Y} \setminus B$ is closed. Therefore $f^{-1}(A)$ is closed. By ❶, $f^{-1}(B)$ is the complement of $f^{-1}(A)$ in $\mathcal{X}$. Hence $f^{-1}(B)$ is open.

The if part follows since B is an arbitrary open set in $\mathcal{Y}$. □

3.3. Exercise.

(a) *Let $\mathcal{X}$ be a metrizable topological space. Show that any closed subset $Q \subset \mathcal{X}$ is an intersection of a sequence of open sets.*

(b) *Construct a topological space $\mathcal{Y}$ with a closed set Q that is not an intersection of any family of open sets.*

B Interior and closure

Let A be an arbitrary subset of a topological space $\mathcal{X}$.

The union of all open subsets of A is called the interior of A and denoted as $\mathring{A}$; for longer expressions we may use A° or $\operatorname{Int} A$.

Note that $\mathring{A}$ is open. Indeed, it is defined as a union of open sets and such a union is open by the definition of a topology; see 2.1. In other words, $\mathring{A}$ is the *maximal* open set in A, as any open subset of A must lie in $\mathring{A}$. In particular,

$$\mathring{\mathring{A}} = \mathring{A}$$

for any set A.

The intersection of all closed subsets containing A is called the closure of A and denoted as $\bar{A}$ or $\operatorname{Cl} A$.

The set $\bar{A}$ is closed. Indeed, it is defined as an intersection of closed sets and such an intersection has to be closed by 3.1. In other words, $\bar{A}$ is the *minimal* closed set that contains A, as any closed subset containing A contains $\bar{A}$. In particular,

$$\bar{\bar{A}} = \bar{A}$$

for any set A.

3.4. Exercise. *Assume A is a subset of a topological space $\mathcal{X}$; consider its complement $B = \mathcal{X} \setminus A$. Show that*

$$\bar{B} = \mathcal{X} \setminus \mathring{A}.$$

Let A and B be subsets of a topological space $\mathcal{X}$. The set A is said to be dense in B if $\bar{A} \supset B$; if $\bar{A} = \mathcal{X}$, then we say that A is everywhere dense.

3.5. Exercise. *Show that A is dense in B if and only if any neighborhood of any point in B intersects A.*

The following exercise is called the Kuratowski problem.

3.6. Advanced exercise.

(a) *Prove that all the inclusions and equalities shown on the diagram hold for any subset A in a topological space $\mathcal{X}$.*

$$\begin{tikzcd}
\bar{\mathring{\bar{A}}} \arrow[rr, hook] & & \bar{A} & & \bar{\mathring{A}} \arrow[ll, hook'] \arrow[rr, equal] & & \bar{\mathring{\bar{\mathring{A}}}} \\
 & & & A \arrow[ul, hook'] & & & \\
\mathring{\bar{\mathring{\bar{A}}}} \arrow[uu, hook] \arrow[rr, equal] & & \mathring{\bar{A}} \arrow[uu, hook] \arrow[uull, hook'] & & \mathring{A} \arrow[uu, hook] \arrow[ul, hook] \arrow[ll, hook'] \arrow[rr, hook] & & \mathring{\bar{\mathring{A}}} \arrow[uu, hook] \arrow[uull, hook]
\end{tikzcd}$$

Conclude that given a set A one can get at most 7 distinct subsets by repeatedly applying closure and interior.

(b) *Give an example of a topological space $\mathcal{X}$ with a subset A such that all the following 7 subsets are distinct:*

$$\bar{\mathring{\bar{A}}},\ \mathring{\bar{A}},\ \bar{A},\ A,\ \mathring{A},\ \bar{\mathring{A}},\ \mathring{\bar{\mathring{A}}}.$$

C Boundary

Let A be an arbitrary subset of a topological space $\mathcal{X}$. The boundary[2] of A (briefly ∂A) is defined as the complement

$$\partial A = \bar{A} \setminus \mathring{A}.$$

3.7. Exercise. *Show that the boundary of any set is closed.*

3.8. Exercise. *Show that the set A is closed if and only if $\partial A \subset A$.*

3.9. Exercise. *Show that for any set A in a topological space we have $\partial(\partial A) \subset \partial A$. Provide an example showing that the inclusion might be strict.*

3.10. Exercise. *Let A be a set in a topological space $\mathcal{X}$. Show that $x \in \partial A$ if and only if any neighborhood of x contains points of A and of its complement $\mathcal{X} \setminus A$.*

D Subspaces

3.11. Proposition. *Let A be a subset of a topological space $\mathcal{Y}$. Then all subsets $V \subset A$ such that $V = A \cap W$ for some open set W in $\mathcal{Y}$ form a topology on A.*

The described topology is called the induced topology on A.

A subset A of a topological space $\mathcal{Y}$ equipped with the induced topology is called a subspace of $\mathcal{Y}$. It is straightforward to check that this notion agrees with the notion introduced in 1C; that is, if $\mathcal{Y}$ is a metric space, then any subset $A \subset \mathcal{Y}$ comes with a metric, and the topology defined by this metric coincides with the induced topology on A.

Proof. We need to check the conditions in 2.1.

[2] Also known as frontier.

First, the whole set A and the empty set are included; indeed, $\varnothing = A \cap \varnothing$ and $A = A \cap \mathcal{Y}$.

Assume $\{V_\alpha\}$ is a family of open sets in A; here α runs in some index set, say $\mathcal{I}$. In other words, for each V_α there is an open set $W_\alpha \subset \mathcal{Y}$ such that $V_\alpha = A \cap W_\alpha$. Note that

$$\bigcup_\alpha V_\alpha = A \cap \left(\bigcup_\alpha W_\alpha\right).$$

Since the union of $\{W_\alpha\}$ is open in $\mathcal{Y}$ (2.1b), the union of $\{V_\alpha\}$ is open in the induced topology on A.

Assume V_1 and V_2 are open in A; that is, $V_1 = A \cap W_1$ and $V_2 = A \cap W_2$ for some open sets $W_1, W_2 \subset \mathcal{Y}$. Note that

$$V_1 \cap V_2 = A \cap (W_1 \cap W_2).$$

Since the intersection $W_1 \cap W_2$ is open in $\mathcal{Y}$ (2.1c), the intersection $V_1 \cap V_2$ is open in the induced topology on A. $\square$

3.12. Exercise. *Let $\mathcal{A}$ be a subspace in $\mathcal{X}$. Given a set $S \subset \mathcal{A}$ denote by $\operatorname{Int}_{\mathcal{A}} S$, $\operatorname{Cl}_{\mathcal{A}} S$, $\operatorname{Int}_{\mathcal{X}} S$, and $\operatorname{Cl}_{\mathcal{X}} S$ its interior and closure in $\mathcal{A}$ and $\mathcal{X}$, respectively. Show that $\operatorname{Int}_{\mathcal{A}} S \supset \operatorname{Int}_{\mathcal{X}} S$ and $\operatorname{Cl}_{\mathcal{A}} S \subset \operatorname{Cl}_{\mathcal{X}} S$. Provide examples showing that these inclusions might be strict.*

Let A be a subset of a topological space $\mathcal{X}$ and let $\mathcal{Y}$ be another topological space. A map $h\colon A \to \mathcal{Y}$ is called continuous if it is continuous with respect to the induced topology on A.

Recall that the restriction of a map $f : \mathcal{X} \to \mathcal{Y}$ to $A \subset \mathcal{X}$ is obtained by keeping the same rule as f but only allowing inputs from A. It is usually denoted by $f|_A$; so $(f|_A)(a) = f(a)$ if $a \in A$ and otherwise it is undefined.

Observe that $(f|_A)^{-1}(W) = A \cap f^{-1}(W)$ for any $W \subset \mathcal{Y}$. It follows that if f is continuous, then so is $f|_A\colon A \to \mathcal{Y}$; here we consider the set A with the induced topology. The following exercise provides a partial converse.

3.13. Exercise! *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a map between topological spaces. Suppose $A, B \subset \mathcal{X}$ are sets such that $A \cup B = \mathcal{X}$ and either*

(a) both A and B are open sets, or

(b) both A and B are closed sets.

Show that f is continuous if and only if so are the restrictions $f|_A$ and $f|_B$.

(c) Show that this is no longer true without the assumptions in (a) or (b).

Chapter 4

Cantor set

The Cantor set is a simple subset of the real line with surprisingly rich properties. It serves as a standard test example in many problems of topology and analysis.

A Isolated points

A point p in a topological space $\mathcal{X}$ is called isolated if the one-point set $\{p\}$ is open in $\mathcal{X}$. Note that a space is discrete if every point in it is isolated.

Further, a point p in a subset $A \subset \mathcal{X}$ is called isolated if it is an isolated point of the subspace A, which means that there is an open set $V \ni p$ that intersects A only at p. For example, in the set $A = \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\} \subset \mathbb{R}$ all points except 0 are isolated.

In 1882, Magnus Mittag-Leffler posed a question, which could be restated as follows: *Does there exist a nowhere dense closed set $\mathcal{C} \subset \mathbb{R}$ without isolated points?* This might be the first question in point-set topology. The question was answered by Georg Cantor in 1883; he constructed an example that now bears his name.

B Cantor set

Let us start with the unit interval $\mathcal{C}_0 = [0, 1]$, and remove its open middle third $(\frac{1}{3}, \frac{2}{3})$; we get a new set $\mathcal{C}_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$ which is a union of two disjoint closed intervals. Let us pass to a new subset $\mathcal{C}_2$ by removing the open middle third of each closed interval in $\mathcal{C}_1$, and repeat the procedure obtaining a nested sequence of closed sets

$$[0, 1] = \mathcal{C}_0 \supset \mathcal{C}_1 \supset \dots$$

each of which is a union of finitely many disjoint closed intervals. Note that $\mathcal{C}_n$ consists of 2^n closed intervals of length $\frac{1}{3^n}$. The intersection

$$\mathcal{C} = \mathcal{C}_0 \cap \mathcal{C}_1 \cap \dots$$

is called the Cantor set.

By construction, $\mathcal{C}$ is a closed nowhere dense set; in particular, it contains no intervals other than points. On the other hand, $\mathcal{C}$ has no isolated points. Indeed, a neighborhood V of any point $x \in \mathcal{C}$ must contain one of the 2^n intervals in $\mathcal{C}_n$ for a sufficiently large integer n; denote it by $[a, b]$. Note that a and b remain endpoints of intervals at every later stage $\mathcal{C}_{n+1}$, $\mathcal{C}_{n+2}, \dots$. Therefore $a, b \in \mathcal{C}$, and since $x \neq a$ or $x \neq b$, we get that x is not isolated.

Alternatively, one can describe $\mathcal{C}$ as all reals $x \in [0, 1]$ that can be written in base 3 only using two digits 0 and 2 (without using 1); that is, $x \in \mathcal{C}$ if there is a sequence $d_1, d_2, \dots$ such that

$$x = (0.d_1 d_2 \dots)_3 := \frac{d_1}{3^1} + \frac{d_2}{3^2} + \dots$$

and each d_i is either 0 or 2.

4.1. Exercise. *Prove the equivalence of the two definitions and use it to show that $\mathcal{C}$ is uncountable.*[1]

C Open versus closed

Observe that any open set $V \subset \mathbb{R}$ is a union of open intervals. Indeed, given a point $x \in V$ there is a unique maximal open interval (a, b) (finite or infinite) such that $x \in (a, b) \subset V$. Note that if two maximal intervals intersect, then they have to coincide (otherwise they would not be maximal). Furthermore, every open interval contains a rational

[1]This exercise requires a bit more set theory than usual; it uses that the set of all subsets of an infinite set is uncountable, which follows from Cantor's power set theorem.

number, and since the set of rational numbers is countable, we have that the set of the maximal intervals is countable as well.

One might naively think that closed sets have a similar feature, but the Cantor set shows that this is not the case — it is not a countable union of closed intervals (including one-point sets).

4.2. Exercise. *Find three disjoint open sets on the real line that have the same nonempty boundary.*

Chapter 5

Maps

Recall that continuous maps were defined in 2D; now we will discuss their relatives.

A Homeomorphisms

A bijection $f\colon \mathcal{X} \to \mathcal{Y}$ between topological spaces is called a homeomorphism if f and its inverse $f^{-1}\colon \mathcal{Y} \to \mathcal{X}$ are continuous.[1]

Topological spaces $\mathcal{X}$ and $\mathcal{Y}$ are called homeomorphic (briefly, $\mathcal{X} \simeq \mathcal{Y}$) if there is a homeomorphism $f\colon \mathcal{X} \to \mathcal{Y}$.

A map $f\colon \mathcal{X} \to \mathcal{Y}$ is called an embedding if f defines a homeomorphism from $\mathcal{X}$ to the subspace $f(\mathcal{X})$ in $\mathcal{Y}$.

5.1. Exercise. *Show that any homeomorphism is a continuous bijection.*

Give an example of a continuous bijection between topological spaces that is not a homeomorphism.

5.2. Exercise. *Show that $x \mapsto e^x$ is a homeomorphism $\mathbb{R} \to (0, \infty)$.*

5.3. Exercise. *Construct a homeomorphism $f\colon \mathbb{R} \to (0, 1)$.*

5.4. Exercise. *Show that $\simeq$ is an equivalence relation; that is, for any topological spaces $\mathcal{X}$, $\mathcal{Y}$, and $\mathcal{Z}$ we have the following:*

(a) $\mathcal{X} \simeq \mathcal{X}$;

(b) if $\mathcal{X} \simeq \mathcal{Y}$, then $\mathcal{Y} \simeq \mathcal{X}$;

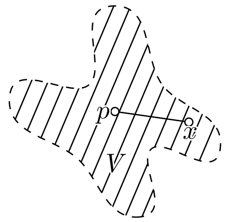
[1] The terms *homomorphism* and *homeomorphism* look similar, and they have similar, but different meanings. Homomorphisms are creatures from abstract algebra; we will need them starting from Chapter 15.

(c) if $\mathcal{X} \simeq \mathcal{Y}$ and $\mathcal{Y} \simeq \mathcal{Z}$, then $\mathcal{X} \simeq \mathcal{Z}$.

5.5. Advanced exercise. *Let $Q \subset \mathbb{R}^2$ be a solid square with side S and vertex v. Show that $Q \setminus S \simeq Q \setminus \{v\}$.*

5.6. Advanced exercise. *Prove that the complement of a circle in $\mathbb{R}^3$ is homeomorphic to $\mathbb{R}^3$ without a line ℓ and a point $p \notin \ell$.*

Recall that a set $V \subset \mathbb{R}^2$ is called star-shaped if there exists a point $p \in V$ such that for all $x \in V$ the line segment from p to x lies in V.



5.7. Advanced exercise. *Show that any nonempty open star-shaped set in the plane is homeomorphic to the open disc.*

5.8. Advanced exercise. *Show that the complements of two countable dense subsets of the plane are homeomorphic.*

B Closed and open maps

5.9. Definition. *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a map between topological spaces; f is called open if, for any open set $V \subset \mathcal{X}$, the image $f(V)$ is open in $\mathcal{Y}$. Similarly, f is called closed if, for any closed set $Q \subset \mathcal{X}$, the image $f(Q)$ is closed in $\mathcal{Y}$.*

Note that a bijective map is open if its inverse is continuous. By 3.2, we also have that a bijective map is closed if its inverse is continuous. Thus we get the following.

5.10. Observation. *A homeomorphism can be defined as a continuous open bijection as well as a continuous closed bijection.*

5.11. Exercise. *Give an example of a map f between topological spaces such that*

(a) f is continuous and open, but not closed,
(b) f is continuous and closed, but not open,
(c) f is closed and open, but not continuous.

5.12. Advanced exercise. *Construct a map $\mathbb{R} \to \mathbb{R}$ that is open, but not continuous.*

Chapter 6

Constructions

In 3D we discussed induced topology. This construction produces a new topological space from the given one. Now we introduce several other constructions of this type.

A Product space

Recall that $\mathcal{X} \times \mathcal{Y}$ denotes the set of all pairs (x, y) such that $x \in \mathcal{X}$ and $y \in \mathcal{Y}$.

Suppose that the sets $\mathcal{X}$ and $\mathcal{Y}$ are equipped with topologies. Let us equip $\mathcal{X} \times \mathcal{Y}$ with a topology by declaring that a set is open in $\mathcal{X} \times \mathcal{Y}$ if it can be presented as a union of sets of the following type:

$$V \times W := \{ (x, y) \in \mathcal{X} \times \mathcal{Y} \, : \, x \in V, \ y \in W \}$$

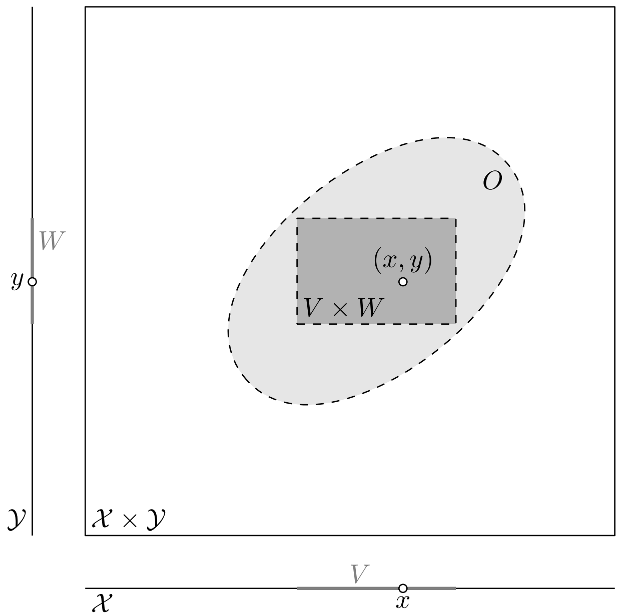
for open sets $V \subset \mathcal{X}$ and $W \subset \mathcal{Y}$. In other words, a subset O is open in $\mathcal{X} \times \mathcal{Y}$ if and only if for any point $(x, y) \in O$ there are open neighborhoods $V \subset \mathcal{X}$ and $W \subset \mathcal{Y}$ of x and y, respectively such that $O \supset V \times W$.

By default, we assume that $\mathcal{X} \times \mathcal{Y}$ is equipped with the product topology; in this case, $\mathcal{X} \times \mathcal{Y}$ is called the p r o d u c t s p a c e.

6.1. Proposition. *The product topology is indeed a topology.*

Proof. Parts *(a)* and *(b)* in 2.1 are evident. It remains to check *(c)*. Consider two sets

$$O = \bigcup_{\alpha \in \mathcal{I}} V_\alpha \times W_\alpha \quad \text{and} \quad O' = \bigcup_{\beta \in \mathcal{J}} V'_\beta \times W'_\beta.$$



We need to show that $O \cap O'$ can be presented as a union of products of open sets; the latter follows from the identity

❶ $$O \cap O' = \bigcup_{\alpha,\beta} (V_\alpha \cap V'_\beta) \times (W_\alpha \cap W'_\beta).$$

Checking ❶ is straightforward. Indeed, $(x, y) \in O \cap O'$ means that $(x, y) \in O$ and $(x, y) \in O'$; the latter means that $x \in V_\alpha$, $y \in W_\alpha$ and $x \in V'_\beta$, $y \in W'_\beta$ for *some* α and β. In other words, $x \in V_\alpha \cap V'_\beta$ and $y \in W_\alpha \cap W'_\beta$ for *some* α and β; the latter means that (x, y) belongs to the right-hand side in ❶. □

By spelling out the above definition, we get the following.

6.2. Observation. *Two maps $f\colon \mathcal{X} \to \mathcal{Y}$ and $g\colon \mathcal{X} \to \mathcal{Z}$ are continuous if and only if $x \mapsto (f(x), g(x))$ is a continuous map $\mathcal{X} \to \mathcal{Y} \times \mathcal{Z}$.*

This observation can be used used for constructing continuous maps; the following exercise provides examples.

6.3. Exercise. *Let f and g be continuous real-valued functions on the topological space $\mathcal{X}$. Show that (a) $h_1 = f + g$, (b) $h_2 = f \cdot g$, and (c) $h_3(x) := \max\{\, f(x), g(x)\,\}$ are continuous.*

6.4. Exercise. *Show that $\partial(A \times B) = (\partial A \times \bar{B}) \cup (\bar{A} \times \partial B)$ for any* *subsets $A \subset \mathcal{X}$, $B \subset \mathcal{Y}$ and their product $A \times B$ in the product space $\mathcal{X} \times \mathcal{Y}$.*

6.5. Exercise. *Construct a function $f\colon [0,1]\times[0,1] \to [0,1]$ such that f is not continuous, but the functions $x \mapsto f(a,x)$ and $x \mapsto f(x,b)$ are continuous for any constants $a, b \in [0,1]$.*

6.6. Exercise. *Show that the projection $(x,y) \mapsto x$ defines an open map $\mathcal{X} \times \mathcal{Y} \to \mathcal{X}$.*

B Base

6.7. Definition. *A family $\mathscr{B}$ of open sets in a topological space $\mathcal{X}$ is called its base if every open set in $\mathcal{X}$ is a union of sets in $\mathscr{B}$.*

By 1.17c, *open balls form a base of a metric space.*

A base completely defines the topology, but a typical topology has many different bases. In metric spaces, for example, the set of all balls with rational radii is a base; another example is the set of all balls with radii smaller than 1.

In many cases, it is convenient (and also economical) to describe the topology by specifying its base. For example, the product topology on $\mathcal{X} \times \mathcal{Y}$ can be redefined as a *topology with a base formed by all products $V \times W$, where V is open in $\mathcal{X}$, and W is open in $\mathcal{Y}$.*

6.8. Observation. *Let $\mathscr{B}$ be a family of open sets in a topological space $\mathcal{X}$. Then $\mathscr{B}$ is a base in $\mathcal{X}$ if and only if for any point $x \in \mathcal{X}$ and any neighborhood $N \ni x$ there is $B \in \mathscr{B}$ such that $x \in B \subset N$.*

Proof; if part. Choose an open set N. For any $x \in N$ choose an element B_x of the base such that $x \in B_x \subset N$. Since

$$N = \bigcup_{x \in N} B_x,$$

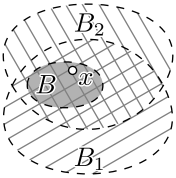
the if part follows.

Only-if part. Now suppose that $\mathscr{B}$ is a base. Then

$$N = \bigcup_{\alpha} B_\alpha,$$

where $B_\alpha \in \mathscr{B}$ for each α. Then for any $x \in N$ there is α such that $B_\alpha \ni x$; in this case, $x \in B_\alpha \subset N$, and the only-if part follows. $\square$

6.9. Proposition. *Let $\mathscr{B}$ be a set of subsets of some set $\mathcal{X}$. Then $\mathscr{B}$ is a base of some topology on $\mathcal{X}$ if and only if it satisfies the following conditions:*



(a) $\mathscr{B}$ covers $\mathcal{X}$; that is, every point $x \in \mathcal{X}$ lies in some set $B \in \mathscr{B}$.

(b) For each pair of sets $B_1, B_2 \in \mathscr{B}$ and each point $x \in B_1 \cap B_2$ there exists a set $B \in \mathscr{B}$ such that

$$x \in B \subset B_1 \cap B_2.$$

Proof. Denote by $\mathscr{O}$ the set of all unions of sets in $\mathscr{B}$. We need to show that $\mathscr{O}$ is a topology on $\mathcal{X}$.

Evidently, the union of any family of sets in $\mathscr{O}$ is in $\mathscr{O}$. Further, $\mathcal{X}$ is in $\mathscr{O}$ by *(a)*. The empty set is in $\mathscr{O}$ since it is a union of the empty family.

It remains to check 2.1c; suppose

$$O = \bigcup_\alpha B_\alpha \quad \text{and} \quad O' = \bigcup_\beta B'_\beta,$$

where α and β run in some index sets, and B_α, $B'_\beta \in \mathscr{B}$ for any α and β. Then $x \in O \cap O'$ if and only if $x \in B_\alpha$ and $x \in B'_\beta$ for some α and β. By *(b)*, we can choose $B \in \mathscr{B}$ so that $x \in B \subset B_\alpha \cap B'_\beta$. Since $B_\alpha \cap B'_\beta \subset O \cap O'$, it follows that

for any $x \in O \cap O'$ there is $B_x \in \mathscr{B}$ such that $x \in B_x \subset O \cap O'$.

Observe that

$$O \cap O' = \bigcup_{x \in O \cap O'} B_x.$$

It follows that $O \cap O' \in \mathscr{O}$ if $O, O' \in \mathscr{O}$. $\square$

6.10. Advanced exercise. *Let $\mathbb{N} = \{1, 2, \dots\}$ be the set of natural numbers, and let $\mathscr{B}$ be a set of all arithmetic progressions in $\mathbb{N}$; that is, $\mathscr{B}$ includes the set $\{a, a+d, a+2\cdot d, \dots\}$ for any $a, d \in \mathbb{N}$.*

Show that $\mathscr{B}$ is a base of some topology on $\mathbb{N}$. Is it true that in this topology the set $\{1\}$ is open and/or closed?

C Prebase

Suppose $\mathscr{P}$ is a family of subsets of $\mathcal{X}$ that covers the whole space; that is, $\mathcal{X}$ is a union of all sets in $\mathscr{P}$. By 6.9, the set of all finite intersections of sets in $\mathscr{P}$ is a base for *some* topology on $\mathcal{X}$. The set $\mathscr{P}$ is called a prebase for this topology.[1]

Note that any base is a prebase, but not the other way around. Spelling the defining properties of open sets (2.1) and the definition of prebase, we get the following.

6.11. Observation. *Let $\mathscr{P}$ be a prebase for the topology on $\mathcal{Y}$. Then a map $f\colon \mathcal{X} \to \mathcal{Y}$ is continuous if and only if $f^{-1}(P)$ is open for any set P in $\mathscr{P}$.*

There are almost no restrictions on a prebase — we may start with any family $\mathscr{P}$ of subsets that covers the whole space $\mathcal{X}$ and define a topology by declaring that *$\mathscr{P}$ is a prebase for the topology.* It defines the weakest topology on $\mathcal{X}$ such that every set of $\mathscr{P}$ is open.

6.12. Exercise! *Let $\mathcal{X}$ and $\mathcal{Y}$ be topological spaces.*

(a) Show that the product topology on $\mathcal{X} \times \mathcal{Y}$ can be redefined as a topology with a prebase formed by all products $\mathcal{X} \times W$ and $V \times \mathcal{Y}$, where V is open in $\mathcal{X}$ and W is open in $\mathcal{Y}$.

(b) Apply part (a) and 6.11 to prove the following: given a map $f\colon \mathcal{X} \to \mathcal{Y}$, consider the map $F\colon \mathcal{X} \to \mathcal{X} \times \mathcal{Y}$ defined by $F\colon x \mapsto (x, f(x))$. Show that f is continuous if and only if F is an embedding.

D Initial topology

Let $\mathcal{X}$ and $\mathcal{Y}$ be topological spaces. Note that the product topology on $\mathcal{X} \times \mathcal{Y}$ is the weakest topology such that projections $(x, y) \mapsto x$ and $(x, y) \mapsto y$ are continuous.

Indeed, these projections are continuous if the inverse images of all open sets in $\mathcal{X}$ and $\mathcal{Y}$ are open in $\mathcal{X} \times \mathcal{Y}$. In other words, the topology on $\mathcal{X} \times \mathcal{Y}$ must contain all sets of the form $V \times \mathcal{Y}$ and $\mathcal{X} \times W$ for open sets $V \subset \mathcal{X}$ and $W \subset \mathcal{Y}$. By 6.12a, these sets form a prebase in $\mathcal{X} \times \mathcal{Y}$ and its topology is the weakest topology that contains these sets.

More generally, given a family of maps $f_\alpha\colon S \to \mathcal{Y}_\alpha$ from a set S to topological spaces $\mathcal{Y}_\alpha$, we can introduce a topology on S by stating that the inverse images $f_\alpha^{-1}(W_\alpha)$ for open sets $W_\alpha \subset \mathcal{Y}_\alpha$ form its prebase. It defines the topology on S induced by the maps f_α; it is the weakest topology on S that makes all maps f_α continuous. For example, if we apply this construction to a single inclusion map $S \hookrightarrow \mathcal{Y}$

[1] Prebase is also known as a subbase, but this is an ill-chosen term. Indeed, prebase is *not* a subset of a base that is itself a base, so subbase breaks the pattern of subspace, subgroup, subsequence, subcover, and so on.

of a subset $S \subset \mathcal{Y}$, then we get the induced topology on S (see 3D).

This construction produces a topology on the source space; for that reason it is also called the initial topology. It can be used to define a topology on an infinite product of spaces, as the induced topology by all its projections. This topology is called the Tychonoff topology.

6.13. Advanced exercise. *Let $\mathcal{O}$ be the topology on $\mathbb{R}$ induced by the maps $x \mapsto (\cos(t{\cdot}x), \sin(t{\cdot}x))$ for all $t \in \mathbb{R}$. Show that the space $(\mathbb{R}, \mathcal{O})$ is not metrizable.*

E Final topology

In the previous section, we defined a natural way to move a topology from the target space to the source of a map. Namely, if $f\colon \mathcal{X} \to \mathcal{Y}$ is a map, and $\mathcal{Y}$ comes with a topology, then a subset $V \subset \mathcal{X}$ is declared to be open if there is an open subset $W \subset \mathcal{Y}$ such that $V = f^{-1}(W)$.

The following exercise describes an analogous construction that moves a topology from source to target. It can be solved by checking the conditions in 2.1 as we did in 3D.

6.14. Exercise. *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a map between two sets. Assume $\mathcal{X}$ is equipped with a topology. Declare a subset $W \subset \mathcal{Y}$ to be open if the subset $V = f^{-1}(W)$ is open in $\mathcal{X}$. Show that it defines a topology on $\mathcal{Y}$.*

The constructed topology on $\mathcal{Y}$ is called the final topology for f. Note that any point $y \in \mathcal{Y}$ outside the image of $\mathcal{X}$ forms an open one-point set $\{y\}$; indeed, if $y \notin f(\mathcal{X})$, then $f^{-1}\{y\} = \varnothing$, and so $\{y\}$ is open.

6.15. Exercise. *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a continuous map between topological spaces.*

(a) Show that the initial topology for f on $\mathcal{X}$ is weaker than its own topology.

(b) Show that the final topology for f on $\mathcal{Y}$ is stronger than its own topology.

6.16. Exercise. *Let $g\colon \mathcal{X} \to \mathcal{Y}$ be a continuous map.*

(a) Suppose $\mathcal{X}$ is equipped with the initial topology. Show that a map $f\colon \mathcal{W} \to \mathcal{X}$ is continuous if and only if the composition $g \circ f\colon \mathcal{W} \to \mathcal{Y}$ is continuous.

(b) Suppose $\mathcal{Y}$ is equipped with the final topology. Show that a map $h\colon \mathcal{Y} \to \mathcal{Z}$ is continuous if and only if the composition $h \circ g\colon \mathcal{X} \to \mathcal{Z}$ is continuous.

6.17. Claim. *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a continuous surjective map. Assume f is (a) open or (b) closed. Then $\mathcal{Y}$ has the final topology induced by f.*

Proof. Choose a set $W \subset \mathcal{Y}$, and let $V = f^{-1}(W)$. Since f is continuous, V is open if W is. Hence the topology on $\mathcal{Y}$ has to be weaker than the final topology.

(a). Since f is open, we have that W is open if V is. Thus the topology on $\mathcal{Y}$ has to be stronger than the final topology, and the result follows.

(b). Set $A = \mathcal{X} \setminus V$ and $B = \mathcal{Y} \setminus W$. If V is open, then A is closed. Since f is surjective, $B = f(A)$. Since f is a closed map, $B = f(A)$ is closed as well. Therefore, $W = \mathcal{Y} \setminus B$ is open. Again, the topology on $\mathcal{Y}$ has to be stronger than the final topology, and the result follows. □

F Quotient topology

The initial topology is used mostly for injective maps; in this case, it is essentially the same as the induced topology. Similarly, the final topology is mostly used for surjective maps. This particular case of the construction is called the quotient topology, which we are about to describe.

Let $\sim$ be an equivalence relation on a topological space $\mathcal{X}$; that is, for any $x, y, z \in \mathcal{X}$ the following conditions hold:

- $x \sim x$;
- if $x \sim y$, then $y \sim x$;
- if $x \sim y$ and $y \sim z$, then $x \sim z$.

Recall that the set

$$[x] = \{\, y \in \mathcal{X} \ :\ y \sim x \,\}$$

is called the equivalence class of x. The set of all equivalence classes in $\mathcal{X}$ will be denoted by $\mathcal{X}/{\sim}$.

Observe that $x \mapsto [x]$ defines a surjective map $\mathcal{X} \to \mathcal{X}/{\sim}$. The corresponding final topology on $\mathcal{X}/{\sim}$ is called the quotient topology. By default, the set $\mathcal{X}/{\sim}$ is equipped with the quotient topology; in this case, it is called a quotient space.

The following exercise ties equivalence relations with maps.

6.18. Exercise. *Show that an arbitrary map $f\colon \mathcal{X} \to \mathcal{Y}$ defines the following equivalence relation on $\mathcal{X}$:*

$$x \sim x' \quad \Leftrightarrow \quad f(x) = f(x').$$

Moreover,

$$f(x) = y \quad \Leftrightarrow \quad [x] = f^{-1}\{y\}.$$

6.19. Exercise. *Construct an equivalence relation* $\sim$ *on* $[0,1]$ *such that the quotient space* $[0,1]/\sim$ *is (a) a connected two-point space or (b) a concrete two-point space.*

Given a set A in a topological space $\mathcal{X}$, the space $\mathcal{X}/A$ is defined as the quotient space $\mathcal{X}/\sim$ for the minimal equivalence relation such that $a \sim b$ for any $a, b \in A$. Intuitively, the space $\mathcal{X}/A$ is obtained from $\mathcal{X}$ by gluing together all points in A.

6.20. Exercise. *Describe the quotient space* $[0,1]/(0,1)$*, where* $[0,1]$ *and* $(0,1)$ *are real intervals; that is, list the points and the open sets of the quotient space.*

One may also check by hand that $[0,1]/\{0,1\} \simeq \mathbb{S}^1$; that is, the unit segment with identified ends is homeomorphic to the circle. However, Theorem 9.15 will provide a very general tool that helps to prove such statements.

G Actions

Let $\mathcal{X}$ be a set. Recall that the identity map $\mathcal{X} \to \mathcal{X}$ takes a point in $\mathcal{X}$ and returns the same point; it will be denoted by $\mathrm{id}_{\mathcal{X}} \colon \mathcal{X} \to \mathcal{X}$.

A subset G of bijections $\mathcal{X} \to \mathcal{X}$ that includes $\mathrm{id}_{\mathcal{X}}$ and is closed under composition and inversion is called a group of symmetries of $\mathcal{X}$. In this case we may say that G acts on $\mathcal{X}$, briefly $G \circlearrowright \mathcal{X}$.

Typically we are interested in group actions that preserve some structure of the set $\mathcal{X}$. If $\mathcal{X}$ is a topological space, then it is natural to assume that each $g \in G$ is a continuous bijection $\mathcal{X} \to \mathcal{X}$. In this case the action $G \circlearrowright \mathcal{X}$ is called continuous. For such an action each $g \in G$ defines a homeomorphism $\mathcal{X} \to \mathcal{X}$. Indeed, if $g \in G$, then $g^{-1} \in G$; therefore g as well as its inverse must be continuous. Actions on topological spaces will always be assumed to be continuous.

Given an action $G \circlearrowright \mathcal{X}$, we can equip the set G with a product defined as composition; that is, if $g, h \in G$ then their product $g \cdot h \in G$ is defined as $(g \cdot h)(x) := g(h(x))$. With this operation, G becomes a group; that is, it meets the following properties:

- (associativity) for any $f, g, h \in G$ we have

$$(f \cdot g) \cdot h = f \cdot (g \cdot h);$$

- (existence of neutral element) there exists an element $1 \in G$ such that for every $g \in G$ we have

$$1{\cdot}g = g{\cdot}1 = g;$$

- (inverse) for each $g \in G$ there exists $g^{-1} \in G$ such that

$$g{\cdot}g^{-1} = g^{-1}{\cdot}g = 1.$$

Checking these properties is straightforward. Associativity follows since it holds for composition, and one can take $1 = \mathrm{id}_{\mathcal{X}}$.

Finally note that an abstract group (that is, a set G with a product that meets the three properties above) acts on itself by $g\colon h \mapsto g{\cdot}h$, so any group acts on some set. Furthermore, we may equip G with the discrete topology and this way we obtain a continuous action $G \circlearrowright G$. (One may think that actions are more fundamental than groups.)

For an action $G \circlearrowright \mathcal{X}$, it is common to denote $g(x)$ by $g{\cdot}x$, where $g \in G$ and $x \in \mathcal{X}$. Note that associativity implies that the expression

$$g{\cdot}h{\cdot}x$$

makes sense for any $g, h \in G$ and $x \in \mathcal{X}$; that is, $g{\cdot}(h{\cdot}x) = (g{\cdot}h){\cdot}x$.

We just defined *left actions*. In practice, left actions are used more often, and unless stated otherwise, by an action we will mean a left action.

The *right actions* are defined similarly but the homeomorphism $\mathcal{X} \to \mathcal{X}$ that corresponds to $g \in G$ is written as $x \mapsto x{\cdot}g$ and so $(x{\cdot}g){\cdot}h = x{\cdot}(g{\cdot}h)$ for any $x \in \mathcal{X}$ and $g, h \in G$. The right action of G on $\mathcal{X}$ is the same thing as a left action of the *opposite group* G^{op}. The group G^{op} has the same set of elements as G, but the product $\bullet$ in G^{op} is taken in the opposite order; that is, $g \bullet h := h{\cdot}g$. For commutative groups, left and right actions differ only by notation.

H Orbit spaces

Choose a continuous action $G \circlearrowright \mathcal{X}$. The set

$$G{\cdot}x := \{\, g{\cdot}x \in \mathcal{X} \;:\; g \in G \,\}$$

is called the G-*orbit* of x (or, briefly, *orbit*).

Set $x \sim y$ if there is $g \in G$ such that $y = g{\cdot}x$. Observe that $\sim$ is an equivalence relation on $\mathcal{X}$. Indeed, $x \sim x$ since $x = 1{\cdot}x$. Further, if $y = g{\cdot}x$, then

$$x = 1{\cdot}x = g^{-1}{\cdot}g{\cdot}x = g^{-1}{\cdot}y;$$

since $g^{-1} \in G$ we get that $x \sim y \Rightarrow y \sim x$. Finally, suppose $x \sim y$ and $y \sim z$; that is, $y = g{\cdot}x$ and $z = h{\cdot}y$ for some $g, h \in G$. Then $z = (h{\cdot}g){\cdot}x$; therefore $x \sim z$.

For the described equivalence relation, the quotient space $\mathcal{X}/{\sim}$ can also be denoted by $\mathcal{X}/G$; it is called the quotient of $\mathcal{X}$ by the action of G.

Note that $[x] = G{\cdot}x$; that is, the orbit of x coincides with its equivalence class for $\sim$. For that reason, $\mathcal{X}/G$ is also called the orbit space.

6.21. Exercise. *Positive real numbers $\mathbb{R}_+$ act on $\mathbb{R}$ by multiplication. Describe the quotient space $\mathbb{R}/\mathbb{R}_+$; that is, list the points and the open sets of the quotient space.*

6.22. Exercise! *Suppose a group G acts[2] on a topological space $\mathcal{X}$ and $f\colon \mathcal{X} \to \mathcal{X}/G$ is the quotient map.*
(a) Show that f is open.
(b) Assume G is finite. Show that f is closed.

[2]Recall that actions on topological spaces are assumed to be continuous.

Chapter 7

Compactness

In this chapter we define compact spaces — especially nice topological spaces that behave in many ways like finite sets.

In practice, compactness is a powerful replacement for "closed and bounded set" in $\mathbb{R}^n$, but the general definition does not resemble it at first glance.

A Definition

We will denote by $\{V_\alpha\} = \{V_\alpha\}_{\alpha \in \mathcal{I}}$ a family of sets, where α runs in an arbitrary index set $\mathcal{I}$.

7.1. Definition. *A family $\{V_\alpha\}$ of open subsets of a topological space $\mathcal{X}$ is called its open cover if it covers the whole $\mathcal{X}$; that is, if every $x \in \mathcal{X}$ belongs to some V_α.*

More generally, $\{V_\alpha\}$ is an open cover of a subset $S \subset \mathcal{X}$ if any $s \in S$ belongs to some V_α.

A subfamily of the cover $\{V_\alpha\}$ that is also a cover is called a subcover of $\{V_\alpha\}$.

Observe that any base of a space is a cover, but not every cover is its base. For example, any space admits a cover by one set (the whole space), but it forms a base only if the topology is concrete.

7.2. Exercise. *Let $\{V_\alpha\}$ be an open cover of a topological space $\mathcal{X}$. Show that $W \subset \mathcal{X}$ is open if and only if each intersection $W \cap V_\alpha$ is open.*

Conclude that a map $f\colon \mathcal{X} \to \mathcal{Y}$ is continuous if for any α the restriction $f|_{V_\alpha}\colon V_\alpha \to \mathcal{Y}$ is continuous.

7.3. Definition. *A topological space $\mathcal{X}$ is called* compact *if any open cover $\{V_\alpha\}$ of $\mathcal{X}$ contains a finite subcover $\{V_{\alpha_1}, \dots, V_{\alpha_n}\}$.*

Analogously, a subset S of a topological space $\mathcal{X}$ is called compact *if any cover of S contains a finite subcover of S.*

Spelling the definitions, we get that *a subset S of a topological space is compact if and only if S equipped with the induced topology is a compact space.*

Clearly, *any finite topological space is compact.* The next exercise provides a source of examples of infinite compact spaces. More interesting examples will appear in Section 7C and further.

7.4. Exercise. *Show that any set equipped with the cofinite topology is compact.*

The solution of the following problem illustrates how one can apply the definition of compactness.

7.5. Problem. *A compact subset S of the real line has to be closed and* bounded*; that is, there is $c \in \mathbb{R}$ such that $|s| \leqslant c$ for any $s \in S$.*

Solution. Suppose $S \subset \mathbb{R}$ is a compact set. Assume S is not closed. Then we can choose a point $s \in \bar{S} \setminus S$ and consider the cover of S by the complements of $[s - \varepsilon, s + \varepsilon]$ for all $\varepsilon > 0$. Since it has a finite subcover, a neighborhood of s does not intersect S, so $s \notin \bar{S}$ — a contradiction.

Applying the same reasoning to the cover of S by all intervals $(-c, c)$ for $c > 0$, one can show that S is bounded. □

7.6. Exercise!

(a) *Construct a topological space with a set that is compact, but not closed.*

(b) *Show that the union of two compact sets is compact.*

(c) *Construct a topological space with two compact sets such that their intersection is not compact.*

(d) *Show that the space in (a) cannot be metrizable. Try to do the same for (c).*

B Finite intersection property

7.7. Proposition. *A topological space $\mathcal{X}$ is compact if and only if for any family of closed sets $\{Q_\alpha\}$ in $\mathcal{X}$ such that*

$$\bigcap_\alpha Q_\alpha = \varnothing$$

there is a finite family $\{Q_{\alpha_1}, \dots, Q_{\alpha_n}\}$ such that

$$Q_{\alpha_1} \cap \dots \cap Q_{\alpha_n} = \varnothing.$$

We say that a family of sets has finite intersection property if any finite subfamily of it has non-empty intersection. According to the proposition, compactness is equivalent to the following property: *any family of closed sets with finite intersection property has a nonempty intersection.*

Proof. By the definition of closed sets, the complements $V_\alpha = \mathcal{X} \setminus Q_\alpha$ are open. Note that

$$\bigcup_\alpha V_\alpha = \mathcal{X} \setminus \left(\bigcap_\alpha Q_\alpha \right);$$

that is, $\{V_\alpha\}$ is an open cover of $\mathcal{X}$ if and only if $\bigcap Q_\alpha = \varnothing$.

Observe that

$$Q_{\alpha_1} \cap \dots \cap Q_{\alpha_n} = \mathcal{X} \setminus (V_{\alpha_1} \cup \dots \cup V_{\alpha_n}).$$

Thus, $\{V_{\alpha_1}, \dots, V_{\alpha_n}\}$ is a subcover if and only if $Q_{\alpha_1} \cap \dots \cap Q_{\alpha_n} = \varnothing$, hence the result. □

7.8. Exercise. *Let $Q_1 \supset Q_2 \supset \dots$ be a nested sequence of closed nonempty sets in a compact space $\mathcal{K}$. Show that there is a point $q \in \mathcal{K}$ such that $q \in Q_n$ for any n.*

7.9. Advanced exercise. *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a closed continuous map. Assume that $\mathcal{Y}$ is compact and every point $y \in \mathcal{Y}$ has a compact inverse image $f^{-1}\{y\} \subset \mathcal{X}$. Show that $\mathcal{X}$ is compact.*

C Real interval

7.10. Theorem. *Any closed interval $[a, b]$ is a compact subset of $\mathbb{R}$.*

Proof. Set $a_0 = a$ and $b_0 = b$, so $[a, b] = [a_0, b_0]$.

Arguing by contradiction, assume that there is an open cover $\{V_\alpha\}$ of $[a_0, b_0]$ that has no finite subcover.

Note that $\{V_\alpha\}$ is also a cover for two intervals

$$[a_0, \tfrac{a_0+b_0}{2}] \quad \text{and} \quad [\tfrac{a_0+b_0}{2}, b_0].$$

Furthermore, if $\{V_\alpha\}$ has a finite subcover of each of these two subintervals, then these two subcovers together give a finite cover of $[a_0, b_0]$. Thus $\{V_\alpha\}$ must have no finite subcover for *at least one* of these subintervals; denote it by $[a_1, b_1]$; so either $a_1 = a_0$ and $b_1 = \frac{a_0+b_0}{2}$ or $a_1 = \frac{a_0+b_0}{2}$ and $b_1 = b_0$.

Continuing in this manner we get a sequence of intervals

$$[a_0, b_0] \supset [a_1, b_1] \supset \dots$$

such that no finite family of sets from $\{V_\alpha\}$ covers any of the intervals $[a_n, b_n]$. In particular,

❶ $$[a_n, b_n] \not\subset V_\alpha \quad \text{for any } n \text{ and } \alpha.$$

Observe that

$$a_0 \leqslant a_1 \leqslant \dots \leqslant b_1 \leqslant b_0 \quad \text{and} \quad b_n - a_n = \tfrac{b-a}{2^n}$$

for any n. Denote by x the least upper bound of $\{a_n\}$. Note that $x \in [a_n, b_n]$ for any n.[1]

Since $\{V_\alpha\}$ is a cover, we can choose $V_\alpha \ni x$. Since V_α is open, it contains an interval $(x-\varepsilon, x+\varepsilon)$ for some $\varepsilon > 0$. Choose a large n so that $\frac{b-a}{2^n} < \varepsilon$. Clearly, $V_\alpha \supset (x-\varepsilon, x+\varepsilon) \supset [a_n, b_n]$; the latter contradicts ❶. □

D Images

7.11. Proposition. *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a continuous map between topological spaces, and let K be a compact set in $\mathcal{X}$. Then the image $Q = f(K)$ is compact in $\mathcal{Y}$.*

Proof. Choose an open cover $\{W_\alpha\}$ of Q. Since f is continuous, $V_\alpha = f^{-1}(W_\alpha)$ is open for each α. Note that $\{V_\alpha\}$ covers K.

Since K is compact, there is a finite subcover $\{V_{\alpha_1}, \dots, V_{\alpha_n}\}$. It remains to observe that $\{W_{\alpha_1}, \dots, W_{\alpha_n}\}$ covers Q. □

7.12. Exercise. *Let f be a continuous real-valued function that is defined on a compact space $\mathcal{K}$.*

(a) Show that f is b o u n d e d*; that is, there is a constant c such that $|f(x)| \leqslant c$ for any $x \in \mathcal{K}$.*

[1] We also have that $a_n \to x$ and $b_n \to x$ as $n \to \infty$, but we will not use it.

(b) *Show that f attains its maximum and minimum; that is, there are points $x_{\min}, x_{\max} \in \mathcal{K}$ such that*

$$f(x_{\min}) \leqslant f(x) \leqslant f(x_{\max})$$

for any $x \in \mathcal{K}$.

E Closed subsets

7.13. Proposition. *A closed set in a compact space is compact.*

Proof. Let Q be a closed set in a compact space $\mathcal{K}$. Since Q is closed, its complement $W = \mathcal{K} \setminus Q$ is open.

Consider an open cover $\{V_\alpha\}_{\alpha \in \mathcal{I}}$ of Q. Add to it W; that is, consider the family of sets that includes W as well as V_α for all $\alpha \in \mathcal{I}$. Note that we get an open cover of $\mathcal{K}$. Indeed, W covers all points in the complement of Q and any point of Q is covered by some V_α.

Since $\mathcal{K}$ is compact, we can choose a finite subcover, say $\{W, V_{\alpha_1}, \dots$ $\dots, V_{\alpha_n}\}$ — without loss of generality, we can assume that it includes W. Observe that $\{V_{\alpha_1}, \dots, V_{\alpha_n}\}$ is a cover of Q, hence the result. □

In the proof, we add an extra open set to the cover, use it, and take it away. This type of reasoning is useful in all branches of mathematics; sometimes it is called the 17 camels trick.[2]

7.14. Exercise. *Show that any closed bounded subset of the real line is compact.*

F One-point compactification

Choose a topological space $\mathcal{X}$ and add to $\mathcal{X}$ one extra point that will be denoted by ∞. Let us define a topology on $\mathcal{X}^\infty = \mathcal{X} \cup \{\infty\}$ such that a subset $V \subset \mathcal{X}^\infty$ is open if V is an open subset of $\mathcal{X}$ (in which case $\infty \notin V$) or $\infty \in V$ and its complement $\mathcal{X} \setminus V$ is a closed *and* compact subset of $\mathcal{X}$.

---

[2]The name comes from the following mathematical parable [25]: *A father left 17 camels to his three sons and, according to the will, the eldest son should be given half of all camels, the middle son the 1/3 part, and the youngest son the 1/9. It was impossible to follow his will until a wise man appeared. He added his own camel, the oldest son took* $18/2 = 9$ *camels, the second son took* $18/3 = 6$ *camels, and the third son* $18/9 = 2$ *camels. The wise man took his camel and went away.*

To show that we have defined a topology, we need to check the conditions in 2.1: *(a)* is straightforward, *(b)* follows since intersection of closed and compact sets is closed and compact (see 7.13), and *(c)* follows since union of two closed sets is closed and union of two compact sets is compact (see 3.1c and 7.6b).

Now, choose an open cover $\{V_\alpha\}$ of $\mathcal{X}^\infty$. Choose α_∞ such that $V_{\alpha_\infty} \ni \infty$. By construction, the complement $K = \mathcal{X}^\infty \setminus V_{\alpha_\infty}$ is compact. Therefore we can choose a finite subcover $\{V_{\alpha_1}, \dots, V_{\alpha_n}\}$ of K. Observe that $\{V_{\alpha_\infty}, V_{\alpha_1}, \dots, V_{\alpha_n}\}$ covers $\mathcal{X}^\infty$; whence $\mathcal{X}^\infty$ is compact.

The space $\mathcal{X}^\infty$ constructed in the proof is called the one-point compactification of $\mathcal{X}$. Observe that the inclusion map $\mathcal{X} \hookrightarrow \mathcal{X}^\infty$ is an embedding. Furthermore, if $\mathcal{X}$ is not compact, then it is dense in $\mathcal{X}^\infty$. In particular, we proved the following statement.

7.15. Claim. *Every topological space is homeomorphic to a subspace of a compact space.*

7.16. Exercise. *Show that the one-point compactification of the real line $\mathbb{R}$ is homeomorphic to the circle $\mathbb{S}^1$.*

G Product spaces

7.17. Theorem. *Assume $\mathcal{X}$ and $\mathcal{Y}$ are compact topological spaces. Then their product space $\mathcal{X} \times \mathcal{Y}$ is compact.*

The following exercise provides a partial converse.

7.18. Exercise. *Suppose that a product space $\mathcal{X} \times \mathcal{Y}$ is nonempty and compact. Show that its factors $\mathcal{X}$ and $\mathcal{Y}$ are compact.*

In the proof, we will need the following definition.

7.19. Definition. *Let $\{V_\alpha\}$ and $\{W_\beta\}$ be two covers of a topological space $\mathcal{X}$. We say that $\{V_\alpha\}$ is inscribed*[3] *in $\{W_\beta\}$ if for any V_α there is W_β such that $V_\alpha \subset W_\beta$.*

7.20. Exercise. *Let $\mathscr{B}$ be a base in a topological space $\mathcal{X}$. Show that for any cover $\{V_\alpha\}$ of $\mathcal{X}$ there is an inscribed cover made from sets in $\mathscr{B}$.*

Suppose that $\{V_\alpha\}$ is inscribed in $\{W_\beta\}$. If $\{V_\alpha\}$ has a finite subcover $\{V_{\alpha_1}, \dots, V_{\alpha_n}\}$, then for each α_i we can choose β_i such that $V_{\alpha_i} \subset W_{\beta_i}$. Note that $\{W_{\beta_1}, \dots, W_{\beta_n}\}$ is a finite subcover of $\{W_\beta\}$. It proves the following:

[3] One can also say that $\{V_\alpha\}$ refines $\{W_\beta\}$.

7.21. Observation. *A space $\mathcal{X}$ is compact if and only if any cover of $\mathcal{X}$ has a finite inscribed cover.*

It is instructive to solve the following exercise before reading the proof of 7.17.

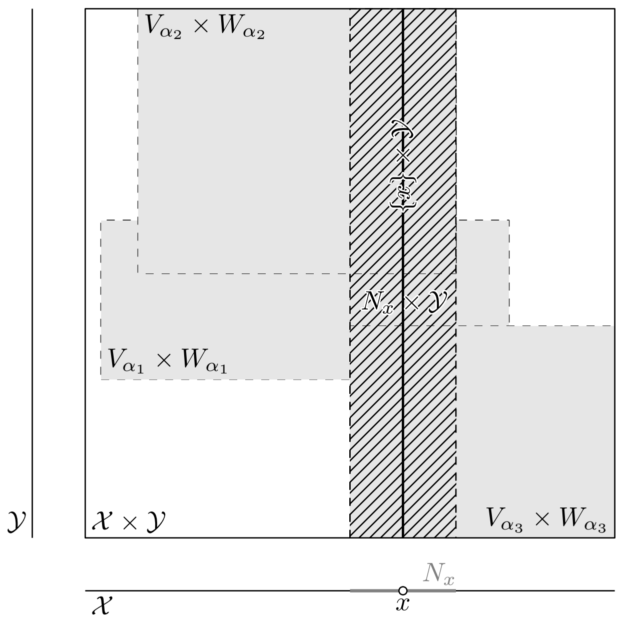
7.22. Exercise. *Find a flaw in the following argument.*

Fake proof of 7.17. Fix an open cover $\{O_\beta\}$ of $\mathcal{X} \times \mathcal{Y}$. Consider all product sets $V_\alpha \times W_\alpha$ such that $V_\alpha \times W_\alpha \subset O_\beta$ for some β; as before, V_α and W_α are open in $\mathcal{X}$ and $\mathcal{Y}$, respectively. Note that $\{V_\alpha \times W_\alpha\}$ is a cover of $\mathcal{X} \times \mathcal{Y}$ that is inscribed in $\{O_\beta\}$. By the observation, it is sufficient to find a finite subcover of $\{V_\alpha \times W_\alpha\}$.

Note that $\{V_\alpha\}$ covers $\mathcal{X}$. Since $\mathcal{X}$ is compact, we can choose its finite subcover $\{V_{\alpha_1}, \dots, V_{\alpha_n}\}$. Similarly, $\{W_\alpha\}$ is a cover of $\mathcal{Y}$; so we can choose its finite subcover $\{W_{\alpha_{n+1}}, \dots, W_{\alpha_{n+m}}\}$. It remains to observe that the products $V_{\alpha_i} \times W_{\alpha_i}$ for all i cover $\mathcal{X} \times \mathcal{Y}$. $\square$

Proof of 7.17. Recall that by definition of the product topology, any open set in $\mathcal{X} \times \mathcal{Y}$ is a union of product sets $V_\alpha \times W_\alpha$, where V_α is open in $\mathcal{X}$ and W_α is open in $\mathcal{Y}$.

Choose an open cover $\{O_\beta\}$ of $\mathcal{X} \times \mathcal{Y}$. Consider all product sets $V_\alpha \times W_\alpha$ such that $V_\alpha \times W_\alpha \subset O_\beta$ for some β (as before, V_α and W_α



are open in $\mathcal{X}$ and $\mathcal{Y}$, respectively). Note that $\{V_\alpha \times W_\alpha\}$ is a cover of $\mathcal{X} \times \mathcal{Y}$ that is inscribed in $\{O_\beta\}$. By 7.21, it is sufficient to find a finite subcover of $\{V_\alpha \times W_\alpha\}$.

Fix $x \in \mathcal{X}$. Note that the subspace $\{x\} \times \mathcal{Y}$ is homeomorphic to $\mathcal{Y}$; see 6.12. In particular, the set $\{x\} \times \mathcal{Y}$ is compact. Therefore, $\{x\} \times \mathcal{Y}$ has a finite cover $\{V_{\alpha_1} \times W_{\alpha_1}, \dots, V_{\alpha_n} \times W_{\alpha_n}\}$; that is,

$$(V_{\alpha_1} \times W_{\alpha_1}) \cup \dots \cup (V_{\alpha_n} \times W_{\alpha_n}) \supset \{x\} \times \mathcal{Y}$$

Consider the set

$$N_x = V_{\alpha_1} \cap \dots \cap V_{\alpha_n};$$

note that N_x is open in $\mathcal{X}$. Since $N_x \subset V_{\alpha_i}$ for any i, we have

$$N_x \times \mathcal{Y} \subset (V_{\alpha_1} \times W_{\alpha_1}) \cup \dots \cup (V_{\alpha_n} \times W_{\alpha_n}).$$

We get that

❷ *every point $x \in \mathcal{X}$ admits an open neighborhood N_x such that $N_x \times \mathcal{Y}$ can be covered by finitely many product sets from $\{V_\alpha \times W_\alpha\}$.*

The sets $\{N_x\}_{x \in \mathcal{X}}$ form a cover of $\mathcal{X}$. Since $\mathcal{X}$ is compact, there is a finite subcover $\{N_{x_1}, \dots, N_{x_m}\}$. Note that

$$\mathcal{X} \times \mathcal{Y} = (N_{x_1} \times \mathcal{Y}) \cup \dots \cup (N_{x_m} \times \mathcal{Y});$$

that is, $\mathcal{X} \times \mathcal{Y}$ can be covered by a finite subfamily of $\{N_x \times \mathcal{Y}\}_{x \in \mathcal{X}}$. Applying ❷, we get that $\mathcal{X} \times \mathcal{Y}$ can be covered by a finite number of product sets from $\{V_\alpha \times W_\alpha\}$. □

7.23. Exercise. *Give two proofs that the circle*

$$\mathbb{S}^1 = \{\, (x, y) \in \mathbb{R}^2 \,:\, x^2 + y^2 = 1 \,\}$$

is compact.

(a) Using 7.11.
(b) Using 7.17 and 7.13.

7.24. Advanced exercise. *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a map between topological spaces. Assume $\mathcal{Y}$ is compact. Show that f is continuous if its graph $\Gamma = \{\, (x, f(x)) \,:\, x \in \mathcal{X} \,\}$ is a closed set in $\mathcal{X} \times \mathcal{Y}$.*

H Remarks

We omit the proof of the following theorem, but it is worth knowing.

7.25. Alexander prebase theorem. *A topological space $\mathcal{X}$ is compact if and only if for some (and therefore any) prebase $\mathscr{P}$ in $\mathcal{X}$, every cover of $\mathcal{X}$ by elements of $\mathscr{P}$ admits a finite subcover.*

Applying this theorem to the prebase from 6.12a gives a short proof of 7.17. Indeed, suppose that a cover of the product space $\mathcal{X} \times \mathcal{Y}$ consists only of products $V_\alpha \times \mathcal{Y}$ and $\mathcal{X} \times W_\beta$, where $V_\alpha \subset \mathcal{X}$ and $W_\beta \subset \mathcal{Y}$ are open sets. Note that $\bigcup V_\alpha = \mathcal{X}$ or $\bigcup W_\beta = \mathcal{Y}$; indeed, if there are $x \in \mathcal{X} \setminus \bigcup V_\alpha$ and $y \in \mathcal{Y} \setminus \bigcup W_\beta$, then (x, y) is not covered by any $V_\alpha \times \mathcal{Y}$ or $\mathcal{X} \times W_\beta$ — a contradiction. Without loss of generality, we can assume that $\{V_\alpha\}$ cover $\mathcal{X}$. Since $\mathcal{X}$ is compact we can choose a finite subcover $\{V_{\alpha_1}, \dots, V_{\alpha_n}\}$. It remains to observe that $\{V_{\alpha_1} \times \mathcal{Y}, \dots, V_{\alpha_n} \times \mathcal{Y}\}$ covers $\mathcal{X} \times \mathcal{Y}$ and to apply the Alexander prebase theorem.

Let us indicate a more serious application of this theorem.

Start with a set S. Denote by Power(S) the set of all subsets of S. Given $s \in S$, define sets

$$\mathfrak{V}_s = \{\, X \subset S \,:\, X \ni s \,\} \quad \text{and} \quad \mathfrak{W}_s = \{\, X \subset S \,:\, X \not\ni s \,\}.$$

Note that $X \subset S$ means that $X \in \mathrm{Power}(S)$, so $\mathfrak{V}_s$ and $\mathfrak{W}_s$ are subsets of Power(S).

Let us equip the set Power(S) with the topology defined by the prebase $\{\mathfrak{V}_s, \mathfrak{W}_s\}_{s \in S}$; now we can call it power space. Applying the argument above, one sees that Power(S) *is compact*. This is a special case of the Tychonoff compactness theorem.

For a finite set F the space Power(F) is a discrete space with $2^{|F|}$ points, which is compact — not a surprise.

For the set of natural numbers $\mathbb{N} = \{1, 2, \dots\}$, a point $p \in \mathrm{Power}(\mathbb{N})$ is a subset $p \subset \mathbb{N}$ and the map

$$p \mapsto 2 \cdot \sum_{n \in p} 3^{-n}$$

defines a homeomorphism from Power$(\mathbb{N})$ to the Cantor set $\mathcal{C} \subset [0, 1]$; see 4B.

The following exercise describes a truly scary example of a compact space.

7.26. Advanced exercise. *Let $\mathcal{A} = \mathrm{Power}(\mathbb{N})$ and $\mathcal{B} = \mathrm{Power}(\mathcal{A})$; as we know, $\mathcal{B}$ is a compact space. Show that the following sequence of points $\mathfrak{p}_1$, $\mathfrak{p}_2, \dots$ in $\mathcal{B}$ does not have a converging subsequence:*

$$\mathfrak{p}_n = \{\, P \in \mathcal{A} \,:\, P \not\ni n \,\}.$$

Chapter 8

Metric spaces revisited

Recall that any metric space comes with a topology. In particular, we may talk about *compact metric spaces.* In this chapter we discuss specific properties of metric spaces that are related to compactness.

A Lebesgue number

8.1. Theorem. *Given an open cover $\{V_\alpha\}$ of a compact metric space $\mathcal{M}$, there is $\varepsilon > 0$ such that every ε-ball in $\mathcal{M}$ is contained in V_α for some α.*

The number ε that meets the condition in the theorem is called the Lebesgue number of the cover.

Proof. Given a point $p \in \mathcal{M}$ we can choose $r = r(p) > 0$ such that the ball $\mathrm{B}(p, 2\cdot r)$ lies in V_α for some α. Observe that all balls $\mathrm{B}(p, r(p))$ form an open cover of $\mathcal{M}$. Since $\mathcal{M}$ is compact, we can choose a finite subcover $\{\mathrm{B}(p_1, r_1), \dots, \mathrm{B}(p_n, r_n)\}$.

Let $\varepsilon = \min\{r_1, \dots, r_n\}$. For any $p \in \mathcal{M}$ we can choose a ball $\mathrm{B}(p_i, r_i) \ni p$. Since $\varepsilon \leqslant r_i$, we have $\mathrm{B}(p, \varepsilon) \subset \mathrm{B}(p_i, 2\cdot r_i)$. Furthermore, since $\mathrm{B}(p_i, 2\cdot r_i)$ lies in some V_α, so does $\mathrm{B}(p, \varepsilon)$. □

8.2. Exercise. *Construct a noncompact metric space $\mathcal{M}$ such that 1 is a Lebesgue number for any open cover of $\mathcal{M}$.*

8.3. Exercise. *Let $\varepsilon > 0$ be a Lebesgue number of an open cover $\{V_\alpha\}$ of a compact metric space $\mathcal{M}$. Show that for any $0 < \varepsilon' < \varepsilon$ there is a finite subcover of $\{V_\alpha\}$ for which ε' is a Lebesgue number.*

Give an example showing that in general there might be no finite subcover of $\{V_\alpha\}$ with Lebesgue number ε.

B Compactness ⇒ sequential compactness

A topological space is called sequentially compact if every sequence in it has a converging subsequence.

8.4. Exercise. *Show that the product of two sequentially compact spaces is sequentially compact.*

For general topological spaces sequential compactness does not imply compactness, nor conversely.[1] The following proposition states that these two notions are equivalent for metric spaces.

8.5. Proposition. *A metric space $\mathcal{M}$ is compact if and only if it is sequentially compact.*

Proof of the only-if part in 8.5. Choose a sequence $x_1, x_2, \ldots \in \mathcal{M}$.

Note that a point $p \in \mathcal{M}$ is a limit of a subsequence of x_n if and only if for any $\varepsilon > 0$, the ball $\mathrm{B}(p, \varepsilon)$ contains infinitely many elements x_n. Indeed, if this property holds, then we can choose i_1 such that $x_{i_1} \in \mathrm{B}(p, 1)$, further $i_2 > i_1$ such that $x_{i_2} \in \mathrm{B}(p, \frac{1}{2})$ and so on; on the n-th step we get $i_n > i_{n-1}$ such that $x_{i_n} \in \mathrm{B}(p, \frac{1}{n})$. The obtained subsequence $x_{i_1}, x_{i_2}, \ldots$ converges to p.

Now assume $\mathcal{M}$ is not sequentially compact; that is, some sequence $x_1, x_2, \ldots \in \mathcal{M}$ has no convergent subsequence. Then for any point p there is $\varepsilon_p > 0$ such that $\mathrm{B}(p, \varepsilon_p)$ contains only finitely many elements of x_n. Note that all balls $\mathrm{B}(p, \varepsilon_p)$ form a cover of $\mathcal{M}$. Since the sequence is infinite, this cover does not have a finite subcover, in particular, $\mathcal{M}$ is not compact. □

The if part of the theorem will be done in 8E after proving a sequence of auxiliary statements. The proof can be divided in three steps. On the first step (8C) we show that any sequentially compact space is *separable*; that is, it has a countable dense subset. On the second step (8D) we show that a separable metric space has a countable base and any topological space with a countable base is *Lindelöf*; that is, every open cover contains a countable subcover. On the last step (8E) we combine these results.

C Nets and separability

Let $\mathcal{M}$ be a metric space. A subset $A \subset \mathcal{M}$ is called an ε-net of $\mathcal{M}$ if for any $p \in \mathcal{M}$ there is $a \in A$ such that $|p - a|_{\mathcal{M}} < \varepsilon$.

[1] An example of a compact, but not sequentially compact, space has been constructed in 7.26. A standard example of a sequentially compact, but not compact, space is $[0, \Omega)$ — the set of all countable ordinals, with the order topology; it is discussed in the classical book of Lynn Arthur Steen and J. Arthur, Jr. Seebach [26].

8.6. Lemma. *Let $\mathcal{M}$ be a sequentially compact metric space. Then for any $\varepsilon > 0$ there is a finite ε-net in $\mathcal{M}$.*

Proof. Choose $\varepsilon > 0$ and apply the following recursive procedure.

Choose a point x_1 in $\mathcal{M}$. Further, choose a point x_2 such that $|x_1 - x_2| > \varepsilon$. Further, choose a point x_3 such that $|x_1 - x_3| > \varepsilon$ and $|x_2 - x_3| > \varepsilon$; and so on. On the n-th step we choose a point x_n such that $|x_i - x_n| > \varepsilon$ for all $i < n$.

This procedure might terminate on the n-th step if there is no point x_n such that $|x_i - x_n| > \varepsilon$ for all $i < n$. In this case, the set $\{x_1, \dots, x_{n-1}\}$ is an ε-net in $\mathcal{M}$ — the lemma is proved.

If the procedure does not terminate, we get an infinite sequence of points $x_1, x_2, \dots$ such that $|x_i - x_j| > \varepsilon$ for all $i \neq j$. Any of its subsequences has the same property; in particular none of its subsequences can converge — a contradiction. □

A topological space is called separable if it contains a countable dense subset.

8.7. Corollary. *Sequentially compact metric spaces are separable.*

Proof. Let $\mathcal{M}$ be a sequentially compact metric space.

By 8.6, for each positive integer n, we can choose a finite $\frac{1}{n}$-net $N_n \subset \mathcal{M}$. It remains to observe that the union $N_1 \cup N_2 \cup \dots$ is countable and dense. □

D Countable base

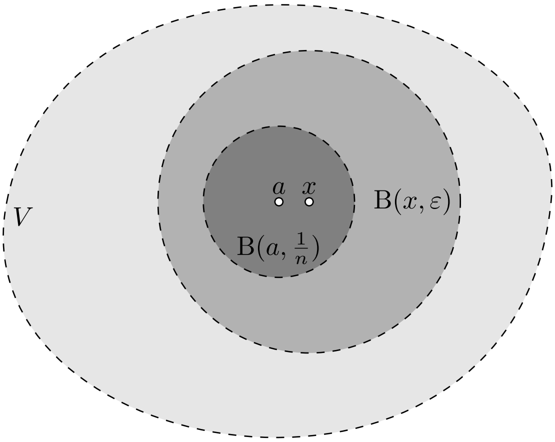
8.8. Proposition. *Any sequentially compact metric space has a countable base.*

Topological spaces that admit a countable base are called second-countable.[2] So the proposition states that *any sequentially compact metric space is second-countable.*

Proof. Let $\mathcal{M}$ be a sequentially compact metric space. By 8.7, we can choose a countable dense subset $A \subset \mathcal{M}$.

Consider the set of all balls $\mathrm{B}(a, \frac{1}{n})$ for $a \in A$ and positive integers n. Note that this set is countable; it remains to show that it forms a base in $\mathcal{M}$.

[2] A space is called first-countable if for each point x there exists a sequence of neighborhoods $N_1, N_2, \dots$ such that for any open set $V \ni x$ we can choose n such that $V \supset N_n \ni x$. For example, any metric space is first-countable; indeed, one can take $N_n = \mathrm{B}(x, \frac{1}{n})$.



Let x be a point in an open set V. Then $\mathrm{B}(x,\varepsilon) \subset V$ for some $\varepsilon > 0$. Choose a positive integer $n > \frac{2}{\varepsilon}$. Since A is everywhere dense, we can choose $a \in A$ so that $|a - x| < \frac{1}{n}$. By the triangle inequality, $x \in \mathrm{B}(a, \frac{1}{n}) \subset \mathrm{B}(x,\varepsilon)$; in particular,

$$x \in \mathrm{B}(a, \tfrac{1}{n}) \subset V.$$

It remains to apply 6.9. $\square$

8.9. Lemma. *Let $\mathcal{X}$ be a topological space with a countable base. Then any open cover of $\mathcal{X}$ has a countable subcover.*

Proof. Choose an open cover $\{V_\alpha\}$. Let $\{B_1, B_2, \dots\}$ be a countable base of $\mathcal{X}$. By 6.8, for any $x \in \mathcal{X}$ we can choose $i = i(x)$ such that $x \in B_i \subset V_\alpha$ for some α. Denote by S the set of all indices that appear as $i(x)$ for some x. Then $\{B_i\}_{i\in S}$ is a countable open cover that is inscribed in $\{V_\alpha\}$. That is, for every B_i there is α_i such that $B_i \subset V_{\alpha_i}$. It remains to observe that $\{V_{\alpha_i}\}_{i\in S}$ is a countable cover of $\mathcal{X}$. $\square$

A topological space is called Lindelöf if every open cover contains a countable subcover; so the lemma says that *any space with a countable base is Lindelöf.*

8.10. Advanced exercise. *The intervals $[a, b)$ form a base of the so-called right half-open interval topology on the real line; let $\mathcal{X}$ be the corresponding topological space.*

(a) Show that $\mathcal{X}$ is Lindelöf.

(b) Show that $\mathcal{X} \times \mathcal{X}$ is not Lindelöf.

E Sequential compactness ⇒ compactness

Proof of the if part in 8.5. Choose an open cover of $\mathcal{M}$. By 8.9, we can assume that the cover is countable; denote it by $\{V_1, V_2, \dots\}$. If it has no finite subcover, then we can choose a sequence of points $x_1, x_2, \dots \in \mathcal{M}$ such that

$$x_n \notin V_1 \cup \ldots \cup V_n$$

for any n.

Since $\mathcal{M}$ is sequentially compact, a subsequence of $x_1, x_2, \dots$ has a limit, say x, and $x \in V_n$ for some n. It follows that $x_i \in V_n$ for an infinite set of indices i, but by construction, $x_i \notin V_n$ for all $i \geqslant n$ — a contradiction. □

8.11. Advanced exercise. *Let $\mathcal{M}$ be a compact metric space and $f\colon \mathcal{M} \to \mathcal{M}$ be a distance-noncontracting map; that is,*

$$❶ \qquad |f(x) - f(y)|_{\mathcal{M}} \geqslant |x - y|_{\mathcal{M}}$$

for any $x, y \in \mathcal{M}$. Show that f is an isometry; that is, f is a bijection and equality always holds in ❶.

F Complete spaces

A sequence $x_1, x_2, \dots$ of points in a metric space is called Cauchy if for any $\varepsilon > 0$ there is n such that $|x_i - x_j| < \varepsilon$ for all $i, j > n$. It is easy to prove that any converging sequence is Cauchy; the converse does not hold in general. A metric space $\mathcal{M}$ is called complete if any Cauchy sequence in $\mathcal{M}$ converges to a point in $\mathcal{M}$.

The Cauchy test (which is assumed to be known from calculus) implies that the real line $\mathbb{R}$ with the standard metric is complete. On the other hand, an open interval $(0, 1)$ isnot complete; indeed, the sequence $x_n = \frac{1}{2^n}$ is Cauchy, but it does not converge in $(0, 1)$ — it converges to 0 in $\mathbb{R}$, but $0 \notin (0, 1)$.

8.12. Proposition. *Any (sequentially) compact metric space $\mathcal{M}$ is complete.*

Proof. Choose a Cauchy sequence $x_1, x_2, \dots \in \mathcal{M}$. By the only-if part of 8.5 (which is proved already), we can assume that the space is sequentially compact; therefore, we can find a converging subsequence; denote its limit by p.

Choose $\varepsilon > 0$ and let n be as in the definition of a Cauchy sequence, so $|x_i - x_j| < \varepsilon$ for all $i, j > n$. Note that the ball $\mathrm{B}(p, \varepsilon)$ contains infinitely many elements of the sequence; in particular, $|x_j - p| < \varepsilon$ for some $j > n$. By the triangle inequality,

$$|x_i - p| \leqslant |x_i - x_j| + |x_j - p| < 2\cdot\varepsilon$$

for any $i > n$. Since $\varepsilon > 0$ is arbitrary, the sequence $x_1, x_2, \ldots$ converges to p. $\square$

8.13. Exercise. *Show that a closed subset of a complete metric space forms a complete subspace.*

8.14. Exercise. *Show that any sequentially compact metric space $\mathcal{M}$ is bounded; that is, $\mathcal{M}$ has finite diameter:*

$$\operatorname{diam} \mathcal{M} = \sup \{\, |x - y|_{\mathcal{M}} \;:\; x, y \in \mathcal{M} \,\} < \infty.$$

Give an example of noncompact, complete, bounded metric space.

8.15. Exercise. *Show that a complete metric space $\mathcal{M}$ is compact if for any $\varepsilon > 0$ there is a compact ε-net in $\mathcal{M}$.*

8.16. Exercise. *Let $\mathcal{M}$ be a complete nonempty metric space, and let $f\colon \mathcal{M} \to \mathcal{M}$ be a contraction map; that is, there is $\lambda < 1$ such that*

$$|f(x) - f(y)|_{\mathcal{M}} \leqslant \lambda\cdot|x - y|_{\mathcal{M}}$$

for any $x, y \in \mathcal{M}$. Show that f has a unique fixed point; that is, a point $p \in \mathcal{M}$ such that $f(p) = p$.

8.17. Exercise. *Let $\mathcal{M}$ be a complete metric space, and let $r\colon \mathcal{M} \to \mathbb{R}$ be a positive continuous function. Show that for any $\varepsilon > 0$, there exists a point $p \in \mathcal{M}$ such that*

$$r(x) > (1 - \varepsilon)\cdot r(p)$$

for every $x \in \mathrm{B}(p, r(p))$.

G An application

8.18. Theorem. *Among all plane curves of given length L, only circles bound the greatest area.*

Since a circle of length L bounds area $\frac{L^2}{4\cdot\pi}$, the statement can be reformulated in a more algebraic manner: *a plane curve γ of given length L bounds area $a \leqslant \frac{L^2}{4\cdot\pi}$, and, in the case of equality, γ is a circle.* This statement is called the isoperimetric inequality.

Given a plane curve γ, denote by F_γ its convex hull; that is, the minimal convex set that includes γ. A figure bounded by γ must lie in F_γ. Furthermore, if γ is a circle then it bounds F_γ. Therefore, it is sufficient to prove the following two statements:

8.19. Lemma. *Among all closed plane curves of length $\leqslant L$ there is a curve γ that maximizes the area of F_γ.*

8.20. Lemma. *If a closed plane curve γ of length $\leqslant L$ maximizes the area of F_γ, then γ must be a circle of length L.*

Note that 8.20 does not say that a circle maximizes the area; it only says that no other curve does. But together with 8.19 it proves the theorem.

Proof of 8.19. A closed plane curve γ with length $\leqslant L$ can be described by a parametrization $\gamma\colon [0,L] \to \mathbb{R}^2$ such that $\gamma(0) = \gamma(L)$ and

$$|\gamma(t_0) - \gamma(t_1)|_{\mathbb{R}^2} \leqslant |t_0 - t_1|.$$

Without loss of generality, we can assume that $\gamma(0) = \gamma(L) = 0 \in \mathbb{R}^2$. Consider the space Γ of all such parametrized curves with the sup-norm metric defined by

❷ $$|\gamma_1 - \gamma_2|_\Gamma = \max\{\, |\gamma_1(t) - \gamma_2(t)|_{\mathbb{R}^2} \;:\; t \in [0,L]\,\}$$

8.21. Exercise. *Prove the following statements:*

(a) *The formula defines a metric on Γ; in particular, the maximum in the right-hand side in ❷ is defined.*
(b) *The space Γ is complete.*
(c) *The space Γ is compact.*

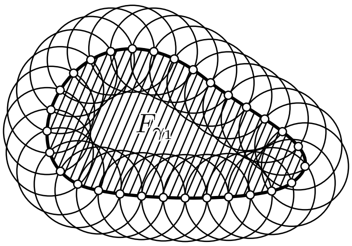
It remains to prove the following sublemma and apply 7.12*b*. □

8.22. Sublemma. *The function $\Gamma \to \mathbb{R}$ defined by $\gamma \mapsto \operatorname{area} F_\gamma$ is continuous.*

Proof. It is sufficient to show that

❸ $$\begin{gathered}|\gamma_1 - \gamma_2|_\Gamma < \tfrac{L}{n}\\ \Downarrow\\ \operatorname{area} F_{\gamma_2} < \operatorname{area} F_{\gamma_1} + \tfrac{4\cdot\pi\cdot L^2}{n}\end{gathered}$$

for any $\gamma_1, \gamma_2 \in \Gamma$ and a positive integer n.



Note that F_{γ_2} lies in an $\frac{L}{n}$-neighborhood of F_{γ_1}. Moreover, for any $p \in F_{\gamma_2} \setminus F_{\gamma_1}$ there is $x \in \partial F_{\gamma_1}$ such that $|p - x| < \frac{L}{n}$. Observe that the length of ∂F_{γ_1} cannot exceed L. Whence we can choose n points $x_1, \dots, x_n$ such that for any $x \in \partial F_{\gamma_1}$ there is x_i such that $|x - x_i| < \frac{L}{n}$. It follows that $|p - x_i| < 2\cdot\frac{L}{n}$ for some i, and

$$F_{\gamma_2} \setminus F_{\gamma_1} \subset \bigcup_{i=1}^{n} \mathrm{B}(x_i, 2\cdot\tfrac{L}{n}).$$

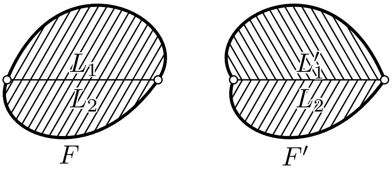
Since $\operatorname{area} \mathrm{B}(x,r)_{\mathbb{R}^2} = \pi\cdot r^2$ we get

$$\operatorname{area}(F_{\gamma_2} \setminus F_{\gamma_1}) \leqslant n\cdot\pi\cdot(2\cdot\tfrac{L}{n})^2,$$

and ❸ follows. □

In this proof, we have introduced the space Γ, proved that it is compact and used that a continuous function on a compact space attains its maximum. This is a very common application of compactness. Note that our space Γ was huge; say, Γ cannot be embedded into $\mathbb{R}^n$ for any n. In particular, one cannot apply standard calculus to Γ.

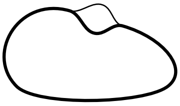
Lemma 8.20 will be proved by means of elementary geometry; it will use the following characterization of a round disc.



Let F be a convex figure; cut it along a chord ℓ into two lenses, L_1 and L_2. Let us denote by L_1' the reflection of L_1 across the perpendicular bisector of the chord. The figure $F' = L_1' \cup L_2$ will be called a flip of F along the chord ℓ.

8.23. Exercise. *Assume that every flip of a convex figure F is convex. Show that F is a round disc.*

Proof of 8.20. If γ maximizes the area, then it must be convex; that is, it must bound a convex figure, say F. If this is not the case, then one can exchange



γ by another curve (as in the picture) enlarging the area while keeping the length the same.

Choose a chord of F and consider its flip F'; it has the same area and perimeter. Therefore, if γ was a maximizer, then so is $\partial F'$. In particular, F' must be convex. It remains to apply 8.23. $\square$

Chapter 9

Hausdorff spaces

Historically, the first definition of a topological space [15, VII § 1] defined what we now call a Hausdorff space; these are especially nice topological spaces that share many features of metric spaces; for example, any convergent sequence in a Hausdorff space has a unique limit.

A Definition

9.1. Definition. *A topological space $\mathcal{X}$ is called Hausdorff if for each pair of distinct points $x, y \in \mathcal{X}$ there are disjoint neighborhoods $V \ni x$ and $W \ni y$.*

9.2. Observation. *Any metrizable space is Hausdorff.*

Proof. It is sufficient to show that a metric space $\mathcal{M}$ is Hausdorff.

If the points $x, y \in \mathcal{M}$ are distinct, then $r = |x - y|_{\mathcal{M}} > 0$. By the triangle inequality $\mathrm{B}(x, \frac{r}{2}) \cap \mathrm{B}(y, \frac{r}{2}) = \varnothing$. Hence the statement follows. □

Recall that a sequence of points in a topological space might have distinct limits (see 2F). For example, for the real line with the cofinite topology most sequences converge to *every* point.

9.3. Exercise. *Show that any converging sequence in a Hausdorff space has a unique limit.*

9.4. Exercise. *Show that a topological space $\mathcal{X}$ is Hausdorff if and only if for any two distinct points $x, y \in \mathcal{X}$ there is an open set $V \ni x$ such that $\bar{V} \not\ni y$.*

9.5. Exercise. *Show that for n distinct points $x_1, \dots, x_n$ in a Hausdorff space one can choose n disjoint open sets $V_1, \dots, V_n$ such that $V_i \ni x_i$ for each i.*

9.6. Exercise. *Show that a topological space $\mathcal{X}$ is Hausdorff if and only if the diagonal*

$$\Delta = \{ (x, x) \in \mathcal{X} \times \mathcal{X} \}$$

is a closed set in the product space $\mathcal{X} \times \mathcal{X}$.

9.7. Exercise. *Let $f \colon \mathcal{X} \to \mathcal{Y}$ be a continuous map between topological spaces. Assume $\mathcal{Y}$ is Hausdorff. Show that the graph*

$$\Gamma = \{ (x, f(x)) \, : \, x \in \mathcal{X} \}$$

is a closed set in $\mathcal{X} \times \mathcal{Y}$.

B Observations

9.8. Observation. *Any one-point set in a Hausdorff space is closed.*

If every one-point set is closed, then the space is called T_1 (see 9E). Therefore the observation above states that *any Hausdorff space is T_1.*

Proof. Let $\mathcal{X}$ be a Hausdorff space and $x \in \mathcal{X}$. By 9.1, given a point $y \neq x$, there are disjoint open sets $V_y \ni x$ and $W_y \ni y$. In particular, $W_y \not\ni x$.

Note that

$$\mathcal{X} \setminus \{x\} = \bigcup_{y \neq x} W_y.$$

It follows that $\mathcal{X} \setminus \{x\}$ is open, and therefore $\{x\}$ is closed. □

Suppose $x \in V$ and $y \in W$ are as in the definition of Hausdorff space (9.1). If x, y lie in some subset A, then $V' = V \cap A \ni x$ and $W' = W \cap A \ni y$ are disjoint neighborhoods in the induced topology on A, and we get the following.

9.9. Observation. *Any subspace of a Hausdorff space is Hausdorff.*

9.10. Exercise. *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are Hausdorff spaces. Show that the product space $\mathcal{X} \times \mathcal{Y}$ is Hausdorff. Prove the converse, assuming $\mathcal{X}$ and $\mathcal{Y}$ are nonempty.*

9.11. Exercise. *Let $f \colon \mathcal{X} \to \mathcal{Y}$ be a continuous injective map. Assume $\mathcal{Y}$ is Hausdorff. Show that $\mathcal{X}$ is Hausdorff.*

C Hausdorff meets compactness

9.12. Proposition. *Any compact subset of a Hausdorff space is closed.*

Note that any one-point set is compact. Therefore the proposition generalizes 9.8. The proof is similar but requires an extra step. Before reading the proof, it is instructive to refresh your memory of 7.6a.

The proof of the proposition is based on the following theorem.

9.13. Theorem. *Let $\mathcal{X}$ be a Hausdorff space and $K \subset \mathcal{X}$ be a compact subset. Then for any point $y \notin K$ there are open sets $V \supset K$ and $W \ni y$ such that $V \cap W = \varnothing$.*

Proof of 9.12 modulo 9.13. For $y \notin K$, let us denote by W_y the open set provided by 9.13; in particular, $W_y \ni y$ and $W_y \cap K = \varnothing$. Note that

$$\mathcal{X} \setminus K = \bigcup_{y \notin K} W_y.$$

It follows that $\mathcal{X} \setminus K$ is open; therefore, K is closed. □

Proof of 9.13. By definition of Hausdorff space, for any point $x \in K$ there is a pair of disjoint open sets $V_x \ni x$ and $W_x \ni y$. Note that $\{V_x\}_{x \in K}$ is a cover of K. Since K is compact we can choose a finite subcover $\{V_{x_1}, \dots, V_{x_n}\}$. Set

$$V = V_{x_1} \cup \dots \cup V_{x_n} \qquad \text{and} \qquad W = W_{x_1} \cap \dots \cap W_{x_n}.$$

It remains to observe that V and W are open, $y \in W$, $K \subset V$, and

$$V \cap W \subset \bigcup_i (V_{x_i} \cap W_{x_i}) = \varnothing.$$ □

9.14. Exercise. *Let $\mathcal{X}$ be a Hausdorff space and $K, L \subset \mathcal{X}$ be two disjoint compact subsets. Show that there are disjoint open sets $V \supset K$ and $W \supset L$.*

D Compact-to-Hausdorff maps

9.15. Observation. *Let f be a continuous map from a compact space $\mathcal{K}$ to a Hausdorff space $\mathcal{Y}$. Then f is a closed map.*

If, in addition, the map f is onto, then $\mathcal{Y}$ is equipped with the quotient topology induced by f.

Proof. Since $\mathcal{K}$ is compact, any closed subset $Q \subset \mathcal{K}$ is compact (7.13). Since the image of a compact set is compact (7.11), we have that $f(Q)$ is a compact subset of $\mathcal{Y}$. Since $\mathcal{Y}$ is Hausdorff, $f(Q)$ is closed (9.12). Hence the first statement follows.

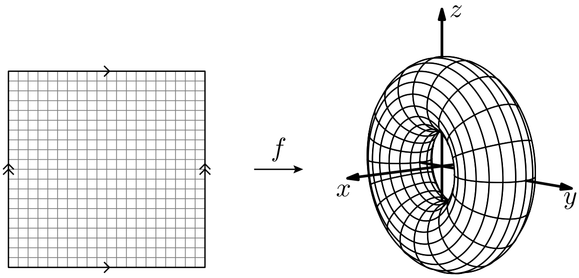
The second statement follows from 6.17. □

Recall that every homeomorphism is a continuous bijection; see 5.1. The following corollary provides a partial converse.

9.16. Corollary. *A continuous bijection from a compact space to a Hausdorff space is a homeomorphism.*

Furthermore, a continuous injective map from a compact space to a Hausdorff space is an embedding.

The corollary can be used to describe the quotient topology. For example, consider the map f, described in the picture; it maps the unit



$f\colon (u,v) \mapsto (a(u), (2+b(u))\cdot b(v), (2+b(u))\cdot a(v))$, where $a(t) = \cos(2\cdot\pi\cdot t)$ and $b(t) = \sin(2\cdot\pi\cdot t)$. Arrows of the same type on line segments indicate that corresponding points on the segments are equivalent.

square $[0,1] \times [0,1]$ to the torus of revolution T. The map f defines an equivalence relation on the square $[0,1] \times [0,1]$:

$$f(u_1,v_1) = f(u_2,v_2) \quad \Leftrightarrow \quad (u_1,v_1) \sim (u_2,v_2);$$

this is the minimal equivalence relation such that $(u,0) \sim (u,1)$ and $(0,v) \sim (1,v)$ for any $u,v \in [0,1]$. Since $[0,1]\times[0,1]$ is compact and $\mathbb{R}^3$ is Hausdorff, we can conclude that f defines a homeomorphism from the quotient space $[0,1] \times [0,1]/\sim$ to T. Note that we did not have to check that the inverse map $T \to [0,1] \times [0,1]/\sim$ is continuous.

Recall that $\mathbb{D}^2$, $\mathbb{S}^1$, and $\mathbb{S}^2$ denote the unit disc, the unit circle, and the unit sphere, respectively. That is,

$$\mathbb{D}^2 = \{\, (x,y) \in \mathbb{R}^2 \,:\, x^2 + y^2 \leqslant 1 \,\},$$
$$\mathbb{S}^1 = \{\, (x,y) \in \mathbb{R}^2 \,:\, x^2 + y^2 = 1 \,\},$$
$$\mathbb{S}^2 = \{\, (x,y,z) \in \mathbb{R}^3 \,:\, x^2 + y^2 + z^2 = 1 \,\}.$$

9.17. Exercise! *Prove the following:*

(a) $\mathbb{S}^1 \simeq [0,1]/\{0,1\}$*. In other words,* $\mathbb{S}^1$ *is homeomorphic to the unit interval with glued ends.*
(b) $\mathbb{D}^2/\mathbb{S}^1 \simeq \mathbb{S}^2$*; here* $\mathbb{S}^1 = \partial\mathbb{D}^2 \subset \mathbb{D}^2$*.*
(c) $(\mathbb{S}^1 \times [0,1])/(\mathbb{S}^1 \times \{0\}) \simeq \mathbb{D}^2$*.*

Recall that a convex body is a compact convex set with nonempty interior in $\mathbb{R}^3$.

9.18. Exercise. *Show that the boundary of a convex body is homeomorphic to* $\mathbb{S}^2$*.*

9.19. Exercise. *Let* Q *be a closed subset of a compact Hausdorff space* $\mathcal{X}$*. Show that the one-point compactification of* $V = \mathcal{X} \setminus Q$ *is homeomorphic to the quotient space* $\mathcal{X}/Q$*.*

9.20. Advanced exercise. *Let* $f\colon \mathcal{X} \to \mathcal{Y}$ *be a continuous closed surjective map between topological spaces. Show that if* $\mathcal{X}$ *is compact and Hausdorff, then so is* $\mathcal{Y}$*.*

E Comments

Recall that any metrizable compact space is Hausdorff and second-countable; the latter means that it has a countable base. The following theorem provides a converse, but we are not going to prove it.

9.21. Theorem. *A compact second-countable Hausdorff space is metrizable.*

This theorem is a corollary of the Urysohn metrization theorem; it gives a satisfactory topological characterization of compact metrizable spaces. There are analogous characterizations of all metrizable spaces; if you are interested in the subject then start with the Nagata–Smirnov theorem. Both theorems are discussed in the book by James Munkres [23].

Let us mention several other conditions that, like Hausdorffness, make a topological space behave more like metrizable spaces.

Recall that a topological space $\mathcal{X}$ is called first-countable if for each point $x \in \mathcal{X}$ there exists a sequence of neighborhoods N_1, $N_2, \dots$ such that for any open set $V \ni x$ we can choose n such that $V \supset N_n \ni x$. Observe that any metric space is first-countable; indeed, one can take $N_n = \mathrm{B}(x, \frac{1}{n})$.

9.22. Exercise. *Let A be a subset of a first-countable space $\mathcal{X}$. Show that $x \in \bar{A}$ if and only if x is a limit of some sequence $x_1, x_2, \dots \in A$.*

Observe that 2.12b implies that this is no longer true without assuming that $\mathcal{X}$ is first-countable.

9.23. Advanced exercise. *Let $\mathcal{X} = \mathbb{R}/\sim$, where $\sim$ is the minimal equivalence relation such that $m \sim n$ for any pair of integers m and n. Show that $\mathcal{X}$ is not metrizable.*

Given a nonempty set A in a metric space $\mathcal{M}$, consider the function $\operatorname{dist}_A\colon \mathcal{M} \to \mathbb{R}$ defined by

$$\operatorname{dist}_A x = \inf\{\, |x-a|_{\mathcal{M}} \;:\; a \in A \,\}.$$

By the triangle inequality, $|\operatorname{dist}_A x - \operatorname{dist}_A y| \leqslant |x-y|_{\mathcal{M}}$. Therefore, dist_A *is a continuous nonnegative function.* Directly from the definition, we have that $x \in \bar{A} \;\Leftrightarrow\; \operatorname{dist}_A x = 0$.

9.24. Exercise. *Consider two disjoint closed sets A and B in a metric space $\mathcal{M}$. Show that the function defined by*

$$f\colon x \mapsto \frac{\operatorname{dist}_A x}{\operatorname{dist}_A x + \operatorname{dist}_B x}$$

is a continuous function $\mathcal{M} \to [0,1]$. Moreover,

$$f(a) = 0 \;\Leftrightarrow\; a \in A \quad \text{and} \quad f(b) = 1 \;\Leftrightarrow\; b \in B.$$

Conclude that there are disjoint open sets $V \supset A$ and $W \supset B$.

The exercise provides a purely topological property of metrizable spaces. Topological spaces that meet the statement about existence of open sets V and W are called T_4. There is a sequence of its related conditions T_0, T_1, T_2, T_3, T_4, and T_5 which I list below for your entertainment. These are the most common conditions, and there are many more; one can easily spend weeks studying the difference between them.

All these conditions hold for metrizable spaces; they were extensively studied. Hausdorffness is T_2, and this is the most important condition in this line. We met T_1 in 9.8; also by 9.13 and 9.14 *any compact Hausdorff space is T_3 and T_4.*

- T_0. For any distinct points $a, b \in \mathcal{X}$ there is an open set V such that $a \in V$ and $b \notin V$ or an open set W such that $a \notin W$ and $b \in W$.
- T_1. For any distinct points $a, b \in \mathcal{X}$ there is an open set V such that $a \in V$ and $b \notin V$.
- T_2. For any distinct points $a, b \in \mathcal{X}$ there are disjoint open sets V and W such that $a \in V$ and $b \in W$.
- T_3. For any point $a \in \mathcal{X}$ and a closed set $B \subset \mathcal{X}$ such that $a \notin B$ there are disjoint open sets V and W such that $a \in V$ and $B \subset W$.
- T_4. For any two disjoint closed sets $A, B \subset \mathcal{X}$ there are disjoint open sets V and W such that $A \subset V$ and $B \subset W$.
- T_5. For any sets $A, B \subset \mathcal{X}$ such that $\bar{A} \cap B = A \cap \bar{B} = \varnothing$ there are disjoint open sets V and W such that $A \subset V$ and $B \subset W$.

Often one includes T_1 in the definition of T_i for $i \geqslant 3$; in this case, one-point sets are closed, and $T_5 \Rightarrow T_4 \Rightarrow T_3 \Rightarrow T_2 \Rightarrow T_1 \Rightarrow T_0$. Spaces that meet T_3 and T_4 are also called r e g u l a r and n o r m a l, respectively; again different authors might define them with or without T_1. I see no reason for a particular choice of terms regular and normal; but it helps to remember that Hausdorff (T_2), regular (T_3), and normal (T_4) do *not* go in the alphabetical order.

9.25. Exercise. *Prove that any metric space is* T_5.

Chapter 10

Topology meets Analysis

Here we discuss two purely topological notions that have applications in mathematical analysis. These results will not be used further in the text.

A Baire theorem

A topological space $\mathcal{X}$ is called B a i r e if any countable family of dense open sets in $\mathcal{X}$ has a dense intersection.

For example, the space of rational numbers $\mathbb{Q}$ is not Baire. Indeed, the complement of a one-point set $\mathbb{Q} \setminus \{q\}$ is dense and open in $\mathbb{Q}$; the intersection of all these complements is empty, and since $\mathbb{Q}$ is countable, this is a countable intersection. On the other hand, the set of integers $\mathbb{Z}$ is a Baire space, since $\mathbb{Z}$ is the only dense open set in $\mathbb{Z}$. The following theorem implies that the real line $\mathbb{R}$ is Baire.

10.1. Theorem. *Any complete metric space is Baire.*

Given two subsets A and B of a topological space, we will write $A \Supset B$ (equivalently, $B \Subset A$) if $\mathring{A} \supset \bar{B}$; that is, the interior of A contains the closure of B.

Proof. Let $V_1, V_2, \dots$ be a sequence of dense open sets in a complete metric space $\mathcal{M}$. We need to show that $G = V_1 \cap V_2 \cap \dots$ intersects any open ball in $\mathcal{M}$.

Choose a ball $B_0 = \mathrm{B}(p_0, r_0)$. Since V_1 is dense and open, we can choose another ball $B_1 = \mathrm{B}(p_1, r_1) \subset B_0 \cap V_1$; moreover, taking the radius r_1 smaller we can achieve that $B_1 \Subset B_0 \cap V_1$ and $r_1 \leqslant \frac{1}{2} \cdot r_0$. Let us continue this process for V_2, $V_3, \dots$; on the n-th step we choose a ball $B_n = \mathrm{B}(p_n, r_n) \Subset B_{n-1} \cap V_n$ and $r_n \leqslant \frac{1}{2} \cdot r_{n-1}$.

If $i < j$, then $|p_i - p_j| < r_i \leqslant \frac{1}{2^i} \cdot r_0$. It follows that $p_0, p_1, \dots$ is a Cauchy sequence. Since $\mathcal{M}$ is complete, the sequence must converge; denote its limit by p_∞. It remains to observe that $p_\infty \in \bar{B}_n$ for any n; in particular, $p_\infty \in G \cap B_0$. □

Theorem 10.1 is the main source of Baire spaces; the following exercise provides another source.

10.2. Exercise. *Let $\mathcal{K}$ be a compact Hausdorff space.*

(a) Given a nonempty open set $V \subset \mathcal{K}$, construct a nonempty open set $W \Subset V$.

(b) Conclude that $\mathcal{K}$ is a Baire space.

10.3. Exercise. *Show that the irrational numbers $\mathbb{R}\backslash\mathbb{Q}$ form a Baire space.*

10.4. Exercise. *Let $B_1, B_2, \dots$ be a sequence of nonempty sets in a Baire space $\mathcal{X}$. Assume $\mathcal{X} = B_1 \cup B_2 \cup \dots$ Show that at least one of the closures $\bar{B}_n$ has nonempty interior.*

10.5. Exercise. *Prove the following statements.*

(a) Any open set in a Baire space is Baire.

(b) Any locally Baire space is Baire; that is, if any point p in a topological space $\mathcal{X}$ has a Baire neighborhood, then $\mathcal{X}$ is Baire.

10.6. Advanced exercise. *Let $Q_1, Q_2, \dots$ be a sequence of nonempty disjoint closed sets in $[0, 1]$. Show that $Q_1 \cup Q_2 \cup \dots \neq [0, 1]$.*

B G-delta sets

Let G be a set in a topological space $\mathcal{X}$. We say that G is a G-delta set if it can be written as a countable intersection of open sets; that is, $G = V_1 \cap V_2 \cap \dots$ for a sequence of open sets $V_1, V_2, \dots \subset \mathcal{X}$. In the definition we can assume that the sequence of open sets is nested; that is, $V_1 \supset V_2 \supset \dots$; if not, pass to the sequence

$$V_1,\ V_1 \cap V_2,\ V_1 \cap V_2 \cap V_3,\ \dots$$

10.7. Exercise. *Show that any closed set in a metrizable space is G-delta.*

10.8. Observation. *The union of two (and therefore finitely many) G-delta sets is G-delta.*

Proof. The observation follows from the identity

$$\Big(\bigcap_i V_i\Big) \cup \Big(\bigcap_j W_j\Big) = \bigcap_{i,j} (V_i \cup W_j);$$

note that the intersection in the right-hand side is countable.

Indeed, we may assume that V_i and W_j are open and our G-delta sets are $G_1 = \bigcap_i V_i$, and $G_2 = \bigcap_j W_j$, so the identity gives two expressions for $G_1 \cup G_2$. Since the set $V_i \cup W_j$ is open for any i and j, the right-hand side says that $G_1 \cup G_2$ is G-delta. □

Recall that *countable union of countable sets is countable.* Since each G-delta set is a countable intersection of open sets, we get the following.

10.9. Observation. *A countable intersection of G-delta sets is G-delta.*

Combining this observation with the definition of Baire space, we get the following.

10.10. Observation. *Let $\mathcal{X}$ be a Baire space. Then a countable intersection of dense G-delta sets in $\mathcal{X}$ is dense.*

10.11. Exercise.

(a) Show that any G-delta set is an intersection of a closed set with a dense G-delta set.

(b) Give an example of a topological space with a G-delta set that is not an intersection of an open set with a dense G-delta set.

Recall that $\mathbb{Q} \subset \mathbb{R}$ denotes the set of rational numbers. Note that the set of irrational numbers $\mathbb{I} = \mathbb{R} \backslash \mathbb{Q}$ is G-delta — it is an intersection of all complements $\mathbb{R} \setminus \{q\}$ for $q \in \mathbb{Q}$. Since $\mathbb{Q}$ and $\mathbb{I}$ are dense, the observation implies that $\mathbb{Q} \subset \mathbb{R}$ is not G-delta. On the other hand, $\mathbb{Q} \subset \mathbb{R}$ is a countable union of one-point sets; each one-point set $\{q\}$ can be written as an intersection of open intervals $(q - \tfrac{1}{n}, q + \tfrac{1}{n})$. Therefore, a countable union of G-delta sets does not have to be G-delta.

10.12. Exercise. *Show that any dense G-delta set in $\mathbb{R}$ is uncountable.*

Let us consider a map $f\colon \mathcal{X} \to \mathcal{Y}$ between topological spaces. We say that f is continuous at $x \in \mathcal{X}$ if for any open set $W \subset \mathcal{Y}$ such that $W \ni f(x)$ there is an open set $V \ni x$ such that $f(V) \subset W$. Observe that f is continuous if and only if it is continuous at every point $x \in \mathcal{X}$.

10.13. Exercise. *Given a function* $f\colon \mathbb{R}\to\mathbb{R}$*, denote by* $\operatorname{Cont} f \subset \mathbb{R}$ *the set of points at which* f *is continuous.*

(a) *Show that* $\operatorname{Cont} f$ *is a G-delta set for any function* $f\colon \mathbb{R}\to\mathbb{R}$.

(b) *Show that there is no function* $f\colon \mathbb{R}\to\mathbb{R}$ *such that* $\operatorname{Cont} f = \mathbb{Q}$.

(c) *Let us enumerate all rational numbers* $r_1, r_2, \dots$ *Consider the function* $f\colon \mathbb{R}\to\mathbb{R}$ *defined by* $f(r_n) = \frac{1}{n}$ *and* $f(x) = 0$ *for any irrational* x*. Show that* $\operatorname{Cont} f = \mathbb{R}\setminus\mathbb{Q}$.

10.14. Advanced exercise. *Show that any G-delta set in a complete metric space is Baire.*

Give an example of a Baire space $\mathcal{X}$ *with a G-delta set that is not Baire.*

C Large or small?

Observation 10.9 can be interpreted the following way: *dense G-delta sets contain all points in a Baire space except a negligible set.* At least, we cannot cover an open set with complements of dense G-delta sets and the intersection of countably many dense G-delta sets is a dense G-delta set.

In spaces with measure, the sets of full measure have analogous features, but as we will see in an incompatible way.

Let us enumerate rational numbers $q_1, q_2, \dots$; consider the unions of open intervals:

$$V_n := \bigcup_i (q_i - \tfrac{1}{2^{n+i}}, q_i + \tfrac{1}{2^{n+i}}).$$

Note that $G = V_1 \cap V_2 \cap \dots$ is a dense G-delta set in $\mathbb{R}$. Observe that the Lebesgue measure[1] of V_n is less than

$$\frac{1}{2^{n-1}} = \frac{2}{2^{n+1}} + \frac{2}{2^{n+2}} + \dots$$

Therefore, G has vanishing measure. So, measure-theoretically, G is nothing, but topologically it is everything.

D Borel sets

Any topological space $\mathcal{X}$ comes with the so-called Borel sets, which form the minimal family of sets that is closed under complement and countable union and includes all open sets in $\mathcal{X}$.

[1] Lebesgue measure of a set in $\mathbb{R}$ can be thaut as its total length, but it is defined on a much larger class of subsets that include all G-delta sets and Borel sets that will be discussed soon.

A family of sets in $\mathcal{X}$ that contains the whole space and is closed under complements and countable unions is called a sigma-algebra, so we can say that Borel sets form the minimal sigma-algebra that includes all open sets.

Since the complement of a countable union is the countable intersection of the complements, a sigma-algebra can be equivalently defined using countable intersection instead of countable union. It follows that Borel sets include all closed sets (since they are complements of open sets) as well as G-delta sets (as countable intersections of open sets) and their complements, which are called F-sigma sets; the F-sigma sets can be defined directly as a union of countable families of closed sets.

The following proposition implies that any Borel set in $\mathbb{R}$ is a negligible modification of an open set.

10.15. Proposition. *For any Borel set B in a Baire space there is a dense G-delta set G and an open set V such that $B \cap G = V \cap G$.*

Proof. Choose a Baire space $\mathcal{X}$. Let us denote by $\mathscr{B}$ the set of all Borel sets in $\mathcal{X}$. Further, denote by $\mathscr{S}$ the set of all subsets of $\mathcal{X}$ that meet the conclusion of the proposition; that is, a subset $S \subset \mathcal{X}$ belongs to $\mathscr{S}$ if there is a dense G-delta set $G \subset \mathcal{X}$ and an open set $V \subset \mathcal{X}$ such that $S \cap G = V \cap G$.

Observe that $\mathscr{S}$ includes all open sets. Let us show that $\mathscr{S}$ is a sigma-algebra; once it is proved, the proposition follows since $\mathscr{B}$ is the minimal sigma-algebra that includes all open sets in $\mathcal{X}$.

Choose $S \in \mathscr{S}$, so there is a dense G-delta set G and an open set V such that $S \cap G = V \cap G$. Let $T = \mathcal{X} \backslash S$. Note that $W = \mathcal{X} \backslash \bar{V}$ is open, $\mathcal{X} \setminus \partial V = V \cup W$ is a dense open set, and therefore $G' = G \cap (V \cup W)$ is a dense G-delta set. It follows that

$$T \cap G' = W \cap G',$$

so $T \in \mathscr{S}$, and we proved that $\mathscr{S}$ is closed under taking complements.

Now, choose a sequence of sets $S_1, S_2, \ldots \in \mathscr{S}$; it comes with a sequence of dense G-delta sets $G_1, G_2, \ldots$ and open sets $V_1, V_2, \ldots$ such that $S_i \cap G_i = V_i \cap G_i$ for each i. Let

$$S = S_1 \cup S_2 \cup \ldots, \quad V = V_1 \cup V_2 \cup \ldots, \quad G = G_1 \cap G_2 \cap \ldots$$

The set V is open as a union of open sets. Furthermore, G is G-delta, and since $\mathcal{X}$ is Baire, it must be dense. Observe that

$$S \cap G = V \cap G;$$

in particular, $S \in \mathscr{S}$ — we proved that $\mathscr{S}$ is closed under countable union, so $\mathscr{S}$ is a sigma-algebra — the proof is finished. $\square$

We say that $L \subset \mathbb{R}$ is a Vitali set if for any $x \in \mathbb{R}$ there is a unique $\ell \in L$ such that $x - \ell \in \mathbb{Q}$; here $\mathbb{Q}$ denotes the set of rational numbers.

To construct a Vitali set, consider the equivalence relation $\sim$ on $\mathbb{R}$ such that $x \sim y \Leftrightarrow x - y \in \mathbb{Q}$ and choose one number from each equivalence class. Be aware that to *choose* a number in every equivalence class one needs the axiom of choice.

Now we are ready to show that not every set in the real line is Borel.

10.16. Claim. *A Vitali set $L \subset \mathbb{R}$ is not Borel.*

Proof. Given a set $S \subset \mathbb{R}$, denote by $x + S$ its shift by $x \in \mathbb{R}$; that is,

$$x + S = \{\, x + s \;:\; s \in S \,\}.$$

Let L be a Vitali set. Note that $L \cap (q + L) = \varnothing$ for any rational $q \neq 0$ and

❶ $$\mathbb{R} = \bigcup_{q \in \mathbb{Q}} (q + L).$$

If L is Borel, then by 10.15 $L \cap G = V \cap G$ for some dense G-delta set G and an open set V.

Note that $V \neq \varnothing$; otherwise ❶ implies that

$$\bigcap_{q \in \mathbb{Q}} (q + G) \subset \mathbb{R} \setminus \bigcup_{q \in \mathbb{Q}} (q + L) = \varnothing,$$

which is impossible since $q + G$ is a dense G-delta set for any $q \in \mathbb{Q}$, the set $\mathbb{Q}$ is countable, and $\mathbb{R}$ is Baire.

Since V is open, it contains an open interval; therefore, we can choose small positive $q \in \mathbb{Q}$ such that $V_q = V \cap (q + V)$ is a nonempty open set. Let $G_q = G \cap (q + G)$; it is a dense G-delta set. Hence $V_q \cap G_q \neq \varnothing$. Therefore

$$\varnothing = L \cap (q + L) \supset V_q \cap G_q \neq \varnothing$$

— a contradiction. $\square$

10.17. Exercise. *Construct a continuous map $f\colon \mathcal{X} \to \mathbb{R}$ and a Borel set $B \subset \mathcal{X}$ such that $f(B)$ is not Borel.*

10.18. Advanced exercise. *Construct a bijection $f\colon [0,1] \to [0,1] \times [0,1]$ such that f and its inverse f^{-1} map Borel sets to Borel sets.*

E An application

Let $\mathcal{C}$ be the set of all continuous functions $[0,1]\to\mathbb{R}$, equipped with the sup-norm metric

$$|f-g|_{\mathcal{C}} := \sup\{\,|f(x)-g(x)| \,:\, x\in[0,1]\,\}.$$

It is straightforward to check that this is indeed a metric; that is, it meets all conditions in 1.1. The following claim is not that obvious.

10.19. Claim. *The space $\mathcal{C}$ is complete.*

Proof. In the space $\mathcal{C}$ every point is a continuous function $[0,1]\to\mathbb{R}$. We need to show that a Cauchy sequence $f_1, f_2, \ldots \in \mathcal{C}$ converges in $\mathcal{C}$.

Choose $x\in[0,1]$. Observe that $f_1(x), f_2(x), \ldots$ is a Cauchy sequence of reals; so, it must converge; let us denote its limit by $f_\infty(x)$.

Evidently,

$$\text{❷}\qquad \sup\{\,|f_n(x)-f_\infty(x)| \,:\, x\in[0,1]\,\}\to 0.$$

Fix $x_0\in[0,1]$ and $\varepsilon>0$. By ❷, we can choose n such that

$$|f_n(x)-f_\infty(x)|<\tfrac{\varepsilon}{3}$$

for any $x\in[0,1]$. By continuity of f_n, we can choose $\delta>0$ such that

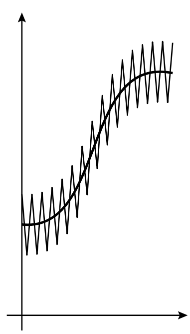
$$|x-x_0|<\delta \quad\Rightarrow\quad |f_n(x)-f_n(x_0)|<\tfrac{\varepsilon}{3}$$

If $|x-x_0|<\delta$, then

$$\begin{aligned}\varepsilon &> |f_\infty(x)-f_n(x)|+|f_n(x)-f_n(x_0)|+|f_n(x_0)-f_\infty(x_0)| \geqslant\\ &\geqslant |f_\infty(x)-f_\infty(x_0)|.\end{aligned}$$

Since $\varepsilon>0$ and x_0 are arbitrary, f_∞ is continuous; that is, $f_\infty\in\mathcal{C}$. Finally by ❷, $f_n\to f_\infty$ in $\mathcal{C}$. □

10.20. Claim. *The set of all continuous functions $[0,1]\to\mathbb{R}$ that are not differentiable at any point contains a dense G-delta set in $\mathcal{C}$.*



Sketch. Given a positive integer n, consider the set V_n of all functions f in $\mathcal{C}$ such that for some integer $m>n$ we have the following inequality

$$|f(\tfrac{k}{m})-f(\tfrac{k-1}{m})|>\tfrac{n}{m}$$

for any integer k from 1 to m.

The set V_n is open and dense in $\mathcal{C}$; to prove the latter statement, take huge value m and approximate a given function by a zig-zag-looking function with large slope as shown in the picture.[2]

Now, assume $f \in V_n$ for every n. Observe that for any $x_0 \in [0,1]$ we can choose a point $x \in [0,1]$ such that

$$|x - x_0| < \tfrac{1}{n}, \quad \text{and} \quad \left|\frac{f(x) - f(x_0)}{x - x_0}\right| > n.$$

It follows that the limit

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

is undefined. Therefore f differentiable at no point $x_0 \in [0,1]$.

It remains to apply 10.1 to $G = V_1 \cap V_2 \cap \dots$ □

We just proved that *most* continuous functions $[0,1] \to \mathbb{R}$ are not differentiable at any point. However, finding one such example is challenging. This is very common in mathematics: as soon as you define a typical object in your favorite theory, you realize that you have never actually seen one. (By the way, did you ever see a typical man?)

Let me list several other statements that can be proved essentially the same way.

- A typical surjective nondecreasing function $f\colon [0,1] \to [0,1]$ is strictly increasing, but its derivative vanishes at every point where it is defined.
- A typical distance-nonexpanding map $f\colon \mathbb{S}^2 \to \mathbb{R}^2$ is length-preserving; that is,

$$\operatorname{length} f \circ \gamma = \operatorname{length} \gamma$$

  for any curve $\gamma\colon [a,b] \to \mathbb{S}^2$.
- A typical compact set in the plane is homeomorphic to the Cantor set; see 4B. Furthermore, a typical compact *connected* set (defined in the next chapter) is homeomorphic to the pseudo-arc, which is a truly weird example in many ways (read about it).

[2]More formally, let us choose a small $\varepsilon > 0$. Given a continuous function f we can choose large m so that if $|x_0 - x_1| \leqslant \frac{1}{m}$, then $|f(x_0) - f(x_1)| < \varepsilon$. Now define a new function h by setting $h(\frac{i}{m}) = f(\frac{i}{m}) + (-1)^i \cdot \varepsilon$ for an integer i from 0 to m and extend h linearly on each interval $[\frac{i-1}{m}, \frac{i}{m}]$. Observe that $h \in \mathcal{C}$ and $|h - f|_{\mathcal{C}} < 2 \cdot \varepsilon$; moreover, given $\varepsilon > 0$ and a positive integer n, one can choose m sufficiently large so that $h \in V_n$.

Chapter 11

Connected spaces

A Definitions

A subset of a topological space is called c l o p e n if it is closed and open at the same time.

11.1. Definition. *A topological space $\mathcal{X}$ is called c o n n e c t e d if it has exactly two clopen sets $\varnothing$ and $\mathcal{X}$.*

A subset of a topological space is called c o n n e c t e d if the corresponding subspace[1] is connected.

According to our definition, *the empty space is not connected.* Not everyone agrees with this convention.[2]

Suppose V is a p r o p e r clopen subset of a topological space $\mathcal{X}$; that is, $V \neq \varnothing$ and $V \neq \mathcal{X}$. Note that its complement $W = \mathcal{X} \setminus V$ is also a proper clopen subset. In particular, there are two open sets $V, W \subset \mathcal{X}$ such that

$$V \neq \varnothing, \quad W \neq \varnothing, \quad V \cup W = \mathcal{X}, \quad \text{and} \quad V \cap W = \varnothing.$$

In this case, we say that V and W s e p a r a t e $\mathcal{X}$.

Spelling out the notion of subspace leads to the following definition: an open s e p a r a t i n g p a i r of a set $A \subset \mathcal{X}$ consists of two open sets $V, W \subset \mathcal{X}$ such that

$$V \cap A \neq \varnothing, \quad W \cap A \neq \varnothing, \quad V \cup W \supset A, \quad \text{and} \quad V \cap W \cap A = \varnothing.$$

[1]Which is the subset equipped with the induced topology.

[2]This convention is similar in spirit to saying that 1 is not prime — if prime meant *no nontrivial divisors*, then 1 would be prime, *but it is not.* Similarly, if a connected space meant *no proper clopen sets*, then the empty set would be connected, but *it is not* (at least in this course).

11.2. Observation. *A nonempty topological space (or a nonempty subset of a topological space) is* disconnected *if and only if it has an open separating pair.*

11.3. Exercise. *Let $f\colon \mathcal{X} \to \mathbb{R}$ be a continuous function defined on a connected space. Suppose that $f(x_1) > 0$ and $f(x_2) < 0$ for some $x_1, x_2 \in \mathcal{X}$. Show that there is $x_3 \in \mathcal{X}$ such that $f(x_3) = 0$.*

11.4. Observation. *Let A be a connected set in a topological space and*

$$A \subset B \subset \bar{A}.$$

Then B is connected.

Proof. Since A is connected, we have $A \neq \varnothing$ and hence $B \neq \varnothing$.

Assume V and W form an open separating pair of B. Since $B \subset \bar{A}$,

$$V \cap B \neq \varnothing \ \Rightarrow V \ \cap A \neq \varnothing,$$

and

$$W \cap B \neq \varnothing \ \Rightarrow W \ \cap A \neq \varnothing.$$

It follows that V and W form an open separating pair of A — a contradiction. $\square$

11.5. Advanced exercise. *Let $Q_1 \supset Q_2 \supset \dots$ be a nested sequence of closed connected sets in a compact space. Show that the intersection $Q_\infty = Q_1 \cap Q_2 \cap \dots$ is connected.*

11.6. Advanced exercise. *Consider the set of positive integers $\mathbb{N} = \{1, 2, \dots\}$. Denote by $\mathcal{B}$ the set of all arithmetic progressions $\{a, a+d, a+2\cdot d, \dots\}$ in $\mathbb{N}$ for all relatively prime positive integers a and d such that $a \leqslant d$.*

(a) Show that $\mathcal{B}$ is a base of some topology on $\mathbb{N}$.

(b) Show that $\mathbb{N}$ equipped with this topology is connected, Hausdorff, and noncompact.

B Key statements

11.7. Proposition. *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a continuous map between topological spaces. Suppose $A \subset \mathcal{X}$ is a connected set. Then the image $B = f(A)$ is a connected set in $\mathcal{Y}$.*

Proof. Since A is connected, $A \neq \varnothing$, and hence $B \neq \varnothing$.

Assume that $B = f(A)$ is disconnected. Choose open sets V and W that separate B; that is,

❶ $\quad V \cap B \neq \varnothing,\ W \cap B \neq \varnothing,\ V \cup W \supset B,$ and $V \cap W \cap B = \varnothing.$

Since f is continuous, $V' = f^{-1}(V)$ and $W' = f^{-1}(W)$ are open sets in $\mathcal{X}$. Note that ❶ implies that V' and W' separate A; that is,

$$V' \cap A \neq \varnothing,\ W' \cap A \neq \varnothing,\ V' \cup W' \supset A, \text{ and } V' \cap W' \cap A = \varnothing.$$

Therefore, A is disconnected — a contradiction. □

11.8. Exercise. *Let $\mathcal{X}$ be a connected space. Show that the quotient space $\mathcal{X}/{\sim}$ is connected for any equivalence relation $\sim$ on $\mathcal{X}$.*

11.9. Proposition. *Assume $\{A_\alpha\}_{\alpha \in \mathcal{I}}$ is a family of connected subsets of a topological space. If $\bigcap_\alpha A_\alpha \neq \varnothing$, then*

$$A = \bigcup_\alpha A_\alpha$$

is connected.

Proof. Assume that A is disconnected; let V and W be an open separating pair.

Choose $p \in \bigcap_\alpha A_\alpha$. Without loss of generality, we can assume that $p \in V$.

In particular, $V \cap A_\alpha \neq \varnothing$ for any α. Since A_α is connected, we have that $W \cap A_\alpha = \varnothing$ for each α; otherwise V and W would separate A_α. Therefore

$$\begin{aligned} W \cap A = W \cap \left(\bigcup_\alpha A_\alpha\right) &= \\ = \bigcup_\alpha (W \cap A_\alpha) &= \\ = \varnothing. \end{aligned}$$

Hence V and W do not separate A — a contradiction. □

11.10. Exercise. *Show that two topological spaces $\mathcal{X}$ and $\mathcal{Y}$ are connected if and only if the product space $\mathcal{X} \times \mathcal{Y}$ is connected.*

C Real interval

11.11. Proposition. *The real interval* $[0,1]$ *is connected.*

The proof reuses the construction in 7C.

Proof. Assume the contrary; choose an open separating pair of $[0,1]$, say V and W. Fix $a_0 \in V$ and $b_0 \in W$; without loss of generality, we can assume that $a_0 < b_0$.

Let us construct a nested sequence of closed intervals

$$[a_0, b_0] \supset [a_1, b_1] \supset \dots$$

such that

❷ $$b_n - a_n = \frac{b_0 - a_0}{2^n}, \quad a_n \in V, \quad \text{and} \quad b_n \in W$$

for any n.

The construction is recursive. Assume $[a_{n-1}, b_{n-1}]$ is already constructed. Let $c = \frac{1}{2}\cdot(a_{n-1} + b_{n-1})$; note that $c \in V$ or $c \in W$. If $c \in V$, then set $a_n = c$ and $b_n = b_{n-1}$; if $c \in W$, then set $a_n = a_{n-1}$ and $b_n = c$. In both cases, the conditions in ❷ hold.

Observe that

$$a_0 \leqslant a_1 \leqslant \dots \leqslant b_1 \leqslant b_0.$$

In particular, the sequence a_n is nondecreasing and bounded above by b_0. Therefore, it must converge; denote its limit by x. Since

$$b_n - a_n = \tfrac{1}{2^n}\cdot(b_0 - a_0),$$

the sequence b_n also converges to x. The point x has to belong to V or W. Since both V and W are open, whichever set contains x contains all a_n and b_n for all large n, which contradicts ❷. □

11.12. Exercise. *Show that the real line* $\mathbb{R}$ *is a connected space. Conclude that any continuous map* $\mathbb{R} \to \mathbb{Z}$ *is constant; here* $\mathbb{Z} \subset \mathbb{R}$ *denotes the set of integers.*

Recall that a set $V \subset \mathbb{R}^n$ is called s t a r - s h a p e d if there exists a point $p \in V$ such that for all $x \in V$ the line segment from p to x lies in V.

11.13. Exercise. *Show that any star-shaped set in* $\mathbb{R}^n$ *is connected.*

D Connected components

Let x be a point in a topological space $\mathcal{X}$. The union of all connected sets containing x is called the connected component of x. Note that the space $\mathcal{X}$ is connected if and only if a connected component of some (and therefore of any) point is the whole space $\mathcal{X}$.

Observe that two connected components either coincide or are disjoint. In other words, lying in the same connected component defines an equivalence relation.

11.14. Exercise.
- *(a) Show that any connected component is a closed set.*
- *(b) Construct an example of a topological space $\mathcal{X}$ and a point $x \in \mathcal{X}$ such that the connected component of x is not open.*
- *(c) Suppose that a space $\mathcal{X}$ has a finite number of connected components. Show that each connected component of $\mathcal{X}$ is open.*

11.15. Advanced exercise. *Let $\mathcal{X}$ be a compact Hausdorff space. Show that the connected component of a point $x \in \mathcal{X}$ is the intersection of all clopen sets containing x. Find examples showing that this is no longer true without assuming compactness or Hausdorffness.*

E Locally connected spaces

A topological space $\mathcal{X}$ is called locally connected if for any point $p \in \mathcal{X}$, any neighborhood of p contains a connected neighborhood; that is, given an open set $W \ni p$ there is a connected open set V such that $p \in V \subset W$. By 11.13, $\mathbb{R}^n$ is locally connected for any n.

11.16. Observation. *Let V be an open set in a locally connected space. Then any connected component of V is open.*

Proof. Let V_0 be a connected component in V. Given a point $x \in V_0$, choose a connected neighborhood $N_x \subset V$ of x. By 11.9, $N_x \subset V_0$, and the observation follows. □

F Cut points

Connectedness is defined in purely topological terms. In particular, the number of connected components is a topological invariant; that is, *if two spaces are homeomorphic, then they have the same number of connected components.* In particular, a connected space is not homeomorphic to a disconnected space.

Let us describe a refined way to apply this observation. Suppose $\mathcal{X}$ is a connected space. A point $x \in \mathcal{X}$ is called a cut point of $\mathcal{X}$ if removing x from $\mathcal{X}$ produces a disconnected space; that is, the subset $\mathcal{X} \setminus \{x\}$ is disconnected.

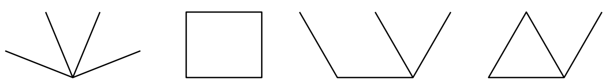
Note that if $f\colon \mathcal{X} \to \mathcal{Y}$ is a homeomorphism, then a point $x \in \mathcal{X}$ is a cut point of $\mathcal{X}$ if and only if $y = f(x)$ is a cut point of $\mathcal{Y}$. Moreover, the number of connected components in $\mathcal{X} \setminus \{x\}$ is the same as in $\mathcal{Y} \setminus \{y\}$. The latter follows since the restriction of f defines a homeomorphism $\mathcal{X} \setminus \{x\} \to \mathcal{Y} \setminus \{y\}$.

For example, consider the circle $\mathbb{S}^1$ and the line segment $[0,1]$. Note that both spaces are connected, compact, Hausdorff, metrizable, and Baire — none of these properties can distinguish these spaces. Now choose the midpoint $\frac{1}{2} \in [0,1]$; observe that its complement $[0,1] \setminus \{\frac{1}{2}\}$ has two connected components. If there is a homeomorphism $f\colon [0,1] \to \mathbb{S}^1$, then the set $\mathbb{S}^1 \setminus \{f(\frac{1}{2})\}$ must have two connected components. On the other hand, $\mathbb{S}^1 \setminus \{x\}$ is connected for any point $x \in \mathbb{S}^1$ (prove it). This leads to a contradiction, proving that $\mathbb{S}^1$ is not homeomorphic to the line segment $[0,1]$.

The same type of argument can be used to solve the following exercises.

11.17. Exercise. *Show that the plane $\mathbb{R}^2$ is not homeomorphic to the real line $\mathbb{R}$.*

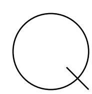
11.18. Exercise. *Show that no two of the following four closed connected sets in the plane are homeomorphic. (Each set is a union of*



(a) (b) (c) (d)

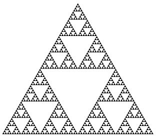
four line segments, and if it looks like they share an endpoint, then this is indeed so.)

11.19. Exercise. *Let Q be the set shown in the picture; it is a union of a circle together with a closed line segment crossing it. Consider the action on Q of the group H of all homeomorphisms $Q \circlearrowleft$. Describe the quotient space Q/H; that is, list its points and the open sets.*



The Sierpiński triangle is constructed in the following way: start with a solid equilateral triangle, subdivide it into four smaller congruent equilateral triangles, and remove the interior of the central one. Repeat this procedure recursively for each of the remaining solid triangles.

11.20. Advanced exercise.



(a) *Prove that the Sierpiński triangle is connected (in particular, nonempty).*

(b) *Describe all the homeomorphisms from the Sierpiński triangle to itself.*

Chapter 12

Path-connected spaces

A Paths

Let $\mathcal{X}$ be a topological space. A continuous map $a\colon [0,1] \to \mathcal{X}$ is called a path. If $x = a(0)$ and $y = a(1)$, we say that a is a path from x to y.

Let us start with several examples and constructions.

- A path a is called constant if it stays at one point. In other words, $a(t) = p$ for some fixed point p; this path can be denoted by e_p.
- Given a path $a\colon [0,1] \to \mathcal{X}$, one can consider the time-reversed path $\bar{a}$, defined by
$$\bar{a}(t) = a(1-t).$$
Note that $\bar{a}$ is continuous since a is.
- Let a and b be paths such that $a(1) = b(0)$. Then we can join them into one path $c\colon [0,1] \to \mathcal{X}$ defined as
$$c(t) = \begin{cases} a(2{\cdot}t) & \text{if } t \leqslant \frac{1}{2}, \\ b(2{\cdot}t - 1) & \text{if } t \geqslant \frac{1}{2}; \end{cases}$$
so, c follows a twice as fast and then b twice as fast. The path c is called the product (or concatenation) of a and b; it is denoted as $c = a * b$. Note that 3.13b implies that $a * b$ is continuous; in other words, $a * b$ is indeed a path.

12.1. Advanced exercise. *Let a be a path from x to y in a Hausdorff space $\mathcal{X}$. Assume $x \neq y$. Show that there is an injective path b from x to y such that $b([0,1]) \subset a([0,1])$.*

B Path-connected components

Consider the following relation on the set of points of a topological space:

$$x \sim y \quad \Leftrightarrow \quad \text{there is a path from } x \text{ to } y.$$

The constructions in 12A imply that $\sim$ is an equivalence relation; in other words, the following properties hold for any points $x, y, z \in \mathcal{X}$:

- $x \sim x$ (since e_x is a path from x to x);
- if $x \sim y$, then $y \sim x$ (indeed, if a is a path from x to y, then $\bar{a}$ is a path from y to x);
- if $x \sim y$ and $y \sim z$, then $x \sim z$ (indeed, if a is a path from x to y, and b is a path from y to z, then $a * b$ is a path from x to z).

The equivalence class of a point x for the equivalence relation $\sim$ is called the path-connected component of x; this is the set of all points in $\mathcal{X}$ that can be reached by a path from x.

C Path-connected spaces

A space $\mathcal{X}$ is called path-connected if it is nonempty and any two points in $\mathcal{X}$ can be connected by a path; that is, for any $x, y \in \mathcal{X}$ there is a path a from x to y. One can say that $\mathcal{X}$ is path-connected if $\mathcal{X}$ is the path-connected component of one (and therefore every) point $x \in \mathcal{X}$.

12.2. Exercise. *Show that the connected two-point space (defined in 2B) is path-connected.*

12.3. Exercise. *Show that the image of a path-connected set under a continuous map is path-connected.*

12.4. Exercise. *Show that the product of path-connected spaces is path-connected.*

12.5. Theorem. *Any path-connected space is connected; the converse does not hold.*

Proof; main part. Let $\mathcal{X}$ be a path-connected space.

By 11.11, the unit interval $[0,1]$ is connected. By 11.7, for any path $a\colon [0,1] \to \mathcal{X}$ the image $a([0,1])$ is connected.

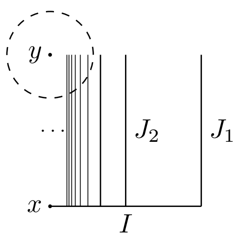
Choose $x \in \mathcal{X}$. Since $\mathcal{X}$ is path-connected,

$$\mathcal{X} = \bigcup_a a([0,1]),$$

where the union is taken over all paths a starting from x. It remains to apply 11.9.

Second part. We need to present an example of a space that is connected, but not path-connected.

Denote by I the closed line segment from $(0,0)$ to $(1,0)$ in $\mathbb{R}^2$. Further, denote by J_n the closed line segment from $(\frac{1}{n},0)$ to $(\frac{1}{n},1)$. Consider the union of all these segments



$$W = I \cup J_1 \cup J_2 \cup \dots$$

and set

$$\dot{W} = W \cup \{y\},$$

where $y = (0,1)$. The space $\dot{W}$ is called the flea and comb; the set W is called the comb, and the point y is called the flea.

Note that $W \subset \dot{W} \subset \bar{W}$. Therefore, by 11.4, $\dot{W}$ is connected.

It remains to show that $\dot{W}$ is not path-connected. Assume the contrary. Let a be a path from $x = (0,0)$ to $y = (0,1)$.

Note that $a^{-1}(\{y\})$ is a closed subset of the compact space $[0,1]$. Therefore $a^{-1}(\{y\})$ is compact (7.13). In particular, the set $a^{-1}(\{y\})$ has a minimum, denote it by t_1. Note that $t_1 > 0$, $a(t_1) = y$ and $a(t) \neq y$ for any $t < t_1$.

Since a is continuous, there is $t_0 < t_1$ such that $|a(t) - y| < \frac{1}{10}$ for any $t \in [t_0, t_1]$. Since $|a(t_0) - y| < \frac{1}{10}$, we have $a(t_0) \in J_n$ for some n (in other words, $a(t_0) \notin I$).

Denote by N the intersection of the $\frac{1}{10}$-neighborhood of y with W. Observe that the intersection $J_n \cap N$ is a connected component of N. By 11.7, $a(t) \in J_n$ for any $t \in [t_0, t_1]$; in particular, $a(t_1) \neq y$ — a contradiction. $\square$

12.6. Exercise. *Describe path-connected components in the flea and comb.*

12.7. Exercise. *Construct a path-connected set in the plane with non-path-connected closure.*

12.8. Exercise. *Assume every path-connected component of a topological space $\mathcal{X}$ is open. Show that $\mathcal{X}$ is connected if and only if $\mathcal{X}$ is path-connected.*

12.9. Advanced exercise. *Let $a, b\colon [0,1] \to [0,1]$ be two continuous functions such that $a(0) = b(0) = 0$ and $a(1) = b(1) = 1$. Consider* *the set*[1]

$$S = \{\, (x,y) \in [0,1] \times [0,1] \;:\; a(x) = b(y) \,\}\,.$$

Note that two points $p = (0,0)$ and $q = (1,1)$ lie in S.

(a) *Show that p and q lie in the same connected component of S.*

(b) *Give an example of a and b such that p and q lie in different path-connected components of S.*

12.10. Advanced exercise. *Consider the following sets in the plane:*

$$A = \left\{\, (x,y) \in \mathbb{R}^2 \;:\; x,y \in \mathbb{Q} \,\right\} \quad \text{and} \quad B = \left\{\, (x,y) \in \mathbb{R}^2 \;:\; x,y \notin \mathbb{Q} \,\right\},$$

where $\mathbb{Q}$ denotes the set of rational numbers. Show that $A \cup B$ is path-connected.

12.11. Advanced exercise. *Construct a bounded open set V in the plane such that its boundary is connected, but totally path-disconnected; that is, each path-connected component in ∂V consists of a single point.*

12.12. Advanced exercise. *Show that a finite topological space is path-connected if it is connected.*

D Locally path-connected spaces

A topological space $\mathcal{X}$ is called locally path-connected if for any point $p \in \mathcal{X}$ and any open set $V \ni p$ there is a path-connected open set W such that $V \supset W \ni p$.

For instance, *Euclidean space is locally path-connected*; this follows since any open ball in a Euclidean space is convex, and therefore path-connected.

12.13. Theorem. *A connected open set in a locally path-connected space is path-connected.*

Proof. Let V be an open subset of a locally path-connected space. Choose a point $p \in V$; denote by $P \subset V$ the path-connected component of p in V.

Choose $q \in V$ and let W_q be the neighborhood provided by the definition of locally path-connected space; that is, $V \supset W_q \ni q$. Since W_q is path-connected, we have $W_q \subset P$ or $W_q \cap P = \varnothing$.

It follows that P and its complement $V \setminus P$ are open. Since V is connected, we get that $V \setminus P = \varnothing$ — hence the result. $\square$

---

[1] This is the fiber-product space of a and b; see 17E.

12.14. Exercise.

(a) *According to the theorem, the flea-and-comb must fail to be locally path-connected at some points. Find the set of such points.*

(b) *Construct a path-connected but not locally path-connected space.*

12.15. Exercise. *Let V be an open set in a locally path-connected space. Show that any path-connected component of V is open.*

Chapter 13

Homotopy

A Homotopy

Two continuous maps $f, g\colon \mathcal{X} \to \mathcal{Y}$ are called homotopic (briefly, $f \sim g$) if f can be continuously deformed into g. Formally, this means that there exists a continuous map

$$H\colon \mathcal{X} \times [0,1] \to \mathcal{Y}$$

such that $H(x,0) = f(x)$ and $H(x,1) = g(x)$ for all $x \in \mathcal{X}$. The map $H\colon \mathcal{X} \times [0,1] \to \mathcal{Y}$ is called a homotopy from f to g.

It is often convenient to think of the homotopy H as a one-parameter family of maps $h_t\colon \mathcal{X} \to \mathcal{Y}$ defined by $h_t(x) = H(x,t)$.

A map is called null-homotopic if it is homotopic to a constant map.

Observe that if $f_t\colon \mathcal{X} \to \mathcal{Y}$ and $g_t\colon \mathcal{Y} \to \mathcal{Z}$ are homotopies, then so is $h_t = g_t \circ f_t$, and this implies the following.

13.1. Observation. *Let $f_0, f_1\colon \mathcal{X} \to \mathcal{Y}$ and $g_0, g_1\colon \mathcal{Y} \to \mathcal{Z}$ be four maps between topological spaces. Assume that $f_0 \sim f_1$ and $g_0 \sim g_1$. Then $g_0 \circ f_0 \sim g_1 \circ f_1$.*

B Linear homotopy

Suppose $\mathcal{C}$ is a convex subset of $\mathbb{R}^n$ with induced topology. Given two continuous maps $f, g\colon \mathcal{X} \to \mathcal{C}$, one can consider the linear homotopy from f to g defined by

$$H(x,t) = (1-t)\cdot f(x) + t\cdot g(x).$$

It follows that any two continuous maps $\mathcal{X} \to \mathcal{C}$ are homotopic; in particular, any continuous map $\mathcal{X} \to \mathcal{C}$ is null-homotopic.

Recall that

$$\mathbb{S}^2 := \{ (x, y, z) \in \mathbb{R}^3 \ : \ x^2 + y^2 + z^2 = 1 \} .$$

13.2. Exercise. *Let $f_0, f_1 \colon \mathcal{X} \to \mathbb{S}^2$ be two continuous maps. Suppose that $f_0(x) + f_1(x) \neq 0$ for any $x \in \mathcal{X}$. Show that $f_0 \sim f_1$.*

C Relative homotopy

Let $H \colon \mathcal{X} \times [0, 1] \to \mathcal{Y}$ be a homotopy, and let $A \subset \mathcal{X}$. Suppose that for any $a \in A$ the point $H(a, t)$ does not depend on t. Then we say that H is a homotopy relative to A. Two maps $f, g \colon \mathcal{X} \to \mathcal{Y}$ are called homotopic relative to A, briefly, $f \sim g$ (rel A), if there is a homotopy relative to A from f to g.

Note also that if $f \sim g$ (rel A), then f agrees with g on A; that is, $f(a) = g(a)$ for any $a \in A$.

D Operations on homotopies

The following operations on homotopies are analogous to the operations on paths discussed in 12A. In fact, a path is a special case of a homotopy — it is a homotopy with a one-point source space.

- the constant homotopy $H(x, t) = f(x)$ joins f with itself;
- if H is a homotopy from f to g, then the time-reversed homotopy defined by

$$\bar{H}(x, t) = H(x, 1 - t)$$

  is a homotopy from g to f;
- Consider two homotopies H and G from f to g and from g to h, respectively. Then their concatenation $F = H * G$ defined by

$$F(x, t) = \begin{cases} H(x, 2{\cdot}t), & t \leqslant \frac{1}{2}, \\ G(x, 2{\cdot}t - 1), & t \geqslant \frac{1}{2}, \end{cases}$$

  is a homotopy from f to h. (By 3.13*b*, F is continuous.)

All these constructions admit straightforward generalizations to relative homotopies.

Recall that

$$\mathbb{S}^1 := \{ (x, y) \in \mathbb{R}^2 \ : \ x^2 + y^2 = 1 \} .$$

13.3. Exercise. *Let $f\colon \mathbb{S}^1 \to \mathbb{S}^1$ be a continuous map such that $f(x) \neq x$ for any x. Show that f is homotopic to the identity map.*

E Homotopy classes

The constructions from the previous section imply that homotopy defines an equivalence relation $\sim$ on the set of continuous maps $\mathcal{X} \to \mathcal{Y}$; in other words, the following properties hold for any continuous maps $f, g, h\colon \mathcal{X} \to \mathcal{Y}$:

- $f \sim f$;
- if $f \sim g$, then $g \sim f$;
- if $f \sim g$ and $g \sim h$, then $f \sim h$.

The equivalence classes for $\sim$ are called homotopy classes; the equivalence class of a map $f\colon \mathcal{X} \to \mathcal{Y}$ will be denoted by $[f]$. Given two spaces $\mathcal{X}$ and $\mathcal{Y}$, the set of all homotopy classes will be denoted by $\mathrm{HClass}(\mathcal{X}, \mathcal{Y})$.

(The same holds for homotopies relative to $A \subset \mathcal{X}$; so $f \sim g$ (rel A) is an equivalence relation. The set of all homotopy classes relative to A can be denoted by $\mathrm{HClass}_A(\mathcal{X}, \mathcal{Y})$.)

Often the set $\mathrm{HClass}(\mathcal{X}, \mathcal{Y})$ admits a direct description; this is precisely its advantage over the vast set of all continuous maps $\mathcal{X} \to \mathcal{Y}$.

Note that any homeomorphism $\mathcal{Y} \to \mathcal{Z}$ defines a bijection between $\mathrm{HClass}(\mathcal{X}, \mathcal{Y})$ and $\mathrm{HClass}(\mathcal{X}, \mathcal{Z})$ for any given space $\mathcal{X}$. So, studying the homotopy classes may help to distinguish topological spaces.

This construction is meaningful even for very simple choices of the source space $\mathcal{X}$. If $\mathcal{X}$ is the one-point space, then $\mathrm{HClass}(\mathcal{X}, \mathcal{Y})$ is also denoted by $\pi_0(\mathcal{Y})$. Note that $\pi_0(\mathcal{Y})$ is the set of path-connected components of $\mathcal{Y}$ — this was the main subject of the previous chapter. The set $\mathrm{HClass}(\mathbb{S}^1, \mathcal{Y})$ is closely related to the fundamental group of $\mathcal{Y}$, which will soon be introduced.

The idea to use homotopy classes to distinguish topological spaces will guide us in the right direction. However, the true reason to consider homotopy classes is different; it will be discussed in Chapter 15.

F Retracts

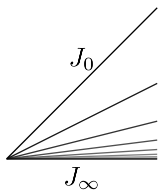
Let A be a subset of a topological space $\mathcal{X}$. A continuous map $r\colon \mathcal{X} \to A$ is called a retraction onto A if $r(a) = a$ for any $a \in A$.[1] In this case A is called a retract of $\mathcal{X}$. If, in addition, $r\colon \mathcal{X} \to A$ is homotopic

[1] The notion of retraction is closely related to projection in linear algebra; say, a continuous map $r\colon \mathcal{X} \to \mathcal{X}$ defines a retraction onto $r(\mathcal{X})$ if and only if $r \circ r = r$.

to the identity map $\mathrm{id}_{\mathcal{X}}$, then A is called a weak deformation retract of $\mathcal{X}$. If $r \sim \mathrm{id}_{\mathcal{X}}$ (rel A), then A is called a deformation retract of $\mathcal{X}$.

Note that a one-point subset is a retract of any space — the retraction maps the whole space to this single point. On the other hand, if a one-point set $\{p\}$ is a deformation retract of the space $\mathcal{X}$, then any point can be connected to p by a path; in particular, $\mathcal{X}$ has to be path-connected.

13.4. Exercise. *Show that a retract of a Hausdorff space has to be a closed subset.*



13.5. Exercise. *Consider the fan F in $\mathbb{R}^2$; it is defined as the union of closed line segments J_n from the origin to the point $(1, \frac{1}{2^n})$ for each integer $n \geqslant 0$ and the line segment J_∞ from the origin to the point $(1, 0)$.*

(a) Show that J_0 is a deformation retract of F.

(b) Show that J_∞ is a weak deformation retract of F, but not a deformation retract of F.

(c) Is it true that F is a retract of the square $[0,1] \times [0,1]$?

G Contractible spaces

A topological space $\mathcal{X}$ is called contractible if the identity map $\mathrm{id}_{\mathcal{X}} \colon \mathcal{X} \to \mathcal{X}$ is homotopic to a constant map; that is, there is a homotopy $h_t \colon \mathcal{X} \to \mathcal{X}$ such that $h_0(x) = x$ and $h_1(x) = p$ for some fixed point $p \in \mathcal{X}$ and all $x \in \mathcal{X}$.

Applying linear homotopy, one can see that any convex subset of $\mathbb{R}^n$ is contractible. This is a good source of contractible spaces.

Soon we will show that the circle $\mathbb{S}^1$ is an example of a path-connected space that is not contractible; in particular, there are non-contractible path-connected spaces.

13.6. Proposition. *A retract of a contractible space is contractible.*

Proof. Let $r \colon \mathcal{X} \to A \subset \mathcal{X}$ be a retraction and let $h_t \colon \mathcal{X} \to \mathcal{X}$ be a homotopy from the identity map to a constant map. Observe that $r \circ h_t$ defines a homotopy from the identity map $A \to A$ to a constant map. □

13.7. Exercise. *Let $\mathcal{X}$ be a contractible space.*

(a) Show that $\mathcal{X}$ is path-connected.

(b) Show that any two continuous maps from a topological space to $\mathcal{X}$ are homotopic.
(c) Show that any two continuous maps from $\mathcal{X}$ to a path-connected space are homotopic.

H Homotopy type

Two topological spaces $\mathcal{X}$ and $\mathcal{Y}$ have the same homotopy type (briefly, $\mathcal{X} \sim \mathcal{Y}$) if there are continuous maps $f\colon \mathcal{X} \to \mathcal{Y}$ and $h\colon \mathcal{Y} \to \mathcal{X}$ such that $h \circ f \sim \mathrm{id}_{\mathcal{X}}$ and $f \circ h \sim \mathrm{id}_{\mathcal{Y}}$. In this case f (as well as h) is called a homotopy equivalence and h is called a homotopy inverse of f.

13.8. Exercise. *Show that $\sim$ defines an equivalence relation on topological spaces. (The corresponding equivalence classes are called homotopy types.)*

13.9. Exercise. *Show that a topological space $\mathcal{X}$ is contractible if and only if it has the homotopy type of the one-point space.*

13.10. Exercise. *Show that a continuous map $f\colon \mathcal{X} \to \mathcal{Y}$ is a homotopy equivalence if there are continuous maps $g, h\colon \mathcal{Y} \to \mathcal{X}$ such that $h \circ f \sim \mathrm{id}_{\mathcal{X}}$ and $f \circ g \sim \mathrm{id}_{\mathcal{Y}}$.*

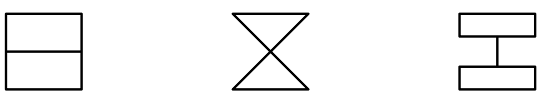
13.11. Proposition. *If A is a weak deformation retract of $\mathcal{X}$, then $\mathcal{X}$ and A have the same homotopy type.*

Proof. Since A is a weak deformation retract, there is a retraction $r\colon \mathcal{X} \to A$ and a homotopy from $\mathrm{id}_{\mathcal{X}}$ to r.

In other words, if $i\colon A \hookrightarrow \mathcal{X}$ is the inclusion map, then $\mathrm{id}_{\mathcal{X}} \sim i \circ r$. Since r is a retraction, $r \circ i = \mathrm{id}_A$, hence the result. □

This proposition can be used together with 13.8 to prove that given spaces have the same homotopy type; here is one example:

13.12. Exercise. *Show that the closed connected sets in the following picture have the same homotopy type. (Each set is a union of several*

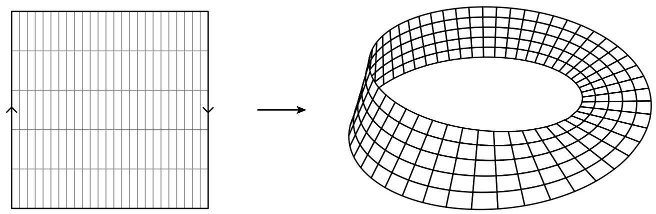


line segments and whenever two of them appear to intersect, they really do intersect.)

The Möbius band can be defined as the quotient space

$$\mathcal{M} := [-1,1] \times [-1,1]/\sim,$$

where $\sim$ is the minimal equivalence relation such that $(-1,t) \sim (1,-t)$.



Applying 9.15, one can show that $\mathcal{M}$ is homeomorphic to the band shown in the picture.

13.13. Exercise. *Show that the cylinder* $\mathbb{S}^1 \times [0,1]$ *and the Möbius band have the same homotopy type.*

13.14. Advanced exercise. *Show that two topological spaces have the same homotopy type if and only if they are homeomorphic to deformation retracts of one common space.*

To gain better intuition for deformation retracts, check the pictures on Ken Baker's webpage about Bing's house [3]. It is a deformation retract of the cube and therefore a contractible space, although it is not easy to construct. The dunce hat is another example of this kind.

Chapter 14

Fundamental group

Let $\mathcal{X}$ be a topological space with a chosen basepoint $x_0 \in \mathcal{X}$. Fix a point $s_0 \in \mathbb{S}^1$, and consider all continuous maps $\mathbb{S}^1 \to \mathcal{X}$ that send s_0 to x_0. Denote by $\pi_1(\mathcal{X}, x_0)$ the set of homotopy classes (rel $\{s_0\}$) of such maps.

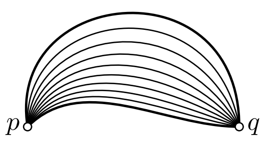
In this chapter we will equip $\pi_1(\mathcal{X}, x_0)$ with multiplication and show that it makes this set into a group. This will justify calling $\pi_1(\mathcal{X}, x_0)$ the *fundamental group* of $\mathcal{X}$ with base point x_0.

The fundamental group is the first meaningful construction in algebraic topology, and the rest of the course will be largely devoted to it.

A Homotopy of paths

Recall that a path is a continuous map defined on the unit interval $[0, 1]$. The term homotopy of paths will be used as a shortcut for *homotopy of paths relative to the ends*; using the notation of 13C, we have the source space $[0, 1]$ with a two-point subset $A = \{0, 1\}$. In the rare case when we need to talk about general homotopy of paths (allowing the endpoints to move), we say free homotopy of paths.

More formally, a homotopy of paths is defined by a continuous map $H\colon [0, 1] \times [0, 1] \to \to \mathcal{X}$ such that $H(0, \tau) = p$ and $H(1, \tau) = q$ for some points p, q in the target space $\mathcal{X}$.

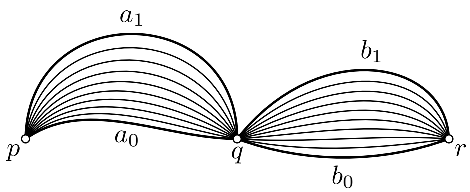


As usual, it is convenient to think that H defines a one-parameter family of paths from p to q; say $a_\tau\colon [0, 1] \to \mathcal{X}$, where $a_\tau(t) := H(t, \tau)$. Two paths $b, c\colon [0, 1] \to \mathcal{X}$ are called homotopic (briefly $b \sim c$) if there is a homotopy $a_\tau\colon[0,1]\to\mathcal{X}$ such that $b=a_0$ and $c=a_1$. Recall that $\sim$ is an equivalence relation. Therefore, we can talk about the equivalence class of a path a, which will be called its homotopy class and denoted by $[a]$.

B Technical claims

14.1. Claim. *Suppose a_0 is a path from p to q, and b_0 is a path from q to r. Suppose $a_0\sim a_1$ and $b_0\sim b_1$, then*

$$a_0*b_0\sim a_1*b_1.$$



Proof. Choose homotopies a_τ from a_0 to a_1, and b_τ from b_0 to b_1. By 3.13*b*, $c_\tau := a_\tau * b_\tau$ is a homotopy from $c_0=a_0*{}*b_0$ to $c_1=a_1*b_1$. ☐

14.2. Observation. *Let $s_0,s_1\colon[0,1]\to[0,1]$ be two paths such that $s_0(0)=s_1(0)$ and $s_0(1)=s_1(1)$. Then $s_0\sim s_1$. Moreover, for any path a we have $a\circ s_0\sim a\circ s_1$.*

Proof. The linear homotopy

$$s_\tau(t)=(1-\tau){\cdot}s_0(t)+\tau{\cdot}s_1(t)$$

provides a homotopy from s_0 to s_1. The conditions $s_0(0)=s_1(0)$ and $s_0(1)=s_1(1)$ imply that $s_\tau(0)$ and $s_\tau(1)$ do not depend on τ; that is, s_τ is a homotopy relative to the endpoints.

To prove the last statement, observe that $a\circ s_\tau$ is a homotopy of paths from $a\circ s_0$ to $a\circ s_1$. ☐

14.3. Claim. *Let $a,b,c\colon[0,1]\to\mathcal{X}$ be paths such that $a(0)=p$, $a(1)=b(0)=q$ and $b(1)=c(0)=r$. Then*

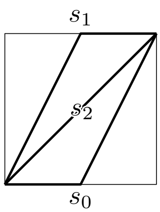
(a) $e_p*a\sim a*e_q\sim a$;

(b) $a*\bar a\sim e_p$ *and* $\bar a*a\sim e_q$;

(c) $(a*b)*c\sim a*(b*c)$.

Proof. Each claim is proved by applying 14.2; each time we rewrite the sides of the equation as $d\circ s_0$ and $d\circ s_1$ for an appropriate path $d\colon[0,1]\to\mathcal{X}$ and a pair of paths $s_0,s_1\colon[0,1]\to[0,1]$ that share the ends.

(a). Note that $e_p * a = a \circ s_0$ and $a * e_q = a \circ s_1$, where



$$s_0(t) := \begin{cases} 0 & \text{if } t \leqslant \frac{1}{2}, \\ 2{\cdot}t - 1 & \text{if } t \geqslant \frac{1}{2}, \end{cases}$$

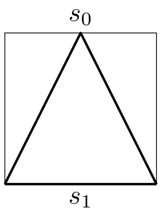
$$s_1(t) := \begin{cases} 2{\cdot}t & \text{if } t \leqslant \frac{1}{2}, \\ 1 & \text{if } t \geqslant \frac{1}{2}. \end{cases}$$

Furthermore, $a = a \circ s_2$ where $s_2(t) := t$, and

$$s_0(0) = s_1(0) = s_2(0) = 0, \qquad s_0(1) = s_1(1) = s_2(1) = 1.$$

It remains to apply 14.2.

(b). Note that $a * \bar{a} = a \circ s_0$ and $e_p = a \circ s_1$, where



$$s_0(t) := \begin{cases} 2{\cdot}t & \text{if } t \leqslant \frac{1}{2}, \\ 2 - 2{\cdot}t & \text{if } t \geqslant \frac{1}{2}, \end{cases} \quad \text{and} \quad s_1(t) := 0$$

Since $s_0(0) = s_1(0) = s_0(1) = s_1(1) = 0$, 14.2 implies $a * \bar{a} \sim e_p$.

By the same argument, we get $\bar{a} * a \sim e_q$.

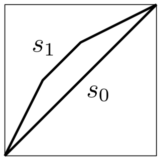
(c). Spelling the definition of product of paths, we get

$$((a * b) * c)(t) := \begin{cases} a(4{\cdot}t) & \text{if } 0 \leqslant t \leqslant \frac{1}{4}, \\ b(4{\cdot}t - 1) & \text{if } \frac{1}{4} \leqslant t \leqslant \frac{1}{2}, \\ c(2{\cdot}t - 1) & \text{if } \frac{1}{2} \leqslant t \leqslant 1, \end{cases}$$

and

$$(a * (b * c))(t) := \begin{cases} a(2{\cdot}t) & \text{if } 0 \leqslant t \leqslant \frac{1}{2}, \\ b(4{\cdot}t - 2) & \text{if } \frac{1}{2} \leqslant t \leqslant \frac{3}{4}, \\ c(4{\cdot}t - 3) & \text{if } \frac{3}{4} \leqslant t \leqslant 1. \end{cases}$$

Therefore, $((a * b) * c)(t) = (a * (b * c)) \circ s_1(t)$, where



$$s_1(t) := \begin{cases} 2{\cdot}t & \text{if } 0 \leqslant t \leqslant \frac{1}{4}, \\ t + \frac{1}{4} & \text{if } \frac{1}{4} \leqslant t \leqslant \frac{1}{2}, \\ \frac{1}{2}{\cdot}t + \frac{1}{2} & \text{if } \frac{1}{2} \leqslant t \leqslant 1. \end{cases}$$

The path $(a * b) * c$ can be written as $((a * b) * c) \circ s_0$ where $s_0(t) := t$. Furthermore, both functions $s_0, s_1 \colon [0,1] \to [0,1]$ are continuous, $s_0(0) = s_1(0) = 0$, and $s_0(1) = s_1(1) = 1$. It remains to apply 14.2. $\square$

14.4. Corollary. *Let a and b be paths from p to q. Show that $a \sim b$ if and only if $a * \bar{b} \sim e_p$.*

Proof. Suppose $a \sim b$. Then 14.1 and 14.2 imply that

$$a * \bar{b} \sim b * \bar{b} \sim e_p.$$

Now suppose $a * \bar{b} \sim e_p$. Applying 14.1 and 14.2 again, we get

$$a \sim a * e_q \sim a * (\bar{b} * b) \sim (a * \bar{b}) * b \sim e_p * b \sim b.$$

□

14.5. Exercise. *Let a, b, and c be paths in a Hausdorff space, and let $a(0) = p$.*

(a) Suppose that $(a*b)*c = a*(b*c)$ and both sides of the equation are defined. Show that $a = b = c = e_p$.

(b) Suppose that $e_p * a = a$. Show that $a = e_p$.

C Fundamental group

Let $\mathcal{X}$ be a topological space. A path $a\colon [0,1] \to \mathcal{X}$ is called a loop with base point $x_0 \in \mathcal{X}$ if $a(0) = a(1) = x_0$. Since $\mathbb{S}^1 \cong [0,1]/\{0,1\}$ (see 9.17a), any loop defines a map $\mathbb{S}^1 \to \mathcal{X}$.

Note that if a and b are loops based at x_0, then their products are defined; moreover, $a*b$, $b*a$, $\bar{a}$, $\bar{b}$ are also loops based at x_0; see 12A.

Recall that homotopy of paths (in particular, loops) does not move their ends and $[a]$ stands for the homotopy class of a path (or loop) a. The multiplication of homotopy classes of loops based at x_0 is defined by

$$[a]\cdot[b] = [a*b];$$

that is, *the product of homotopy classes of loops a and b is the homotopy class of the product $a * b$.*

Observe that the product is well-defined; that is, if $[a_0] = [a_1]$ and $[b_0] = [b_1]$, then $[a_0 * b_0] = [a_1 * b_1]$. In other words, if $a_0 \sim a_1$ and $b_0 \sim b_1$, then $a_0 * b_0 \sim a_1 * b_1$. The latter is stated in Claim 14.1.

Denote by $\pi_1(\mathcal{X}, x_0)$ the set of all homotopy classes of loops based at x_0.

14.6. Theorem. *The set $\pi_1(\mathcal{X}, x_0)$ with the introduced multiplication is a group.*

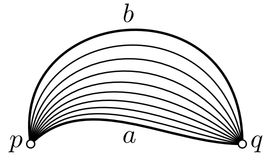
The group $\pi_1(\mathcal{X}, x_0)$ is called the fundamental group of $\mathcal{X}$ with base point x_0.

Proof. Recall that e_{x_0} denotes the constant loop at x_0 in $\mathcal{X}$; that is, $e_{x_0}(t) = x_0$ for any t. We will show that the homotopy class $[e_{x_0}]$ is the neutral element of $\pi_1(\mathcal{X}, x_0)$ and $[\bar{a}] = [a]^{-1}$, where $\bar{a}$ denotes the time-reversed path of a.

By 14.3,
(a) $a * e_{x_0} \sim e_{x_0} * a \sim a$,
(b) $a * \bar{a} \sim \bar{a} * a \sim e_{x_0}$,
(c) $(a * b) * c \sim a * (b * c)$
for any loops a, b, and c based at x_0. And these interrelations can be rewritten as
(a) $[a]{\cdot}[e_{x_0}] = [e_{x_0}]{\cdot}[a] = [a]$,
(b) $[a]{\cdot}[\bar{a}] = [\bar{a}]{\cdot}[a] = [e_{x_0}]$,
(c) $([a]{\cdot}[b]){\cdot}[c] = [a]{\cdot}([b]{\cdot}[c])$,
and they imply the conditions in the definition of a group with neutral element $[e_{x_0}]$ and $[a]^{-1} := [\bar{a}]$. □

14.7. Exercise. *Let $\mathcal{G}$ be a topological group; that is, a group equipped with a topology such that the maps $\mathcal{G} \times \mathcal{G} \to \mathcal{G}$ and $\mathcal{G} \to \mathcal{G}$ defined by $(g, h) \mapsto g{\cdot}h$ and $g \mapsto g^{-1}$ are continuous. Show that the fundamental group $\pi_1(\mathcal{G}, 1)$ is abelian; in other words, $a_1 * a_2 \sim a_2 * a_1$ for any two loops a_1 and a_2 with base point $1 \in \mathcal{G}$.*

D Simply-connected spaces



A nonempty topological space $\mathcal{X}$ is called simply-connected if for any pair of points $p, q \in \mathcal{X}$ there is a unique homotopy class of paths from p to q. In other words, there is a path from p to q and any two such paths are homotopic.

Note that any simply-connected space is path-connected.

14.8. Exercise. *Show that a retract of a simply-connected space is simply-connected.*

Recall that a group is called trivial if it contains only one element, which is necessarily the neutral element. If the fundamental group $\pi_1(\mathcal{X}, x_0)$ is trivial, it is common to write $\pi_1(\mathcal{X}, x_0) = 0$ altho this makes no sense — in general the group $\pi_1(\mathcal{X}, x_0)$ is not commutative, and so it would be more reasonable to write $\pi_1(\mathcal{X}, x_0) = \{1\}$, meaning that 1 is the only element of $\pi_1(\mathcal{X}, x_0)$.

14.9. Proposition. *A path-connected space $\mathcal{X}$ is simply-connected if and only if $\pi_1(\mathcal{X}, x_0) = 0$ for some (and therefore any) base point $x_0 \in \mathcal{X}$.*

Since $(a*b)*c \neq a*(b*c)$, one has to fully parenthesize products of paths. However, I will omit parentheses when any parenthesization does the job. (Since $(a*b)*c \sim a*(b*c)$ this is the case if we consider loops up to homotopy.) On a more formal basis, one can assume that the product is taken in the usual order; that is,

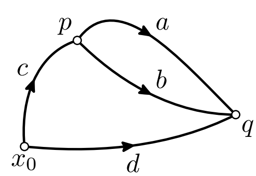
$$a*b*c*d := ((a*b)*c)*d.$$

Proof. Choose a base point $x_0 \in \mathcal{X}$. If $\mathcal{X}$ is simply-connected, then any loop based at x_0 is homotopic to e_{x_0}. Hence the only-if part follows.

Now suppose $\pi_1(\mathcal{X}, x_0) = 0$. Choose points $p, q \in \mathcal{X}$. Let a and b be paths from p to q and let c and d be paths from x_0 to p and to q, respectively; all these paths exist since $\mathcal{X}$ is path-connected.

Since $\pi_1(\mathcal{X}, x_0) = 0$, the loops $c*a*\bar{d}$, e_{x_0}, and $c*b*\bar{d}$ are homotopic; that is,

$$c*a*\bar{d} \sim e_{x_0} \sim c*b*\bar{d}.$$



Applying 14.3, we get

$$\begin{aligned} a &\sim \bar{c}*c*a*\bar{d}*d \sim \\ &\sim \bar{c}*c*b*\bar{d}*d \sim \\ &\sim b \end{aligned}$$

— hence the result. □

14.10. Exercise! *Show that a path-connected space $\mathcal{X}$ is simply-connected if and only if every continuous map $f\colon \mathbb{S}^1 \to \mathcal{X}$ is null-homotopic.*

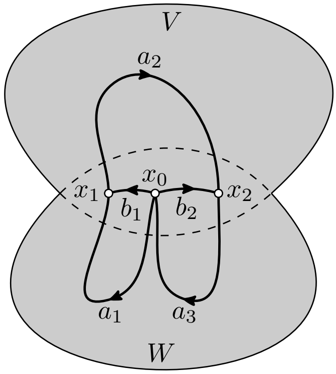
Conclude that any contractible space is simply-connected.

14.11. Proposition. *Suppose that V and W are open simply-connected subsets of a topological space $\mathcal{X}$ such that $\mathcal{X} = V \cup W$, and the set $V \cap W$ is path-connected. Then $\mathcal{X}$ is simply-connected.*

This statement is a special case of van Kampen's theorem. Roughly speaking, the general theorem allows one to recover the fundamental group of $\mathcal{X} = V \cup W$ from the fundamental groups of the open sets V, W, and $V \cap W$, together with the way $V \cap W$ sits inside V and W. See also 19.12.

Recall that if the product of paths is not parenthesized, then it is taken in the usual order; that is, $a*b*c*d := ((a*b)*c)*d$.

Proof. Let a be a loop in $\mathcal{X}$ based at $x_0 \in V \cap W$. Note that we can choose a sequence $0 = t_0 < t_1 < \ldots < t_n = 1$ such that each image $a([t_{i-1}, t_i])$ lies in V or W, and each subdivision point $a(t_i)$ lies in $V \cap W$.



Indeed, the two open sets $a^{-1}(V)$ and $a^{-1}(W)$ cover $[0,1]$; let $\varepsilon > 0$ be a Lebesgue number of this cover. So if $|t_i - t_{i-1}| < \varepsilon$, then $a([t_{i-1}, t_i]) \subset V$ or $a([t_{i-1}, t_i]) \subset W$ for each i. If $a(t_i) \notin V \cap W$ for some i, then it can be removed from the sequence keeping the property; after several such removals we get the needed subdivision.

Reparametrizing $[t_{i-1}, t_i]$ by $[0,1]$, we get paths $a_1, \dots, a_n$ such that

$$a \sim a_1 * \cdots * a_n$$

and each a_i lies completely in V or W and has endpoints in $V \cap W$. Let us connect x_0 to every point $x_i = a_i(1)$ for $i < n$ by a path b_i in $V \cap W$. By 14.3,

$$a_1 * \ldots * a_n \sim (a_1 * \bar{b}_1) * (b_1 * a_2 * \bar{b}_2) * \ldots * (b_{n-1} * a_n).$$

Observe that each loop $a_1 * \bar{b}_1$, $b_1 * a_2 * \bar{b}_2, \dots, b_{n-1} * a_n$ lies entirely in V or W. Since V and W are simply-connected,

$$a_1 * \bar{b}_1 \sim e_{x_0}, \quad b_1 * a_2 * \bar{b}_2 \sim e_{x_0}, \quad \dots, \quad b_{n-1} * a_n \sim e_{x_0}.$$

Hence $a_1 * \ldots * a_n \sim e_{x_0}$, and the statement follows. □

14.12. Exercise. *Show that (a) $\mathbb{S}^2$ and (b) the union of a sphere and a disc bounded by the equator of the sphere are simply-connected.*

Chapter 15

Topology meets algebra

In this chapter we show that the fundamental-group construction respects the maps between topological spaces[1] and behaves predictably when one changes the base point of the space. A bit later, we will calculate the fundamental group of the circle — this will provide the first example of a space with a nontrivial fundamental group. After that, the results of this chapter will lead to applications.

A A group theory recap

Recall that groups were defined in 6G.

We will denote by $\cdot$ the group operation and by 1 the neutral element of the group; we also may use 1_G to emphasize that this is the neutral element of the group G.

Homomorphisms. Let G and H be groups. A map $\varphi\colon G \to H$ is called a homomorphism if

$$\varphi(g\cdot h) = \varphi(g)\cdot\varphi(h)$$

for any $g, h \in G$.

15.1. Exercise. *Show that any homomorphism maps the neutral element to the neutral element.*

A bijective homomorphism is called an isomorphism; an isomorphism from a group G to itself is called an automorphism of G.

15.2. Exercise. *Let h be an element of a group G. Show that the map $g \mapsto h^{-1}\cdot g\cdot h$ is an automorphism.*

[1] In a more fancy language, it is functorial.

Give an example of a group G and an automorphism of G that cannot be written this way.

The automorphisms $g \mapsto h^{-1}\cdot g\cdot h$ as in the exercise are called inner.

Subgroups, cosets, and index. A subset H of a group G is called a subgroup (briefly $H < G$) if H is a group with the product induced from G; equivalently, if the following conditions hold: $1 \in H$, and if $g, h \in H$, then $g^{-1}, g\cdot h \in H$.

For example, the neutral element forms a one-element subgroup, which is called trivial.

Given $g \in G$, the sets

$$g\cdot H = \{\, g\cdot h \,:\, h \in H \,\} \quad \text{and} \quad H\cdot g = \{\, h\cdot g \,:\, h \in H \,\}$$

are called left and right cosets of H. The number of left cosets is called the index of H in G; the map $g \mapsto g^{-1}$ (which is not a homomorphism in general) sends left cosets to right cosets, so if you count right cosets you will also get the index of H in G.

15.3. Exercise. *Let G be a group. Show that $g \mapsto g^{-1}$ is a homomorphism if and only if G is commutative.*

Kernels, images and normal subgroups. Given a group homomorphism $\varphi\colon G \to H$, its kernel and image are defined by

$$\begin{aligned}\operatorname{Ker}\varphi &:= \varphi^{-1}(\{1_H\}) = \{g \in G \mid \varphi(g) = 1_H\},\\ \operatorname{Im}\varphi &:= \varphi(G) = \{\varphi(g) \in H \mid g \in G\}.\end{aligned}$$

It is easy to see that the kernel of φ is a subgroup of G and the image of φ is a subgroup of H.

15.4. Exercise. *Show that a group homomorphism is injective if and only if it has trivial kernel.*

Any subgroup $K < H$ is an image of some homomorphism to H; in fact the inclusion map $K \hookrightarrow H$ is a homomorphism with image K.

On the other hand, not every subgroup N of G can appear as a kernel of some homomorphism from G; if it can, then the subgroup is called normal, written as $N \triangleleft G$.

If $N = \operatorname{Ker}\varphi$ is a normal subgroup of G, then

$$\varphi(g\cdot n) = \varphi(g)\cdot\varphi(n) = \varphi(g) = \varphi(n)\cdot\varphi(g) = \varphi(n\cdot g)$$

for any $g \in G$ and $n \in N$. It follows that the left cosets of N coincide with its right cosets; that is,

➊ $$g{\cdot}N = N{\cdot}g$$

for any $g \in G$. On the other hand, if $N < G$ satisfies ➊, then N defines an equivalence relation on G such that $g \sim h$ if there is $n \in N$ such that $g = n{\cdot}h$. The equivalence class of an element $g \in G$ is $[g] = g{\cdot}N = N{\cdot}g$. The quotient set $H = G/N := G/\sim$, equipped with the product $[g]{\cdot}[h] := [g{\cdot}h]$, becomes a group; moreover, the quotient map $G \to H$ defined by $g \mapsto [g]$ is a homomorphism with kernel N. Thus ➊ provides an equivalent definition of normal subgroups.

15.5. Exercise. *Consider the group S_3 of all permutations of the set $\{1, 2, 3\}$ (that is, the set of all bijections from $\{1, 2, 3\}$ to itself). Let us use long notation for its 6 elements:*

$$1_{S_3} = [123], \qquad [231], \qquad [312], \qquad [132], \qquad [321], \qquad [213].$$

For example, $[132]$ stands for the permutation that maps $1 \mapsto 1$, $2 \mapsto 3$ and $3 \mapsto 2$.

(a) *List all subgroups of S_3 and find the index of each; determine which of them are normal.*
(b) *List all homomorphisms $S_3 \to S_3$ and find the kernel and the image of each; which of them are (inner) automorphisms?*

B Additive versus multiplicative

The equivalence class of an integer m modulo n will be denoted by $[m]_n$; that is,

$$[m]_n = \{\, m + k{\cdot}n \;:\; k \in \mathbb{Z} \,\};$$

here $\mathbb{Z}$ denotes the set of integers. The set of equivalence classes of integers modulo n will be denoted by $\mathbb{Z}_n$. The operation

$$[m_1]_n + [m_2]_n := [m_1 + m_2]_n$$

defines a commutative group structure on $\mathbb{Z}_n$ with neutral element $[0]_n$; it is a c y c l i c g r o u p of o r d e r[2] n, generated by the element $[1]_n$.

Recall that all cyclic groups of order n are isomorphic, so it is acceptable to denote a cyclic group of order n by $\mathbb{Z}_n$.

We will mostly need two groups:

[2]The order of a group is the number of elements in the group; it is either a positive integer or infinity.

- $\mathbb{Z}$ is the additive group of integers and
- $\mathbb{Z}_2$ is the integers modulo two.

Both of these groups are commutative. For commutative groups, one often uses additive notation; so the group operation is denoted by $+$ and the neutral element is denoted by 0. It works perfectly in $\mathbb{Z}$.

Any group can appear as a fundamental group of a topological space (we will not prove it, but it is not hard to prove). For non-commutative groups, additive notation should not be used. So we will use multiplicative notation (as in the previous section) for general groups, but once we know that the group is commutative, we will switch to the additive notation; in particular, the following changes will occur:

$$1 \rightsquigarrow 0, \quad \alpha\cdot\beta \rightsquigarrow \alpha+\beta, \quad \text{and} \quad \alpha^n \rightsquigarrow n\cdot\alpha$$

— do not be confused by that.

15.6. Exercise. *Describe all homomorphisms $\mathbb{Z} \to S_3$ and find the image group for each.*

C Induced homomorphism

A pointed space, briefly $(\mathcal{X}, x_0)$, is a topological space $\mathcal{X}$ with a choice of base point $x_0 \in \mathcal{X}$; so the fundamental group is a construction that produces a group $\pi_1(\mathcal{X}, x_0)$ from a pointed space $(\mathcal{X}, x_0)$.

A pointed map (briefly $f\colon (\mathcal{X}, x_0) \to (\mathcal{Y}, y_0)$) is a map $f\colon \mathcal{X} \to \mathcal{Y}$ with a choice of base points $x_0 \in \mathcal{X}$ and $y_0 \in \mathcal{Y}$ such that $y_0 = f(x_0)$.

Let $f\colon (\mathcal{X}, x_0) \to (\mathcal{Y}, y_0)$ be a continuous pointed map. Suppose that a_0 and a_1 are loops based at x_0 in $\mathcal{X}$. Then $(f \circ a_0)$ and $(f \circ a_1)$ are loops based at y_0 in $\mathcal{Y}$. Moreover,

$$f \circ (a_0 * a_1) = (f \circ a_0) * (f \circ a_1).$$

Indeed, applying the definition of the product of paths and composition of maps to $f \circ (a_0 * a_1)(t)$ and $(f \circ a_0) * (f \circ a_1)(t)$ we get exactly the same expression:

$$\begin{cases} (f \circ a_0)(2\cdot t) & \text{if } t \leqslant \frac{1}{2}, \\ (f \circ a_1)(2\cdot t - 1) & \text{if } t \geqslant \frac{1}{2}. \end{cases}$$

Further, note that

$$a_0 \sim a_1 \quad \Rightarrow \quad f \circ a_0 \sim f \circ a_1.$$

Indeed, if a_τ is a homotopy from a_0 to a_1, then $f \circ a_\tau$ is a homotopy from $f \circ a_0$ to $f \circ a_1$.

It follows that the map $f\colon (\mathcal{X}, x_0) \to (\mathcal{Y}, y_0)$ induces a homomorphism

$$f_*\colon \pi_1(\mathcal{X}, x_0) \to \pi_1(\mathcal{Y}, y_0)$$

that is defined by $f_*\colon [a] \mapsto [f \circ a]$. (The homomorphism f_* maps the homotopy class of the loop a to the homotopy class of the loop $f \circ a$.)

Evidently, the identity map of $(\mathcal{X}, x_0)$ induces the identity homomorphism on $\pi_1(\mathcal{X}, x_0)$. Furthermore, for any continuous pointed maps

$$(\mathcal{X}, x_0) \xrightarrow{f} (\mathcal{Y}, y_0) \xrightarrow{g} (\mathcal{Z}, z_0),$$

we have $g_* \circ f_* = (g \circ f)_*$. These two observations will be extremely useful.

15.7. Exercise! *Let A be a retract of $\mathcal{X}$ with retraction $r\colon \mathcal{X} \to$ $\to A$. Choose $x_0 \in A$; let $i\colon A \to \mathcal{X}$ be the inclusion map. Show the following.*

(a) $r_*\colon \pi_1(\mathcal{X}, x_0) \to \pi_1(A, x_0)$ is a surjective homomorphism.
(b) $i_*\colon \pi_1(A, x_0) \to \pi_1(\mathcal{X}, x_0)$ is an injective homomorphism.
(c) If A is a weak deformation retract, then r_* and i_* are isomorphisms.

D About notation f_*

The induced homomorphism $f_*\colon \pi_1(\mathcal{X}, x_0) \to \pi_1(\mathcal{Y}, y_0)$ has the same direction as the original map $f\colon (\mathcal{X}, x_0) \to (\mathcal{Y}, y_0)$; for that reason it is also called pushforward. The symbol f_* (read "f-star") can be used for *any* induced map of this type; for example, one can write f_* for the induced map from the set of path-connected components of $\mathcal{X}$ to the set of path-connected components of $\mathcal{Y}$. The associated constructions (fundamental group or set of path-connected components) are called covariant.

If you see f_*, then it is induced by f and goes in the same direction; you have to guess from the context the source and target of f_*.

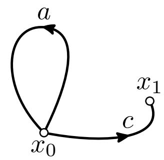
For some other constructions, the induced map might go in the opposite direction. In this case, the induced map is denoted by f^*, and is called pullback. The associated constructions are called contravariant.

As an example of a contravariant construction, consider the so-called first cohomotopy group $\pi^1(\mathcal{X}, x_0)$, which is the set of homotopy classes of pointed continuous maps $(\mathcal{X}, x_0) \to (\mathbb{S}^1, s_0)$. (The set $\pi^1(\mathcal{X}, x_0)$ has a natural structure of a commutative group, but this is not important for us.) Then any pointed continuous map $f\colon(\mathcal{X}, x_0) \to (\mathcal{Y}, y_0)$ has a pullback $f^*\colon \pi^1(\mathcal{Y}, y_0) \to \pi^1(\mathcal{X}, x_0)$. Indeed, for a map $h\colon(\mathcal{Y}, y_0) \to (\mathbb{S}^1, s_0)$, the composition $h\circ f$ provides a good choice of map $(\mathcal{X}, x_0) \to (\mathbb{S}^1, s_0)$, so we can define $f^*[h] := [h\circ f]$. On the other hand, there is no good choice of map $(\mathcal{Y}, y_0) \to (\mathbb{S}^1, s_0)$ for a given map $(\mathcal{X}, x_0) \to (\mathbb{S}^1, s_0)$.

E Transport of base point

The following theorem provides a more precise version of 14.9.

15.8. Theorem. *Let c be a path from x_0 to x_1 in a topological space $\mathcal{X}$. Then $[a] \mapsto [\bar c * a * c]$ defines an isomorphism of fundamental groups $\varphi_c\colon \pi_1(\mathcal{X}, x_0) \to \pi_1(\mathcal{X}, x_1)$.*



Notice that if c is a loop based at x_0, then φ_c is an inner automorphism of $\pi_1(\mathcal{X}, x_0)$.

Recall that we assume that the product of paths is taken in the usual order; that is, $a * b * c * d := ((a * b) * c) * d$.

Proof. Suppose a_τ is a homotopy of loops at x_0. Then $\bar c * a_\tau * c$ is a homotopy of loops at x_1. It follows that the map

❷ $$\varphi_c\colon [a] \mapsto [\bar c * a * c]$$

is defined. Indeed, if $a_0 \sim a_1$, then $\bar c * a_0 * c \sim \bar c * a_1 * c$, which means that $[\bar c * a_0 * c] = [\bar c * a_1 * c]$; so the right-hand side in ❷ does not depend on the choice of loop in the homotopy class $[a]$.

Further, if a and b are loops based at x_0, then 14.3 implies that

$$\begin{aligned}(\bar c * a * c) * (\bar c * b * c) &\sim \bar c * a * (c * \bar c) * b * c \sim\\ &\sim \bar c * (a * e_{x_0}) * b * c \sim\\ &\sim \bar c * (a * b) * c.\end{aligned}$$

In other words,

$$\varphi_c([a]\cdot[b]) = \varphi_c[a]\cdot\varphi_c[b] \quad \text{for any} \quad [a],[b] \in \pi_1(\mathcal{X}, x_0);$$

that is, $\varphi_c\colon \pi_1(\mathcal{X}, x_0) \to \pi_1(\mathcal{X}, x_1)$ is a homomorphism.

The same argument shows that $\varphi_{\bar c}\colon \pi_1(\mathcal{X}, x_1) \to \pi_1(\mathcal{X}, x_0)$ defined by

$$\varphi_{\bar c}\colon [d] \mapsto [c * d * \bar c]$$

is a homomorphism.

Finally, note that

$$c * (\bar{c} * a * c) * \bar{c} \sim (c * \bar{c}) * a * (c * \bar{c}) \sim e_{x_0} * a * e_{x_0} \sim a$$

for any loop a based at x_0. Therefore, $\varphi_{\bar{c}}$ is a left inverse of φ_c. The same argument shows that $\varphi_{\bar{c}}$ is also a right inverse of φ_c. It follows that φ_c is an isomorphism. □

According to the theorem, the fundamental group[3] of a path-connected space does not depend on its base point. Therefore, for a path-connected space $\mathcal{X}$ we do not need to specify the base point of its fundamental group. So, if we think about the group up to isomorphism, then it is safe to write $\pi_1\mathcal{X}$ instead of $\pi_1(\mathcal{X}, x_0)$.

15.9. Exercise.

(a) *Let $f_\tau\colon \mathcal{X} \to \mathcal{Y}$ be a homotopy. Consider the path defined by $c(\tau) = f_\tau(x_0)$. Show that*

$$\varphi_c \circ f_{0*} = f_{1*};$$

here $f_{i*}\colon \pi_1(\mathcal{X}, x_0) \to \pi_1(\mathcal{Y}, f_i(x_0))$ denotes the homomorphism induced by f_i.

(b) *Conclude that if two path-connected topological spaces $\mathcal{X}$ and $\mathcal{Y}$ have the same homotopy type, then their fundamental groups are isomorphic.*

F Product space

Let G and H be two groups. Consider the set of all pairs (g, h) with $g \in G$ and $h \in H$; it is straightforward to check that the operation

$$(g_1, h_1)\cdot(g_2, h_2) := (g_1\cdot g_2, h_1\cdot h_2)$$

defines a group structure on the set of such pairs. The resulting group is called the product of G and H and denoted by $G \times H$.

15.10. Exercise. *Let $\mathcal{X}$ and $\mathcal{Y}$ be two path-connected topological spaces. Choose points $x_0 \in \mathcal{X}$ and $y_0 \in \mathcal{Y}$. Consider the projections and their induced homomorphisms:*

$$\begin{aligned} f&\colon \mathcal{X} \times \mathcal{Y} \to \mathcal{X}, & f_*&\colon \pi_1(\mathcal{X} \times \mathcal{Y}, (x_0, y_0)) \to \pi_1(\mathcal{X}, x_0),\\ g&\colon \mathcal{X} \times \mathcal{Y} \to \mathcal{Y}, & g_*&\colon \pi_1(\mathcal{X} \times \mathcal{Y}, (x_0, y_0)) \to \pi_1(\mathcal{Y}, y_0). \end{aligned}$$

[3] more precisely, its *isomorphism class*

Define $\varphi\colon \pi_1(\mathcal{X}\times\mathcal{Y},(x_0,y_0)) \to \pi_1(\mathcal{X},x_0)\times\pi_1(\mathcal{Y},y_0)$ *by*

$$\varphi\colon \alpha \mapsto (f_*(\alpha), g_*(\alpha))$$

for any $\alpha \in \pi_1(\mathcal{X}\times\mathcal{Y},(x_0,y_0))$*. Prove the following.*

(a) φ *is a homomorphism.*

(b) φ *is a monomorphism, which is injective homomorphism; that is, if* $\varphi(\alpha)=\varphi(\beta)$ *for* $\alpha,\beta \in \pi_1(\mathcal{X}\times\mathcal{Y},(x_0,y_0))$*, then* $\alpha=\beta$*.*

(c) φ *is an epimorphism, which is surjective homomorphism; that is, for any* $\gamma \in \pi_1(\mathcal{X},x_0)\times\pi_1(\mathcal{Y},y_0)$ *there is* $\alpha \in \pi_1(\mathcal{X}\times\mathcal{Y},(x_0,y_0))$ *such that* $\varphi(\alpha)=\gamma$*.*

Conclude that $\pi_1(\mathcal{X}\times\mathcal{Y})$ *is isomorphic to* $\pi_1(\mathcal{X})\times\pi_1(\mathcal{Y})$*.*

15.11. Advanced exercise. *Let* $\mathcal{X}$ *be a path-connected topological space. Consider the quotient space* $\mathcal{Y}=(\mathcal{X}\times\mathcal{X})/\sim$ *where* $\sim$ *is the minimal equivalence relation such that* $(x,y)\sim(y,x)$*. Show that the fundamental group of* $\mathcal{Y}$ *is commutative.*

Chapter 16

Fundamental group of a circle

This chapter computes the fundamental group of the circle, which is the first example of a space with a nontrivial fundamental group. In the proof one has to show the nonexistence of homotopies between certain loops; this part requires new ideas.

A The statement

16.1. Theorem. *The fundamental group of the circle $\mathbb{S}^1$ is isomorphic to the additive group of integers $\mathbb{Z}$.*

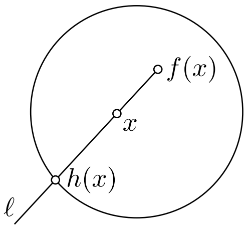
The proof is done in several steps. First we introduce the degree of a loop $a\colon [0,1] \to \mathbb{S}^1$, briefly $\deg a$, as the number of times it goes around $\mathbb{S}^1$ counterclockwise (each clockwise turn is counted with a minus sign). After giving a formal definition of degree, we prove that it depends only on the homotopy class of the loop and then show that $\deg\colon \pi_1\,\mathbb{S}^1 \to \mathbb{Z}$ is an isomorphism.

Let us give one application of the theorem before proving it.

16.2. Brouwer fixed point theorem in the plane. *Any continuous map $f\colon \mathbb{D}^2 \to \mathbb{D}^2$ has a fixed point.*

Proof. Arguing by contradiction, assume $x \neq f(x)$ for all $x \in \mathbb{D}^2$. Given $x \in \mathbb{D}^2$, consider the half-line ℓ from $f(x)$ to x.[1]

[1] Remember that ℓ goes backwards, from $f(x)$ to x, not the other way around.



The half-line ℓ intersects $\mathbb{S}^1 = \partial\mathbb{D}^2$ at a single point; denote it by $h(x)$. Note that the map $h\colon \mathbb{D}^2 \to \mathbb{S}^1$ is continuous and $h(x) = x$ if $x \in \mathbb{S}^1$; that is, h defines a retraction $\mathbb{D}^2 \to \to \mathbb{S}^1$.

Note that both spaces $\mathbb{S}^1$ and $\mathbb{D}^2$ are path-connected. By 16.1, $\mathbb{S}^1$ is not simply-connected. By 15.7a, the same holds for $\mathbb{D}^2$. The latter contradicts 14.10. $\square$

16.3. Exercise. *Apply the theorem to show the following.*

(a) $\mathbb{R}^3 \not\simeq \mathbb{R}^2$.

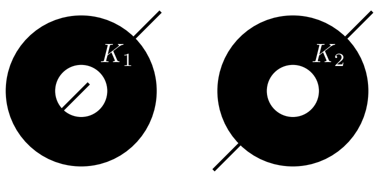
(b) $\mathbb{D}^2 \not\simeq \mathbb{S}^2$.

(c) *Given $x \in \mathbb{D}^2$, the complement $\mathbb{D}^2 \setminus \{x\}$ is simply-connected if and only if $x \in \mathbb{S}^1 = \partial\mathbb{D}^2$. Conclude that any homeomorphism $\mathbb{D}^2 \to \mathbb{D}^2$ maps $\partial\mathbb{D}^2$ to itself.*

(d) *Show that the cylinder $\mathbb{S}^1 \times [0,1]$ is not homeomorphic to the Möbius band (defined in 13H).*

(e) *The complement of any finite nonempty set $F \subset \mathbb{R}^2$ is not contractible.*

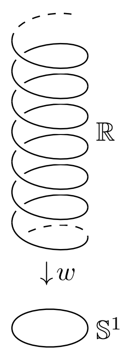
16.4. Advanced exercise. *Consider two plane sets K_1 and K_2, shown in the picture.*



Each set is a closed annulus with two attached line segments. In K_1 one segment is attached from the inside and the other from the outside; in K_2 both segments are attached from the outside.

(a) *Show that $K_1 \not\simeq K_2$.*

(b) *Show that $K_1 \times [0,1] \simeq K_2 \times [0,1]$.*



B Wrapping map

Let us identify $\mathbb{R}^2$ with the complex plane $\mathbb{C}$; namely, we will encode a point $(x,y) \in \mathbb{R}^2$ by the complex number $z = x + i{\cdot}y$; here i denotes the imaginary unit. In particular,

$$\mathbb{S}^1 = \left\{\, z = x + i{\cdot}y \in \mathbb{C} \;:\; |z| := \sqrt{x^2+y^2} = 1 \,\right\}.$$

Consider the wrapping map $w\colon \mathbb{R} \to \mathbb{S}^1 \subset \mathbb{C}$ defined by

$$w\colon x \mapsto \exp(2\cdot\pi\cdot i\cdot x) = \cos(2\cdot\pi\cdot x) + i\cdot\sin(2\cdot\pi\cdot x).$$

The last equality follows from Euler's formula $\exp(i\cdot x) = \cos x + {} + i\cdot\sin x$, which can be taken as a definition of $\exp(i\cdot x)$.

Let $a\colon \mathcal{X} \to \mathbb{S}^1$ be a continuous map. A continuous map $\tilde a\colon \mathcal{X} \to \mathbb{R}$ will be called a l i f t of a if $a = w\circ\tilde a$. In other words, if the diagram c o m m u t e s, which means that composing the maps following directed paths in the diagram with the same initial and terminal lead to the same result.

$$\begin{tikzcd} & \mathbb{R} \arrow[d, "w"] \\ \mathcal{X} \arrow[ur, "\tilde a"] \arrow[r, "a"'] & \mathbb{S}^1 \end{tikzcd}$$

16.5. Proposition. *Let $a_0\colon [0,1] \to \mathbb{S}^1$ be a path and let $x_0 \in \mathbb{R}$ be a point such that $w(x_0) = a_0(0)$. Then there is a unique lift $\tilde a_0$ of a_0 such that $\tilde a_0(0) = x_0$.*

Moreover, given a homotopy of paths a_τ from a_0 to a_1 there is a unique lifted homotopy of paths $\tilde a_\tau$ such that $\tilde a_0(0) = x_0$. In particular, if $a_0 \sim a_1$ then $\tilde a_0(1) = \tilde a_1(1)$.

16.6. Lemma. *Let $W = \mathbb{S}^1 \setminus \{s_0\}$ for some $s_0 = w(x_0) \in \mathbb{S}^1$. Then $w^{-1}(W)$ is a disjoint union of open intervals $V_n = (x_0 + n, x_0 + n + 1)$ for $n \in \mathbb{Z}$ and w defines a homeomorphism $V_n \to W$ for each n.*

In particular, given a connected space $\mathcal{X}$, a continuous map $a\colon \mathcal{X} \to {} \to W$, and points $p \in \mathcal{X}$ and $q \in \mathbb{R}$ such that $w(q) = a(p)$, there is a unique lift $\tilde a\colon \mathcal{X} \to \mathbb{R}$ of a such that $\tilde a(p) = q$.

Proof. Observe that the restriction $w_n = w|_{V_n}$ is a homeomorphism $V_n \to W$. Indeed, w_n is a continuous bijection. Furthermore, by 9.15 $w|_{\bar V_n}$ is a closed map. Therefore, w_n is a continuous bijection $V_n \to W$ that sends closed sets in the induced topology of V_n to closed sets in the induced topology of W; hence w_n is a homeomorphism.

Note that the intervals V_n are connected components of $w^{-1}(W)$. In particular, $q \in V_n$ for some n. Since $\mathcal{X}$ is connected, $\tilde a(\mathcal{X}) \subset {} \subset V_n$ for any lift $\tilde a$ of a such that $\tilde a(p) = q$. Since $w_n = w|_{V_n}$ is a homeomorphism, we have $\tilde a = w_n^{-1} \circ a$, which proves both existence and uniqueness. □

Proof of 16.5. Consider two open subsets $W_1 = \mathbb{S}^1 \setminus \{1\}$ and $W_2 = {} = \mathbb{S}^1 \setminus \{-1\}$. Note that $\mathbb{S}^1 = W_1 \cup W_2$. Therefore,

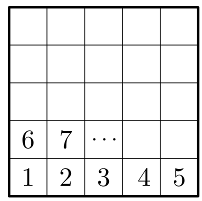
$$[0,1] = a_0^{-1}(W_1) \cup a_0^{-1}(W_2).$$

The open sets $a_0^{-1}(W_1)$ and $a_0^{-1}(W_2)$ cover $[0,1]$; let $\varepsilon > 0$ be a Lebesgue number (see 8.1). Consider a partition $0 = t_0 < t_1 <$

$\dots < t_n = 1$ of $[0,1]$ into equal intervals shorter than ε. Note that $a_0([t_i, t_{i+1}]) \subset W_1$ or $a_0([t_i, t_{i+1}]) \subset W_2$ for each i.

By 16.6, we can lift $a_0|_{[t_0,t_1]}$ to $\tilde{a}_0\colon [t_0, t_1] \to \mathbb{R}$ such that $\tilde{a}_0(0) = x_0$; let $x_1 = \tilde{a}_0(t_1)$. By the same argument, we can extend $\tilde{a}_0$ to $[t_0, t_2]$ via a lift of $a_0|_{[t_1,t_2]}$ such that $\tilde{a}_0(t_1) = x_1$. The continuity of $\tilde{a}_0$ follows from 3.13*b*. Repeating this argument n times gives us a lift of a_0, and it is unique by construction.

The proof of the last statement is similar. Choose a homotopy $H\colon [0,1] \times [0,1] \to \mathbb{S}^1$ from a_0 to a_1. Applying 8.1 again, we can subdivide $[0,1] \times [0,1]$ into small squares by horizontal and vertical lines so that H maps each small square into W_1 or into W_2. Then we can build the lift $\tilde{H}$ square by square, from the bottom up in the order shown in the picture.



When we extend $\tilde{H}$ over a small square $\square$, the value of $\tilde{H}$ is already defined at its lower left corner, say v. So we can apply 16.6. The part of $\square$ on which $\tilde{H}$ was already defined consists of one or two sides adjacent to v. In both cases, this set is connected, and by 16.6 the extension agrees with the already constructed part. Again, the continuity of $\tilde{H}$ follows from 3.13*b*.

It remains to show that for the constructed lift $\tilde{a}_\tau(t) := \tilde{H}(t,\tau)$ the point $\tilde{a}_\tau(1)$ is constant, so $\tilde{a}_\tau$ is a true homotopy of paths (not just a free homotopy). Observe that $(w \circ \tilde{a}_\tau)(1) = a_\tau(1)$ which is a constant point. Therefore the continuous function $\tau \mapsto \tilde{a}_\tau(1) - \tilde{a}_0(1)$ takes integer values, and hence it must be a constant function. Since $\tilde{a}_0(1) - \tilde{a}_0(1) = 0$, we have $\tilde{a}_\tau(1) - \tilde{a}_0(1) = 0$ for any τ; hence the last statement follows. $\square$

16.7. Exercise. *Suppose $a\colon [0,1] \to \mathbb{S}^1$ is a loop that misses one point in $\mathbb{S}^1$. Show that any lift of a to $\mathbb{R}$ is a loop.*

Construct a loop $b\colon [0,1] \to \mathbb{S}^1$ such that $b([0,1]) \supset \mathbb{S}^1$, but its lift in $\mathbb{R}$ is a loop.

C Degree

Recall that we consider $\mathbb{S}^1$ as a subset of the complex plane $\mathbb{C}$.

Let $a\colon [0,1] \to \mathbb{S}^1$ be a loop based at 1, and let $\tilde{a}\colon [0,1] \to \mathbb{R}$ be its unique lift with $\tilde{a}(0) = 0$ provided by 16.5. Since $w^{-1}\{1\} = \mathbb{Z}$, we have $\tilde{a}(1) \in \mathbb{Z}$. The integer $\tilde{a}(1)$ will be called the degree of the loop a; it will be denoted by $\deg a$.

16.8. Theorem. *The map* $[a] \mapsto \deg a$ *defines an isomorphism*

$$\pi_1(\mathbb{S}^1, 1) \to \mathbb{Z}.$$

Proof. Suppose a_0 and a_1 are two homotopic loops in $\mathbb{S}^1$ with base point 1, and let $\tilde{a}_0$ and $\tilde{a}_1$ be their lifts to $\mathbb{R}$ based at 0. By 16.5,

$$\deg a_0 = \tilde{a}_0(1) = \tilde{a}_1(1) = \deg a_1.$$

It follows that the degree depends only on the homotopy class of a loop. In other words, $[a] \mapsto \deg a$ is a well-defined map $\pi_1(\mathbb{S}^1, 1) \to \mathbb{Z}$.

Choose a pair of loops a and b based at 1 in $\mathbb{S}^1$. Let $\tilde{a}$ and $\tilde{b}$ be their lifts such that $\tilde{a}(0) = \tilde{b}(0) = 0$. Observe that the path $\tilde{b}_a \colon t \mapsto \deg a + \tilde{b}(t)$ is a lift of b that starts at $\deg a = \tilde{a}(1)$. Therefore $\tilde{a} * \tilde{b}_a$ is a lift of $a * b$ that starts at 0. Hence

$$\deg(a * b) = (\tilde{a} * \tilde{b}_a)(1) = \tilde{b}_a(1) = \deg a + \deg b$$

for any two loops a and b. Since $[a]\cdot[b] = [a * b]$, we proved that $\deg \colon \pi_1(\mathbb{S}^1, 1) \to \mathbb{Z}$ is a homomorphism.

Surjectivity of $\deg \colon \pi_1(\mathbb{S}^1, 1) \to \mathbb{Z}$ follows since the loop $a_n \colon t \mapsto w(n\cdot t)$ has degree n for any $n \in \mathbb{Z}$.

Let a and b be two loops in $\mathbb{S}^1$ with base point 1, and let $\tilde{a}$ and $\tilde{b}$ be their lifts such that $\tilde{a}(0) = \tilde{b}(0) = 0$. If $\deg a = \deg b$, then $\tilde{a}(1) = \tilde{b}(1)$. Therefore there is a linear homotopy of paths $\tilde{a}_\tau$ from $\tilde{a}$ to $\tilde{b}$. Note that $a_\tau = w \circ \tilde{a}_\tau$ is a homotopy from a to b; therefore, $a \sim b$. It follows that the homomorphism $\deg \colon \pi_1(\mathbb{S}^1, 1) \to \mathbb{Z}$ is injective. $\square$

16.9. Exercise. *Let* $w \colon \mathbb{R} \to \mathbb{S}^1$ *be the wrapping map. Show that there is no continuous map* $f \colon \mathbb{S}^1 \to \mathbb{R}$ *such that* $w \circ f = \mathrm{id}_{\mathbb{S}^1}$.

D Fundamental theorem of algebra

16.10. Theorem. *Every non-constant complex polynomial has a complex root.*

Proof. Choose a polynomial

$$p(z) = c_n\cdot z^n + c_{n-1}\cdot z^{n-1} + \dots + c_0.$$

If p is not constant, we can assume that $n \geqslant 1$ and $c_n \neq 0$.

Choose $r \geqslant 0$, and let

$$\textbf{❶} \qquad a_r(t) := \frac{p(r\cdot w(t))}{|p(r\cdot w(t))|} \Big/ \frac{p(r\cdot w(0))}{|p(r\cdot w(0))|};$$

here $w\colon t \mapsto \exp(2\cdot\pi\cdot i\cdot t)$ is the wrapping map. If p has no roots, then the denominators in ❶ do not vanish; therefore the map $(t, r) \mapsto a_r(t)$ is continuous. Furthermore, by ❶ we have $a_r(0) = a_r(1) = 1$ and $|a_r(t)| \equiv 1$; so a_r is a loop in $\mathbb{S}^1$ based at 1.

Observe that a_0 is a constant loop and $a_r \sim a_0$ for any r. In particular, $\deg a_r = 0$ for any r.

On the other hand, consider the loop $a_\infty(t) := w(n\cdot t)$; note that $\deg a_\infty = n$.

It is straightforward to check that $\frac{p(z)}{z^n} \to c_n$ and $\frac{|p(z)|}{|z^n|} \to |c_n|$ as $|z| \to \infty$. By ❶, we have $|a_r(t) - a_\infty(t)| < 1$ for any large r. Applying linear homotopy with radial projection to $\mathbb{S}^1$, we get the following homotopy from a_r to a_∞; namely,

$$H(t, s) = \frac{(1-s)\cdot a_r(t) + s\cdot a_\infty(t)}{|(1-s)\cdot a_r(t) + s\cdot a_\infty(t)|}.$$

In particular, $a_r \sim a_\infty$ and $\deg a_r = n$ for any large r — a contradiction. □

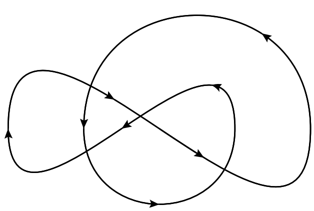
16.11. Exercise. *Consider a loop $a\colon [0,1] \to \mathbb{C}$ and a point $z \notin a([0,1])$. The the* winding number *of the loop a around point z is defined as*

$$\operatorname{wind}_a(z) := \deg b_z,$$

where

$$b_z(t) := \frac{(a(t) - z)}{|a(t) - z|} \Big/ \frac{(a(0) - z)}{|a(0) - z|}.$$

(a) *Show that the function wind_a is constant in every connected component of $\mathbb{C} \setminus a([0,1])$.*



(b) *Redraw the loop and write its winding number in each connected component of the complement.*

Chapter 17

Coverings

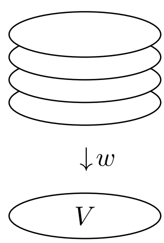
Now we will study covering maps — a class of maps that mimic properties of the wrapping map $w\colon \mathbb{R} \to \mathbb{S}^1$ from 16B. We used w to calculate the fundamental group of $\mathbb{S}^1$. The covering maps will be used to calculate fundamental groups of other spaces. More importantly, these maps give another connection between algebra and geometry that has applications from both sides.

A Definitions

17.1. Definition. *Let $w\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$ be a continuous map.*

An open set $V \subset \mathcal{Z}$ is called evenly covered by w if

$$w^{-1}(V) = \bigcup_{\alpha\in\mathcal{I}} \tilde{V}_\alpha,$$



where $\tilde{V}_\alpha$ are pairwise disjoint open sets in $\tilde{\mathcal{Z}}$ such that the restriction $w|_{\tilde{V}_\alpha}$ is a homeomorphism $\tilde{V}_\alpha \to V$ for each $\alpha \in \mathcal{I}$. The sets $\tilde{V}_\alpha$ are called slices over V.

If for every point $z \in \mathcal{Z}$ there is an open set $V \ni z$ that is evenly covered by w, then w is called a covering with total space $\tilde{\mathcal{Z}}$ and base $\mathcal{Z}$.

Note that $w\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$ is a covering if $\mathcal{Z}$ admits an open cover by evenly covered sets.[1] Following the proof of 16.5, one sees that the wrapping map $w(x) = \exp(2{\cdot}\pi{\cdot}i{\cdot}x)$ defines a covering map $\mathbb{R} \to \mathbb{S}^1$. Keep this example in your head while reading further.

[1] This phrase uses "cover" with two different meanings from 17.1 and 7.1.

The following observation will be used without mentioning.

17.2. Observation. *Let V be an open set that is evenly covered by a continuous map w. Then*

(a) any open set $W \subset V$ is also evenly covered by w, and

(b) if V is connected, then its slices are connected components of $w^{-1}(V)$.

17.3. Observation. *Any covering map is open.*

Proof. Let $w\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$ be a covering map. Choose an open set $\tilde{V} \subset \tilde{\mathcal{Z}}$ and a point $\tilde{z} \in \tilde{V}$.

Choose an evenly covered open set $W \ni z = w(\tilde{z})$. The point $\tilde{z}$ must lie in one of its slices, say $\tilde{W}_\alpha$. Let $\tilde{W}'_\alpha = \tilde{W}_\alpha \cap \tilde{V}$. Since $w|_{\tilde{W}_\alpha}$ is a homeomorphism, $W' = w(\tilde{W}'_\alpha)$ is open in W; since W is open, we also have that W' is open in $\mathcal{Z}$. That is, any $z \in w(\tilde{V})$ admits an open neighborhood $W' \subset w(\tilde{V})$; hence $w(\tilde{V})$ is open. □

17.4. Exercise! *Let $w\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$ be a covering map.*

(a) Show that the inverse image $w^{-1}\{p\}$ of any one-point set $\{p\}$ is a discrete subset of $\tilde{\mathcal{Z}}$.

(b) Show that the image $w(\tilde{\mathcal{Z}})$ is a clopen set in $\mathcal{Z}$. Conclude that, if $\mathcal{Z}$ is connected, and $\tilde{\mathcal{Z}} \neq \varnothing$, then w is surjective.

Most interesting examples of coverings have connected total space $\tilde{\mathcal{Z}}$ and base $\mathcal{Z}$; we will call them connected coverings. By 17.4*b*, any connected covering is surjective.[2]

17.5. Exercise. *Let $v\colon \tilde{\mathcal{X}} \to \mathcal{X}$ and $w\colon \tilde{\mathcal{Y}} \to \mathcal{Y}$ be two covering maps. Show that the map $\tilde{\mathcal{X}} \times \tilde{\mathcal{Y}} \to \mathcal{X} \times \mathcal{Y}$ defined by $(\tilde{x}, \tilde{y}) \mapsto (v(\tilde{x}), w(\tilde{y}))$ is a covering.*

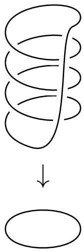
B Trivial coverings

Observe that any homeomorphism is a covering. Further, suppose $\mathcal{Z}$ is an arbitrary topological space and $\mathcal{I}$ is a discrete space. Then the projection $\mathcal{Z} \times \mathcal{I} \to \mathcal{Z}$ defined by $(z, i) \mapsto z$ is a covering. (Informally speaking, we take several disjoint copies of space $\mathcal{Z}$ and map a point in each copy to the corresponding point in $\mathcal{Z}$.)

A covering $w\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$ is called trivial if there is a discrete space $\mathcal{I}$ and a homeomorphism $h\colon \mathcal{Z} \times \mathcal{I} \to \tilde{\mathcal{Z}}$ such that $w \circ h$ is the projection $\mathcal{Z} \times \mathcal{I} \to \mathcal{Z}$. In other words, a covering $w\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$ is trivial if and only if the whole base $\mathcal{Z}$ is evenly covered by w.

[2] Sometimes coverings are defined to be connected or surjective.

Note that any trivial connected covering must be a homeomorphism. Since $\mathbb{R}$ and $\mathbb{S}^1$ are connected, the wrapping map $w(x) = \exp(2\cdot\pi\cdot i\cdot x)$ provides an example of a nontrivial covering map $\mathbb{R} \to \mathbb{S}^1$. The map $\mathbb{S}^1 \to \mathbb{S}^1$ defined by $z \mapsto z^n$ for an integer $n > 1$ gives another example of a nontrivial covering (which is an example of an n-fold covering; see next section). In this case, the total space and base are homeomorphic, but the map is not a homeomorphism. The case $n = 4$ is in the picture.



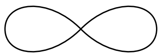
C n-fold coverings

Let $w\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$ be a covering. If for some integer $n \geqslant 0$ the inverse image of every point $z \in \mathcal{Z}$ has exactly n points, then we say that w is an n-fold covering.

17.6. Exercise. *Let $w\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$ be a connected covering, and let $n \geqslant 1$ be an integer. Assume that an inverse image of some point $z \in \mathcal{Z}$ has exactly n points. Show that w is n-fold.*

D Figure eight

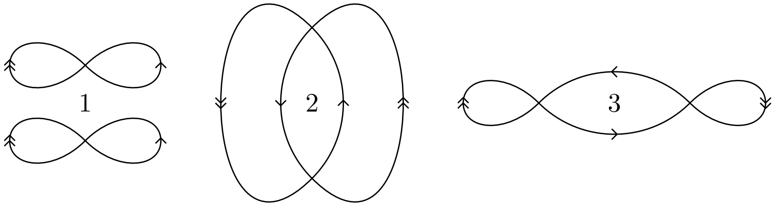
Let us denote by $\mathcal{F}_8$ the figure eight; it is the space obtained from two line segments by identifying all four endpoints. Formally speaking,



$$\mathcal{F}_8 \simeq ([0,1] \cup [2,3])/\{0,1,2,3\}.$$

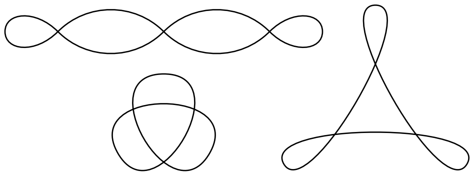
From 9.15, it follows that $\mathcal{F}_8$ is homeomorphic to the subset shown in the picture.

The next picture shows total spaces of three 2-fold coverings of $\mathcal{F}_8$; the paths identified by the covering map are marked by corresponding



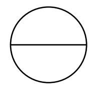
arrows. The first example is the total space of the trivial covering; it has two connected components, each homeomorphic to $\mathcal{F}_8$.

17.7. Exercise. *The following picture shows total spaces of some 3-fold coverings of $\mathcal{F}_8$. Redraw them and mark identified paths by*

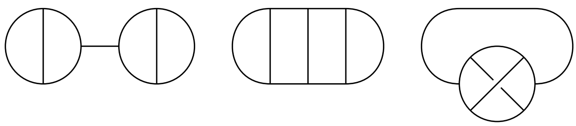


corresponding arrows.

17.8. Exercise. *Let Θ be the space homeomorphic to the set in the picture; it is the union of a circle and its diameter with induced topology from the plane.*



Which of the following spaces cover Θ? If so, then label paths identified by the covering map by three types of arrows; if not, then explain why. (If it looks like one path goes under another, then it



does.)

E Fibered product

Given two continuous maps $v\colon \mathcal{X} \to \mathcal{Z}$ and $w\colon \mathcal{Y} \to \mathcal{Z}$, consider the set

$$\mathcal{P} = \{\, (a,b) \in \mathcal{X} \times \mathcal{Y} \;:\; v(a) = w(b) \,\}$$

equipped with the induced topology from $\mathcal{X} \times \mathcal{Y}$. Define $\tilde{v}\colon \mathcal{P} \to \mathcal{Y}$ and $\tilde{w}\colon \mathcal{P} \to \mathcal{X}$ as the restrictions to $\mathcal{P}$ of the projections $\mathcal{X} \times \mathcal{Y} \to \mathcal{Y}$ and $\mathcal{X} \times \mathcal{Y} \to \mathcal{X}$.

$$\begin{tikzcd} \mathcal{P} \arrow[r, "\tilde{v}"] \arrow[d, "\tilde{w}"'] & \mathcal{Y} \arrow[d, "w"] \\ \mathcal{X} \arrow[r, "v"'] & \mathcal{Z} \end{tikzcd}$$

The space $\mathcal{P}$ equipped with the maps $\tilde{v}\colon \mathcal{P} \to \mathcal{Y}$ and $\tilde{w}\colon \mathcal{P} \to \mathcal{X}$ is called the fibered product of the maps $v\colon \mathcal{X} \to \mathcal{Z}$ and $w\colon \mathcal{Y} \to \mathcal{Z}$. The map $\tilde{v}$ is called the pullback of v along w; similarly $\tilde{w}$ is a pullback of w along v. This is a natural choice of space and maps that makes the diagram commute.

For pointed maps $v\colon (\mathcal{X}, x_0) \to (\mathcal{Z}, z_0)$ and $w\colon (\mathcal{Y}, y_0) \to (\mathcal{Z}, z_0)$, we can take $p_0 = (x_0, y_0) \in \mathcal{P}$, so the pullbacks $\tilde{v}\colon (\mathcal{P}, p_0) \to (\mathcal{Y}, y_0)$ and $\tilde{w}\colon (\mathcal{P}, p_0) \to (\mathcal{X}, x_0)$ are also pointed maps.

If $\mathcal{Z}$ is a one-point space (denoted by $*$ on the diagram), then the maps v and w are uniquely defined. In this case, $\mathcal{P} = \mathcal{X} \times \mathcal{Y}$ and the maps $\tilde{v}$ and $\tilde{w}$ are projections; so the fibered product generalizes the usual product.

$$\begin{tikzcd} \mathcal{X} \times \mathcal{Y} \arrow[r] \arrow[d] & \mathcal{Y} \arrow[d] \\ \mathcal{X} \arrow[r] & * \end{tikzcd}$$

17.9. Exercise. *Let $w\colon \mathcal{X} \to \mathcal{Z}$ be a continuous map. Assume $V \subset \subset \mathcal{Z}$ is an open set and $\tilde{w}$ is the pullback of w along the inclusion map $V \hookrightarrow \mathcal{Z}$. Show that V is evenly covered by w if and only if $\tilde{w}$ is a trivial covering.*

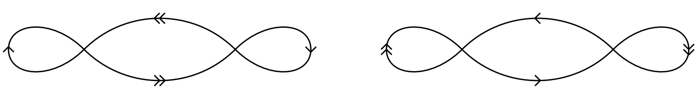
17.10. Exercise! *Show that the pullback $\tilde{v}$ of a covering v is a covering; moreover, if v is (a) trivial or (b) m-fold, then so is $\tilde{v}$.*

17.11. Exercise.

$$\begin{tikzcd} \mathcal{P} \arrow[r, "\tilde{v}"] \arrow[d, "\tilde{w}"'] \arrow[dr, "f"] & \mathcal{Y} \arrow[d, "w"] \\ \mathcal{X} \arrow[r, "v"'] & \mathcal{Z} \end{tikzcd}$$

(a) *Suppose that the maps v and w in the fibered-product diagram are coverings. Show that $f = = v \circ \tilde{w} = w \circ \tilde{v}$ is a covering.*

(b) *The picture shows 2-fold coverings v and w of the figure eight*



encoded with arrow notation as in 17D. Draw the space of their fibered product and draw the corresponding arrows for the covering $f = v \circ \tilde{w} = w \circ \tilde{v}$.

F Lifts

Earlier in 16B, we defined lifts along the wrapping map $\mathbb{R} \to \mathbb{S}^1$. Now we repeat the same definition for coverings.

Let $w\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$ be a covering, and let $f\colon \mathcal{Y} \to \mathcal{Z}$ be a continuous map. A continuous map $\tilde{f}\colon \mathcal{Y} \to \tilde{\mathcal{Z}}$ is called a lift of f (along w) if $w \circ \tilde{f} = f$, which means that the diagram commutes.

$$\begin{tikzcd} & \tilde{\mathcal{Z}} \arrow[d, "w"] \\ \mathcal{Y} \arrow[ur, "\tilde{f}"] \arrow[r, "f"'] & \mathcal{Z} \end{tikzcd}$$

In particular, if $\mathcal{Y} = [0,1]$, then f is a path, and the path $\tilde{f}\colon [0,1] \to \to \tilde{\mathcal{Z}}$ is its lift. Furthermore, taking $\mathcal{Y} = [0,1] \times [0,1]$, we can talk about lifts of homotopies of paths.

17.12. Lemma. *Let $w\colon (\tilde{\mathcal{Z}}, \tilde{z}_0) \to (\mathcal{Z}, z_0)$ be a pointed covering map, and let $a_0\colon [0,1] \to \mathcal{Z}$ be a path such that $a_0(0) = z_0$. Then there is a unique lift $\tilde{a}_0$ of a_0 such that $\tilde{a}_0(0) = \tilde{z}_0$.*

Moreover, for any homotopy of paths a_τ from a_0 to a_1 there is a unique lifted homotopy of paths $\tilde a_\tau$ such that $\tilde a_0(0) = \tilde z_0$ (and therefore $\tilde a_\tau(0) = \tilde z_0$ for any τ). In particular, if $a_0 \sim a_1$, then $\tilde a_0(1) = \tilde a_1(1)$.

This lemma is a direct generalization of 16.5. We omit the proof since it goes along the same lines. In fact, the definition of a covering map was made just to make this lemma hold.

17.13. Corollary. *Let $w\colon \tilde{\mathcal Z} \to \mathcal Z$ be a covering, and let $f\colon \mathcal Y \to \mathcal Z$ be a continuous map. Assume $\mathcal Y$ is path-connected and $\tilde f_1, \tilde f_2\colon \mathcal Y \to \tilde{\mathcal Z}$ are two lifts of f along w. Show that if $\tilde f_1(y_0) = \tilde f_2(y_0)$ for some $y_0 \in \mathcal Y$, then $\tilde f_1(y) = \tilde f_2(y)$ for any $y \in \mathcal Y$.*

Proof. Since $\mathcal Y$ is path-connected, we can connect y_0 to y by a path. It remains to apply 17.12. □

17.14. Theorem. *Let $w\colon (\tilde{\mathcal Z}, \tilde z_1) \to (\mathcal Z, z_0)$ be a covering. Assume that $\tilde{\mathcal Z}$ is simply-connected. Then there is a bijection $\alpha \leftrightarrow \tilde z_\alpha$ between elements of the fundamental group $\pi_1(\mathcal Z, z_0)$ and points in the inverse image $w^{-1}\{z_0\} \subset \tilde{\mathcal Z}$.*

Moreover, the bijection can be constructed in the following way: Choose a loop $a\colon [0,1] \to \mathcal Z$ based at z_0; let $\alpha = [a] \in \pi_1(\mathcal Z, z_0)$. Consider a lift $\tilde a$ such that $\tilde a(0) = \tilde z_1$ and let $\tilde z_\alpha = \tilde a(1)$.

Proof. Note that $w \circ \tilde a(1) = a(1) = z_0$; therefore $\tilde z_\alpha \in w^{-1}\{z_0\}$ as required.

Let us show that $\alpha \mapsto \tilde z_\alpha$ is well-defined; that is, the point $\tilde a(1)$ does not depend on the choice of the loop a, assuming that $\alpha = [a]$. Indeed, if $\alpha = [b]$ for some loop b, then $a \sim b$, and by the second statement in the lemma, we get $\tilde a(1) = \tilde b(1)$.

Since $\tilde{\mathcal Z}$ is simply-connected, there is a path $\tilde a$ from $\tilde z_1$ to any point in $w^{-1}\{z_0\}$. Note that $a = w \circ \tilde a$ is a loop in $\mathcal Z$ with base point z_0, and $\tilde a$ is its lift as in the construction. This implies that the map $\alpha \mapsto \tilde z_\alpha$ is onto; that is, any point in $w^{-1}\{z_0\}$ is $\tilde z_\alpha$ for some $\alpha \in \pi_1(\mathcal Z, z_0)$.

It remains to check that the map $\alpha \mapsto \tilde z_\alpha$ is injective. Suppose $\tilde a(1) = \tilde b(1)$ for two paths starting at $\tilde z_1$. Since $\tilde{\mathcal Z}$ is simply-connected, we have that $\tilde a \sim \tilde b$; that is, there is a homotopy $\tilde a_\tau$ from $\tilde a$ to $\tilde b$. Observe that $a_\tau = w \circ \tilde a_\tau$ is a homotopy from $a = w \circ \tilde a$ to $b = w \circ \tilde b$; that is, $a \sim b$, which is equivalent to $[a] = [b] \in \pi_1(\mathcal Z, z_0)$. □

17.15. Exercise. *Let $w\colon (\tilde{\mathcal Z}, \tilde z_0) \to (\mathcal Z, z_0)$ be a pointed covering map.*[3]

[3]Recall that this means that w is a covering and $w(\tilde z_0) = z_0$.

(a) *Show that* $w_*\colon \pi_1(\tilde{\mathcal{Z}}, \tilde{z}_0) \to \pi_1(\mathcal{Z}, z_0)$ *is a monomorphism (which means injective homomorphism).*

(b) *Let* b *be a loop in* $\mathcal{Z}$ *with base point* z_0*, and let* $\tilde{b}$ *be its lift in* $\tilde{\mathcal{Z}}$ *such that* $\tilde{b}(0) = \tilde{z}_0$*. Show that* $\tilde{b}$ *is a loop if and only if* $[b] \in \operatorname{Im} w_*$*.*

(c) *Show that the following construction defines a right action of* $\pi_1(\mathcal{Z}, z_0)$ *on* $w^{-1}\{z_0\}$*.*[4] *Given* $\alpha = [a] \in \pi_1(\mathcal{Z}, z_0)$ *and a point* $p \in w^{-1}\{z_0\}$*, consider a lift* $\tilde{a}$ *of* a *such that* $\tilde{a}(0) = p$ *and define* $p{\cdot}\alpha := \tilde{a}(1)$*.*

17.16. Exercise. *Suppose that the maps* v *and* w *in the pointed fibered-product diagram are coverings. Show that*

$$\begin{array}{ccc} (\mathcal{P}, p_0) & \xrightarrow{\tilde{v}} & (\mathcal{Y}, y_0) \\ {\scriptstyle \tilde{w}}\downarrow & \overset{f}{\searrow} & \downarrow{\scriptstyle w} \\ (\mathcal{X}, x_0) & \xrightarrow[v]{} & (\mathcal{Z}, z_0) \end{array}$$

$$\operatorname{Im} f_* = \operatorname{Im} v_* \cap \operatorname{Im} w_*.$$

---

[4]This is the so-called monodromy action.

Chapter 18

Normal coverings

A normal covering is a covering that can be realized as a quotient map by a group action. This class of coverings is particularly important in many applications.

A Freely discontinuous action

A continuous action $G \circlearrowright \mathcal{X}$ is called freely discontinuous if for any $x \in \mathcal{X}$ there is an open neighborhood $V \ni x$ such that $V \cap g \cdot V = \varnothing$ for any $g \neq 1$. In this case, we also have

$$g_1 \cdot V \cap g_2 \cdot V = g_1 \cdot (V \cap g_1^{-1} \cdot g_2 \cdot V) = \varnothing$$

for any two distinct elements $g_1, g_2 \in G$.

18.1. Proposition. *Let $G \circlearrowright \mathcal{X}$ be a continuous action. The quotient map $w\colon \mathcal{X} \to \mathcal{X}/G$ is a covering if and only if the action is freely discontinuous.*

If a covering w can be realized as a quotient map by an action of some group G, then we say that w is a normal covering[1] or G-covering; the latter term is used if one needs to specify the group.

Note that the wrapping map $w\colon \mathbb{R} \to \mathbb{S}^1$ constructed in 16B is a normal $\mathbb{Z}$-covering, where $\mathbb{Z}$ denotes the additive group of integers; it is the quotient map for a freely discontinuous action $\mathbb{Z} \circlearrowright \mathbb{R}$ by integer shifts $x \mapsto x + n$, where $n \in \mathbb{Z}$.

[1] If $w\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$ is a normal covering and $w(\tilde{z}_0) = z_0$, then $w_*(\pi_1(\tilde{\mathcal{Z}}, \tilde{z}_0))$ is a normal subgroup of $\pi_1(\mathcal{Z}, z_0)$ — this is the reason for the name. Normal coverings are also called regular and Galois.

Proof. By the definition of the quotient map, $w\colon \mathcal{X} \to \mathcal{X}/G$ is continuous and surjective; moreover, by 6.22a, w is open. Let V be an open neighborhood of $x \in \mathcal{X}$ that meets the condition in the definition of freely discontinuous action.

Since w is open, $W = w(V)$ is an open neighborhood of $w(x) = [x] = G\cdot x$. Furthermore,

$$w^{-1}(W) = G\cdot V = \bigcup_{g\in G} g\cdot V$$

and the restriction $w|_{g\cdot V}$ is a homeomorphism $g\cdot V \to W$; indeed, $w|_{g\cdot V}\colon g\cdot V \to W$ is open, continuous, and bijective. We have proved that w is a covering.

On the other hand, if the quotient map $w\colon \mathcal{X} \to \mathcal{X}/G$ is a covering, then every point $w(x) \in \mathcal{X}/G$ admits an evenly covered neighborhood W; its slice $V \ni x$ meets the condition in the definition of freely discontinuous action. $\square$

18.2. Exercise. *Let $G \curvearrowright \mathcal{X}$ be a freely discontinuous action. Assume $\mathcal{X}$ is compact. Show that G is finite.*

18.3. Advanced exercise. *Give an example of a freely discontinuous action $G \curvearrowright \mathcal{X}$ on a Hausdorff space $\mathcal{X}$ such that the quotient space $\mathcal{X}/G$ is not Hausdorff.*

B Finite group

An action $G \curvearrowright \mathcal{X}$ is called free if $g\cdot x \neq x$ for any $x \in \mathcal{X}$ and $g \neq 1$.

Note that any freely discontinuous action is free, but not the other way around; for example, the action $\mathbb{R} \curvearrowright \mathbb{R}$ by shifts $x \mapsto x + c$ is free, but not freely discontinuous.

18.4. Corollary. *Let $G \curvearrowright \mathcal{X}$ be a continuous free action of a finite group G. Assume that $\mathcal{X}$ is Hausdorff. Then the quotient map $\mathcal{X} \to \mathcal{X}/G$ is a covering.*

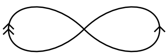
Proof. Choose $x \in \mathcal{X}$ and $g \in G$. Since $G \curvearrowright \mathcal{X}$ is free, if $g \neq 1$, then $g\cdot x \neq x$. Since $\mathcal{X}$ is Hausdorff, we can choose disjoint open sets $U_g \ni x$ and $V_g \ni g\cdot x$. Let $W_g = U_g \cap g^{-1}\cdot V_g$. Note that $W_g \ni x$ and $g\cdot W_g \ni g\cdot x$ are disjoint open sets. Indeed, $W_g \cap g\cdot W_g \subset U_g \cap V_g = \varnothing$.

Now consider the finite intersection of open sets

$$W = \bigcap_{g\neq 1} W_g \ni x.$$

Note that W is open and $x \in W$. The sets $g \cdot W$ are disjoint for all $g \in G$; that is, the action $G \circlearrowright \mathcal{X}$ is freely discontinuous. It remains to apply 18.1. $\square$

18.5. Exercise. *Recall that $\mathcal{F}_8$ denotes the figure eight shown in the picture. Using the arrow notation as in 17D, draw a connected total space of*



(a) a normal $\mathbb{Z}_4$-covering of $\mathcal{F}_8$;
(b) a normal $\mathbb{Z}_2 \times \mathbb{Z}_2$-covering of $\mathcal{F}_8$;
(c) a non-normal 4-fold covering of $\mathcal{F}_8$.

C Real projective plane

Consider the action $\mathbb{Z}_2 \circlearrowright \mathbb{S}^2$ generated by the involution $x \mapsto -x$; note that this action is free. The quotient space $\mathbb{S}^2/\mathbb{Z}_2$ is called the real projective plane and denoted by $\mathbb{R}\mathrm{P}^2$. Since $\mathbb{Z}_2$ is finite, by 18.4, the quotient map $\mathbb{S}^2 \to \mathbb{R}\mathrm{P}^2$ is a covering. By 14.12, $\mathbb{S}^2$ is simply-connected. By 17.14, $\pi_1(\mathbb{R}\mathrm{P}^2)$ has exactly two elements, and therefore it is isomorphic to $\mathbb{Z}_2$ — this is the only group with two elements up to isomorphism. (One could also deduce that $\pi_1(\mathbb{R}\mathrm{P}^2) \cong \mathbb{Z}_2$ from 18.14*b* below.)

One way or another, we proved the following.

18.6. Claim. $\pi_1(\mathbb{R}\mathrm{P}^2) \cong \mathbb{Z}_2$.

18.7. Exercise. *Show that any 2-fold covering is normal.*

18.8. Advanced exercise. *Construct a finite topological space with fundamental group isomorphic to $\mathbb{Z}_2$.*

D Borsuk–Ulam theorem

18.9. Theorem. *There is no odd continuous map $\tilde{h}\colon \mathbb{S}^2 \to \mathbb{S}^1$; that is, no continuous map $\tilde{h}\colon \mathbb{S}^2 \to \mathbb{S}^1$ meets the condition $\tilde{h}(-\tilde{x}) = -\tilde{h}(\tilde{x})$ for any $\tilde{x} \in \mathbb{S}^2$.*

The proof relies on the following exercise and lemma.

18.10. Exercise! *Let $\mathbb{Z}_2 \circlearrowright \mathbb{S}^1$ be the action generated by the involution $x \mapsto -x$. Show that $\mathbb{S}^1/\mathbb{Z}_2 \simeq \mathbb{S}^1$ and the quotient map defines a 2-fold cover $\mathbb{S}^1 \to \mathbb{S}^1$.*

A group homomorphism $f\colon G \to H$ is called trivial if it maps every element of G to the neutral element of H.

18.11. Lemma. *Any homomorphism $\mathbb{Z}_2 \to \mathbb{Z}$ is trivial.*

Since $\mathbb{Z}_2$ has only two elements $[0]_2$ and $[1]_2$, the homomorphism in the lemma is trivial if and only if $[1]_2 \mapsto 0$.

Proof. Assume $\varphi\colon \mathbb{Z}_2 \to \mathbb{Z}$ is a nontrivial homomorphism. Then $\varphi[1]_2 = n$ for an integer $n \neq 0$. But then

$$0 = \varphi([0]_2) = \varphi([1]_2 + [1]_2) = \varphi([1]_2) + \varphi([1]_2) = n + n \neq 0$$

— a contradiction. □

Most of the proof of 18.9 will be defining and explaining the following commutative diagram.

$$\begin{tikzcd}
(\mathbb{S}^2, \tilde{x}_0) \arrow[rr, "\tilde{h}"] \arrow[dd, "v"'] & & (\mathbb{S}^1, \tilde{y}_0) \arrow[dd, "w"] \\
& {[0,1]} \arrow[ul, "\tilde{a}"'] \arrow[ur, "\tilde{b}"] \arrow[dl, "a"] \arrow[dr, "b"'] & \\
(\mathbb{RP}^2, x_0) \arrow[rr, dashed, "h"'] & & (\mathbb{S}^1, y_0)
\end{tikzcd}$$

Proof of 18.9. Assume that $\tilde{h}$ exists.

Choose $\tilde{x}_0 \in \mathbb{S}^2$ and a path $\tilde{a}\colon [0,1] \to \mathbb{S}^2$ from $\tilde{x}_0$ to $-\tilde{x}_0$. Let $v\colon \mathbb{S}^2 \to \mathbb{RP}^2$ be the quotient map for the equivalence relation $\tilde{x} \sim -\tilde{x}$. Recall that v is a 2-fold covering; see 18C. Note that $a = v \circ \tilde{a}$ is a loop in $\mathbb{RP}^2$ based at $x_0 = v(\tilde{x}_0) = v(-\tilde{x}_0)$, and $\tilde{a}$ is its lift.[2]

The path $\tilde{b} = \tilde{h} \circ \tilde{a}$ runs from $\tilde{y}_0 = \tilde{h}(\tilde{x}_0)$ to $-\tilde{y}_0 = \tilde{h}(-\tilde{x}_0)$. Let $w\colon \mathbb{S}^1 \to \mathbb{S}^1$ be the quotient map for the equivalence relation $\tilde{y} \sim -\tilde{y}$, which is also a 2-fold covering map (see 18.10); note that $b = w \circ \tilde{b}$ is a loop in $\mathbb{S}^1$ based at $y_0 = w(\tilde{y}_0)$ and $\tilde{b}$ is its lift to $\mathbb{S}^1$. Since $\tilde{b}(1) \neq \tilde{b}(0)$, 17.14 implies that $\beta = [b] \neq 0 \in \mathbb{Z} = \pi_1(\mathbb{S}^1)$.

Recall that $v\colon \mathbb{S}^2 \to \mathbb{RP}^2$ and $w\colon \mathbb{S}^1 \to \mathbb{S}^1$ are quotient maps that identify antipodal points. Since $\tilde{h}(-\tilde{x}) = -\tilde{h}(\tilde{x})$, it maps equivalence classes in $\mathbb{S}^2$ to equivalence classes in $\mathbb{S}^1$. In particular, there is a map $h\colon \mathbb{RP}^2 \to \mathbb{S}^1$ such that $h \circ v = w \circ \tilde{h}$.[3]

We have finished describing the diagram and now the proof starts. Consider the induced homomorphism

$$h_*\colon \pi_1(\mathbb{RP}^2, x_0) \to \pi_1(\mathbb{S}^1, y_0).$$

[2] By 17.14, we have $\alpha = [a] = [1]_2 \in \mathbb{Z}_2 \cong \pi_1(\mathbb{RP}^2, x_0)$, but it will not be used in the proof directly.

[3] The dashed arrow on the diagram encodes the claim that h exists.

Since $\pi_1(\mathbb{RP}^2, x_0) \cong \mathbb{Z}_2$ and $\pi_1(\mathbb{S}^1, y_0) \cong \mathbb{Z}$, the lemma implies that h_* is trivial. On the other hand, $h_*[a] = [b]$. From above, $[b] \neq 0 \in \mathbb{Z}$ — a contradiction. □

18.12. Exercise.

(a) *Show that for every continuous map $f\colon \mathbb{S}^2 \to \mathbb{R}^2$ there exists a pair of antipodal points in $\mathbb{S}^2$ that map to a single point.*

(b) *Suppose that a sandwich is made from three ingredients, say, cheese, ham, and bread. Show that one can cut the sandwich by a plane so that each ingredient is evenly divided.*[4]

(c) *Let A, B, and C be closed subsets of $\mathbb{S}^2$ such that $A \cup B \cup C = \mathbb{S}^2$. Apply (a) to the map $f\colon \mathbb{S}^2 \to \mathbb{R}^2$ defined by*

$$f\colon x \mapsto (\mathrm{dist}_A x, \mathrm{dist}_B x)$$

to conclude that one of the sets A, B, or C contains a pair of antipodal points in $\mathbb{S}^2$. Give an example showing that the analogous statement does not hold for four closed sets A, B, C, and D.

E Monodromy

18.13. Proposition. *Given a pointed G-covering $w\colon (\tilde{\mathcal{Z}}, \tilde{z}_1) \to (\mathcal{Z}, z_0)$, there is a unique homomorphism $\pi_1(\mathcal{Z}, z_0) \to G$ that will be written as $\alpha \mapsto m_\alpha$ such that $m_{[a]}\cdot\tilde{z}_1 = \tilde{a}(1)$ for any loop $a\colon [0,1] \to \mathcal{Z}$ based at z_0 and its lift $\tilde{a}\colon [0,1] \to \tilde{\mathcal{Z}}$ such that $\tilde{a}(0) = \tilde{z}_1$.*

The homomorphism $\alpha \mapsto m_\alpha$ is called monodromy homomorphism of the pointed G-covering $w\colon (\tilde{\mathcal{Z}}, \tilde{z}_1) \to (\mathcal{Z}, z_0)$. The following proof reuses the argument in 16.8.

Proof. Note that $w^{-1}\{z_0\} = G\cdot\tilde{z}_1$.

Choose $\alpha = [a] \in \pi_1(\mathcal{Z}, z_0)$, and let $\tilde{a}$ be a lift of the loop a such that $\tilde{a}(0) = \tilde{z}_1$. Note that $(w \circ \tilde{a})(1) = a(1) = z_0$, so $\tilde{a}(1) = m_\alpha\cdot\tilde{z}_1$ for some $m_\alpha \in G$.

By 17.12, $\tilde{a}(1)$ and therefore m_α depend only on the homotopy class $\alpha = [a]$. Thus we get a well-defined map $\alpha \mapsto m_\alpha$ from $\pi_1(\mathcal{Z}, z_0)$ to G.

To show that $\alpha \mapsto m_\alpha$ is a homomorphism, we need to check that

❶ $$m_{\alpha\cdot\beta} = m_\alpha\cdot m_\beta$$

[4] A picky reader can assume that each ingredient occupies a convex polyhedron in the space and has constant density.

for any $\alpha, \beta \in \pi_1(\mathcal{Z}, z_0)$.

Choose loops a and b based at $z_0 \in \mathcal{Z}$ such that $\alpha = [a]$ and $\beta = [b]$. Let $\tilde{a}$ and $\tilde{b}$ be the lifts of a and b that start at $\tilde{z}_1$. Note that $m_\alpha \cdot \tilde{b}$ is a lift of b that starts at $\tilde{a}(1) = m_\alpha \cdot \tilde{z}_1$. Therefore $\tilde{a} * (m_\alpha \circ \tilde{b})$ is a lift of $a * b$ that starts at $\tilde{z}_1$. It follows that

$$m_{\alpha\cdot\beta} \cdot \tilde{z}_1 = (\tilde{a} * (m_\alpha \cdot \tilde{b}))(1) = (m_\alpha \cdot \tilde{b})(1) = m_\alpha \cdot m_\beta \cdot \tilde{z}_1.$$

Since the action $G \curvearrowright \tilde{\mathcal{Z}}$ is free, $m_{\alpha\cdot\beta} \cdot \tilde{z}_1 = m_\alpha \cdot m_\beta \cdot \tilde{z}_1$ implies ❶. □

18.14. Exercise! *Let $\alpha \mapsto m_\alpha$ be a monodromy homomorphism of a pointed G-covering $w \colon (\tilde{\mathcal{Z}}, \tilde{z}_1) \to (\mathcal{Z}, z_0)$.*

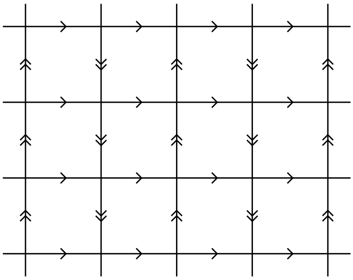
(a) Assume $\tilde{\mathcal{Z}}$ is path-connected. Show that $\alpha \mapsto m_\alpha$ is an epimorphism (which is a surjective homomorphism).

(b) Assume $\tilde{\mathcal{Z}}$ is simply-connected. Show that $\alpha \mapsto m_\alpha$ is an isomorphism (which is a bijective homomorphism).

(c) Assume w is a trivial covering. Show that $\alpha \mapsto m_\alpha$ is a trivial homomorphism.

F Klein bottle

Consider the following tessellation of $\mathbb{R}^2$ into unit squares. Let $G \curvearrowright \mathbb{R}^2$

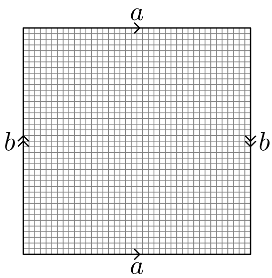


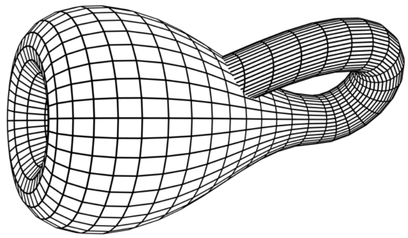
be the action that preserves this tessellation with the arrows on the sides. Formally speaking, the action includes the motions

$$(x, y) \mapsto (m + x, n + (-1)^m \cdot y)$$

for all integers m and n.

The quotient space $\mathbb{R}^2/G$ is called the Klein bottle. This space can also be obtained as the quotient of the square $[-1, 1] \times [-1, 1]$ by the minimal equivalence relation such that $(x, -1) \sim (x, 1)$ and $(-1, y) \sim (1, -y)$. The picture on the right might help to visualize this space, but this is not an embedding of $\mathcal{K}$, and in fact there is no embedding of $\mathcal{K} \hookrightarrow \mathbb{R}^3$.





18.15. Exercise. *Let $G \circlearrowleft \mathbb{R}^2$ be the described action, and let $\mathcal{K} = \mathbb{R}^2/G$ be the Klein bottle.*

(a) Show that the action $G \circlearrowleft \mathbb{R}^2$ is freely discontinuous.

(b) Apply 18.14b to show that $\pi_1(\mathcal{K}) \cong G$.

(c) Show that G and therefore $\pi_1(\mathcal{K})$ are noncommutative.

(d) Construct a 2-fold covering of $\mathcal{K}$ by the torus $\mathbb{S}^1 \times \mathbb{S}^1$.

Chapter 19

Universal covering

A Reasonable spaces

The term *reasonable space* will be used as a shortcut for *locally simply-connected space*. This term will be used only in these notes; it is not standard and can mean something else elsewhere.

So, a topological space $\mathcal{Z}$ is *reasonable* if for any open set $V \subset \mathcal{Z}$ and any $p \in V$ there is an open simply-connected set W such that $V \supset W \ni p$.

For instance, Euclidean space is reasonable; it follows since any open ball in a Euclidean space is convex; in particular, contractible and therefore simply-connected.

Since any simply-connected set is path-connected, any reasonable space is locally path-connected and locally connected; see 12D and 11E. In particular, by 11.16 we have that every connected component of a reasonable space is clopen.

19.1. Exercise! *Let $w\colon \mathcal{Y} \to \mathcal{Z}$ be a covering. Show the following:*

(a) If $\mathcal{Z}$ is reasonable then so is $\mathcal{Y}$, and the restriction of w to any connected component of $\mathcal{Y}$ is a covering.

(b) If w is surjective and $\mathcal{Y}$ is reasonable, then so is $\mathcal{Z}$.

In this chapter we will sketch a proof of the following theorem.

$$\begin{tikzcd} & (\mathcal{Y}, y_0) \arrow[d, "w"] \\ (\tilde{\mathcal{Z}}, \tilde{z}_1) \arrow[ur, "v"] \arrow[r, "u"] & (\mathcal{Z}, z_0) \end{tikzcd}$$

19.2. Theorem. *Any reasonable connected space $\mathcal{Z}$ admits a covering $u\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$ with a simply-connected total space $\tilde{\mathcal{Z}}$.*

Moreover such a covering has the following universal property: given a connected pointed covering $w\colon (\mathcal{Y}, y_0) \to (\mathcal{Z}, z_0)$ and any choice of point

$\tilde{z}_1 \in \tilde{\mathcal{Z}}$ such that $u(\tilde{z}_1) = z_0$ there is a pointed covering $v\colon (\tilde{\mathcal{Z}}, \tilde{z}_1) \to$ $\to (\mathcal{Y}, y_0)$ which makes the diagram commute.

The first part of the theorem will be provided by 19.14, and the second part will follow from 19.11*b* and 19.13; see 19E.

The theorem can be used together with 17.14 and 18.14*b*; it is useful since most spaces that appear in applications are reasonable.

The covering u as in the theorem is called universal. Note that the universal property can be applied to the covering $u\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$ with different choices of base points. It defines a homeomorphism $h\colon \tilde{\mathcal{Z}} \circlearrowleft$ and all such homeomorphisms define a group action on $\tilde{\mathcal{Z}}$, which meets the definition of a normal covering; applying 18.14*b*, we get the following.

19.3. Observation. *A universal covering of a connected reasonable space $\mathcal{Z}$ is a normal $\pi_1\mathcal{Z}$-covering.*

Note that the covering v in 19.2 is universal, and by the observation, it has to be normal. This leads to the following classification of coverings up to equivalence. Two pointed coverings $v\colon (\mathcal{X}, x_0) \to (\mathcal{Z}, z_0)$ and $w\colon (\mathcal{Y}, y_0) \to (\mathcal{Z}, z_0)$ are equivalent if there is a pointed homeomorphism $h\colon (\mathcal{X}, x_0) \to (\mathcal{Y}, y_0)$ that makes the diagram commute.

$$\begin{tikzcd}
(\mathcal{X}, x_0) \arrow[rr, dashed, "h"] \arrow[dr, "v"'] & & (\mathcal{Y}, y_0) \arrow[dl, "w"] \\
& (\mathcal{Z}, z_0) &
\end{tikzcd}$$

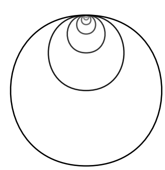
19.4. Proposition. *Let $\mathcal{Z}$ be a connected reasonable space. Then a connected pointed covering $v\colon (\mathcal{X}, x_0) \to (\mathcal{Z}, z_0)$ is uniquely defined up to equivalence by the subgroup $H = \operatorname{Im} v_* < \pi_1(\mathcal{Z}, z_0)$.*

This classification follows from 17.15*a*, 18.13, and 19.2; we will not prove it formally, but it is a good exercise in concentration.

In the following two sections we show how the theorem might fail to be true for nonreasonable spaces.

B Hawaiian earring and its relatives

Let us denote by $\mathcal{H}$ the Hawaiian earring; it is a topological space homeomorphic to the union of a nested sequence of circles in the plane shown in the picture; the radii of circles converge to zero and the circles have one common point.



19.5. Claim. *The Hawaiian earring $\mathcal{H}$ is not reasonable; it is path-connected, locally path-connected, but not locally simply-connected.*

Proof. It is straightforward to check path-connectedness and local path-connectedness of $\mathcal{H}$. Let us show that $\mathcal{H}$ is not locally simply-connected.

Let p be the common point of the circles in $\mathcal{H}$. An arbitrary neighborhood V of p contains a circle, say S. Consider the map $r\colon V \to S$ that is the identity on S and maps $V \setminus S$ to p. Note that r is a retraction.

Recall that S has a nontrivial fundamental group; see 16.1. By 15.7a, V is not simply-connected. Hence the statement follows. □

19.6. Exercise. *Let $\mathcal{H}$ be the Hawaiian earring, and let $\mathcal{F}_\infty$ be the bouquet of countably many circles; that is, a topological space obtained by gluing together a sequence of circles along a single point. In other words,*

$$\mathcal{F}_\infty = (\mathbb{S}^1 \times \mathbb{N})/(\{p\} \times \mathbb{N}),$$

where $\mathbb{N}$ denotes the set of positive integers.

Construct a continuous bijection $\mathcal{F}_\infty \to \mathcal{H}$. Show that $\mathcal{F}_\infty \not\simeq \mathcal{H}$.

19.7. Exercise. *Show that the Hawaiian earring $\mathcal{H}$ does not admit a covering that meets the universal property.*

Moreover, for any covering $w\colon \mathcal{X} \to \mathcal{H}$ with connected $\mathcal{X}$ there is another covering $v\colon \mathcal{Y} \to \mathcal{H}$ with connected $\mathcal{Y}$ that cannot be included in the commutative diagram.

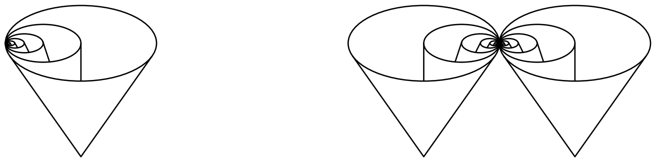
$$\begin{tikzcd} & \mathcal{Y} \arrow[d, "w"] \\ \mathcal{X} \arrow[ur, dotted, "\nexists"] \arrow[r, "u"] & \mathcal{H} \end{tikzcd}$$

The following exercise implies that composition of coverings might fail to be a covering; as we will see in 19.11a, reasonable spaces behave better.

19.8. Advanced exercise. *Construct a $\mathbb{Z}$-covering of the Hawaiian earring $w\colon \mathcal{Y} \to \mathcal{H}$ and a $\mathbb{Z}_2$-covering $v\colon \mathcal{X} \to \mathcal{Y}$ such that the composition $w \circ v\colon \mathcal{X} \to \mathcal{H}$ is not a covering.*

The cone over the Hawaiian earring provides an example of a contractible space (in particular, simply-connected) that is not locally simply-connected. To construct this space, think that $\mathcal{H}$ lies in the (x, y)-plane in $\mathbb{R}^3$; choose a tip of the cone, say q, under the (x, y)-plane and consider the union of line segments from q to the points in $\mathcal{H}$.

The so-called Griffiths space $\mathcal{G}$ (right picture) is obtained by gluing together two copies of such a cone at the common point of the circles in $\mathcal{H}$. It gives an example of a space that is path-connected, locally path-connected, and not simply-connected. However, $\mathcal{G}$ does not have nontrivial coverings. In fact, the trivial covering $\mathrm{id}_\mathcal{G}\colon \mathcal{G} \to \mathcal{G}$

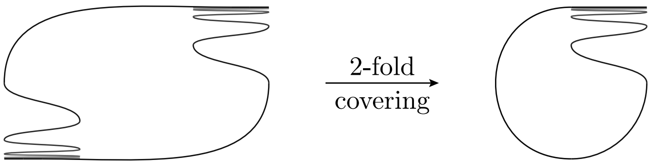


meets the universal property in 19.2, but its total space is not simply-connected.

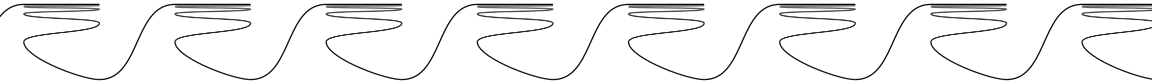
19.9. Very advanced exercise. *Show that the Griffiths space has nontrivial fundamental group.*

C Polish circle

The Polish circle $\mathcal{P}$ is an example of a topological space shown on the right side of the picture. It is a simply-connected space that is not locally path-connected. On the left side of the picture, you see the total space of its connected 2-fold cover. Thus, $\mathcal{P}$ is a simply-connected



space that admits nontrivial connected coverings. The total space of the trivial covering $\mathrm{id}_{\mathcal{P}}\colon \mathcal{P} \to \mathcal{P}$ is simply-connected, but this covering does not meet the universal property in 19.2. Below you see the total space of its covering that meets the universal property in 19.2, but this space is not path-connected and therefore not simply-connected.



D Reasonable coverings

From now on we restrict attention to reasonable spaces. The following claim is one of their key features. Note that it does not hold for the Polish circle, so the claim does not hold without assuming reasonability. (This time local path-connectedness will suffice.)

19.10. Claim. *Let $w\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$ be a covering map. Assume $\mathcal{Z}$ is a reasonable simply-connected space. Then w is trivial.*

Proof. By 19.1, $\tilde{\mathcal{Z}}$ is reasonable, so each connected component of $\tilde{\mathcal{Z}}$ is clopen, and it is sufficient to show that the restriction of w to any connected component of $\tilde{\mathcal{Z}}$ is a homeomorphism.

In other words, we need to show that w is a homeomorphism, assuming that $\tilde{\mathcal{Z}}$ is connected.

By 17.3, w is open. By 17.4, $w(\tilde{\mathcal{Z}})$ is a clopen subset of $\mathcal{Z}$. Since $\mathcal{Z}$ is simply-connected, w is surjective.

It remains to show that w is injective; indeed, an open bijective continuous map must be a homeomorphism.

Choose two points $\tilde{z}_0, \tilde{z}_1 \in \tilde{\mathcal{Z}}$ that map to a single point, say $z_0 = w(\tilde{z}_0) = w(\tilde{z}_1)$. Since $\tilde{\mathcal{Z}}$ is path-connected, we can choose a path $\tilde{a}$ from $\tilde{z}_0$ to $\tilde{z}_1$. Note that $a = w \circ \tilde{a}$ is a loop in $\mathcal{Z}$. Since $\mathcal{Z}$ is simply-connected, $a \sim e_{z_0}$. Now, 17.12 implies that $\tilde{z}_0 = \tilde{a}(0) = \tilde{a}(1) = \tilde{z}_1$, which proves that w is injective. □

Note that the statement at the end of this proof also follows from 17.15*b*.

19.11. Exercise! *Assume that each map in the commutative diagram is continuous, v is surjective*[1]*, and the space $\mathcal{Z}$ is reasonable. Show that*

$$\begin{tikzcd} & \mathcal{Y} \arrow[dr, "w"] & \\ \mathcal{X} \arrow[ur, two heads, "v"] \arrow[rr, "u"] & & \mathcal{Z} \end{tikzcd}$$

(a) if v and w are coverings, then so is u,
(b) if u and w are coverings, then so is v,
(c) if u and v are coverings, then so is w.

19.12. Exercise! *Let V and W be two connected open sets in a reasonable space $\mathcal{Z}$. Assume $\mathcal{Z} = V \cup W$ and the intersection $V \cap W$ has at least two connected components. Construct the total space $\tilde{\mathcal{Z}}$ of a nontrivial 2-fold covering of $\mathcal{Z}$ by gluing together two copies of V and two copies of W. Conclude that $\mathcal{Z}$ is not simply-connected.*

E Lifts again

The following theorem and 19.11*b* provide the second part of 19.2; that is, a covering of a connected reasonable space with a simply-connected total space must meet the universal property.

[1] Double arrow $\twoheadrightarrow$ is used to denote surjective maps.

19.13. Theorem. *Let $w\colon (\tilde{\mathcal{Z}}, \tilde{z}_0) \to (\mathcal{Z}, z_0)$ be a pointed covering, and let $f\colon (\mathcal{Y}, y_0) \to (\mathcal{Z}, z_0)$ be a continuous pointed map. Assume $\mathcal{Y}$ is reasonable and simply-connected. Then there exists a unique pointed lift $g\colon (\mathcal{Y}, y_0) \to (\tilde{\mathcal{Z}}, \tilde{z}_0)$.*

$$\begin{tikzcd} & (\tilde{\mathcal{Z}}, \tilde{z}_0) \arrow[d, "w"] \\ (\mathcal{Y}, y_0) \arrow[ur, dashed, "g"] \arrow[r, "f"'] & (\mathcal{Z}, z_0) \end{tikzcd}$$

Before reading the proof, refresh the definition of fibered product from 17E.

Proof. The uniqueness follows from 17.13.

$$\begin{tikzcd} (\tilde{\mathcal{Y}}, \tilde{y}_0) \arrow[r, "\tilde{f}"] \arrow[d, "\tilde{w}"', shift right] & (\tilde{\mathcal{Z}}, \tilde{z}_0) \arrow[d, "w"] \\ (\mathcal{Y}, y_0) \arrow[r, "f"'] \arrow[u, dotted, "s"', shift right] & (\mathcal{Z}, z_0) \end{tikzcd}$$

Consider the fibered product of f and w, described by the diagram. By 17.10, $\tilde{w}\colon \tilde{\mathcal{Y}} \to \mathcal{Y}$ is a covering. Since $\mathcal{Y}$ is reasonable and simply-connected, by 19.10, $\tilde{w}$ is trivial. In particular, there is a continuous pointed map $s\colon (\mathcal{Y}, y_0) \to (\tilde{\mathcal{Y}}, \tilde{y}_0)$ such that $\tilde{w} \circ s = \mathrm{id}_{\mathcal{Y}}$ (the map s is called a pointed section of $\tilde{w}$).

Note that the diagram commutes except $s \circ \tilde{w} \neq \mathrm{id}_{\tilde{\mathcal{Y}}}$. It remains to observe that $g = \tilde{f} \circ s$ is the required lift of f. $\square$

F Existence

19.14. Theorem. *Any connected reasonable space $\mathcal{Z}$ admits a covering $u\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$ with simply-connected total space $\tilde{\mathcal{Z}}$.*

Sketch of proof. We will construct a *set* $\tilde{\mathcal{Z}}$ with a map $u\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$, and then equip $\tilde{\mathcal{Z}}$ with a topology and check that u meets the conditions in the theorem.

The set and the map. Choose a point $z_0 \in \mathcal{Z}$. Denote by $\mathrm{Path}(\mathcal{Z}, z_0)$ the set of all paths that start at z_0; that is, a path $a\colon [0,1] \to \mathcal{Z}$ belongs to $\mathrm{Path}(\mathcal{Z}, z_0)$ if $a(0) = z_0$.

Let $\tilde{\mathcal{Z}}$ be the set of all homotopy classes in $\mathrm{Path}(\mathcal{Z}, z_0)$; that is,

$$\tilde{\mathcal{Z}} = \mathrm{Path}(\mathcal{Z}, z_0)/\sim,$$

where $\sim$ stands for the equivalence relation defined by the homotopy of paths; see 14A. Recall that if $a_0 \sim a_1$ for $a_0, a_1 \in \mathrm{Path}(\mathcal{Z}, z_0)$, then $a_0(1) = a_1(1)$. Therefore, $u\colon [a] \mapsto a(1)$ is a well-defined map $u\colon \tilde{\mathcal{Z}} \to \mathcal{Z}$.

The topology. So far $\tilde{\mathcal{Z}}$ is just a set; this step equips it with a topology.

Let $V\subset\mathcal{Z}$ be an open simply-connected set. Choose a path $a\in$ $\in\mathrm{Path}(\mathcal{Z},z_0)$ that ends in V. Consider the set $\tilde V_{[a]}\subset\tilde{\mathcal{Z}}$ defined by

$$\tilde V_{[a]}=\left\{\,[a*b]\in\tilde{\mathcal{Z}}\;:\;b\text{ is a path in }V\text{ that starts at }a(1)\,\right\}.$$

Note that $\tilde V_{[a]}$ depends only on the homotopy class of a.

The required topology on $\tilde{\mathcal{Z}}$ can be described by the following property: for any open simply-connected set $V\subset\mathcal{Z}$, and any path a from z_0 to a point in V, the set $\tilde V_{[a]}$ is open and u defines a homeomorphism $\tilde V_{[a]}\to V$.[2] Checking that this property uniquely defines a topology is generously left to the reader.

If a and b are two paths from z_0 to a point in V, then

$$\tilde V_{[a]}\cap\tilde V_{[b]}\neq\varnothing\quad\Rightarrow\quad\tilde V_{[a]}=\tilde V_{[b]};$$

again, the proof is left to the reader. In particular, V is evenly covered by u and the sets $\tilde V_{[a]}$ are the slices over V. Since $\mathcal{Z}$ is reasonable, it admits an open cover by simply-connected sets, and this implies that u is a covering.

Let $\tilde z_0=[e_{z_0}]$; that is, the point $\tilde z_0\in\tilde{\mathcal{Z}}$ corresponds to the constant path e_{z_0} in $\mathcal{Z}$. Note that $u(\tilde z_0)=e_{z_0}(1)=z_0$.

Given a path a and $s\in[0,1]$, let $a_s\colon[0,1]\to\mathcal{Z}$ be the path defined by $a_s(t)=a(s{\cdot}t)$; note that a_s runs from $a(0)$ to $a(s)$.[3]

Choose $\tilde z=[a]\in\tilde{\mathcal{Z}}$; here $a\in\mathrm{Path}(\mathcal{Z},z_0)$. By 17.12, there is a unique lift $\tilde a$ of a such that $\tilde a(0)=\tilde z_0$. Observe that $\tilde a(s)=[a_s]$ for any $s\in[0,1]$. In particular, $\tilde a$ is a path from $\tilde z_0$ to $\tilde z$; since $\tilde z$ is arbitrary, we get that $\tilde{\mathcal{Z}}$ is path-connected.

Choose a loop $\tilde a\colon[0,1]\to\tilde{\mathcal{Z}}$ based at $\tilde z_0$. Let $a=u\circ\tilde a$; it is a loop based at z_0. From above, we have $\tilde a(s)=[a_s]$. Note that $\tilde a$ is a loop, $a=a_1$ and $e_{z_0}=a_0$; therefore,

$$[a]=\tilde a(1)=\tilde a(0)=[e_{z_0}].$$

In other words, $a\sim e_{z_0}$. By 17.12, the corresponding homotopy can be lifted to $\tilde{\mathcal{Z}}$ and it contracts $\tilde a$. That is, any loop in $\tilde{\mathcal{Z}}$ is contractible; by 14.9, $\tilde{\mathcal{Z}}$ is simply-connected. □

Recall that topological groups were defined in 14.7.

19.15. Exercise. *Let $u\colon\tilde{\mathcal{G}}\to\mathcal{G}$ be the universal covering of a reasonable topological group $\mathcal{G}$. Show that $\tilde{\mathcal{G}}$ has the structure of a topological group such that u becomes a homomorphism of groups; that is, $u(\tilde g{\cdot}\tilde h)=u(\tilde g){\cdot}u(\tilde h)$ for any $\tilde g$ and $\tilde h$ in $\tilde{\mathcal{G}}$.*

---

[2]In this case, the sets $\tilde V_{[a]}$ for every open simply-connected set $V\subset\mathcal{Z}$ and every path $a\in\mathrm{Path}(\mathcal{Z},z_0)$ that ends in V form a base of the topology on $\tilde{\mathcal{Z}}$.

[3]Note that a_s is not a homotopy of paths, but it is a free homotopy; see 14A.

G An application

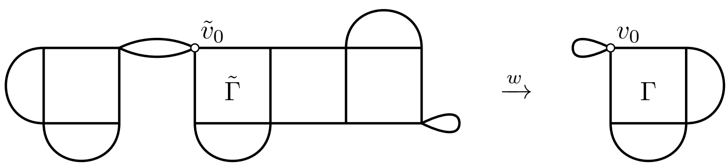
A graph is a topological space built from a discrete set of points called vertices and copies of the interval $[0, 1]$ called edges, where each edge is attached by identifying its two endpoints with vertices.

Groups that are isomorphic to the fundamental groups of graphs are called free.

Free groups are important in group theory. They are usually defined in purely algebraic terms. The following statement, if proved in a purely algebraic way, is not trivial, but as we are about to see, a topological proof takes a few lines. In a perfect world, I should present both definitions and prove their equivalence, but I will skip this part — please forgive this shortcut.

19.16. Theorem. *Any subgroup of a free group is free.*

Proof. Choose a free group $F = \pi_1(\Gamma, v_0)$, where Γ is a graph and v_0 is one of its vertices. Note that any graph is a reasonable space. By 19.4, for any subgroup $H < F$ there is a pointed covering $w\colon (\tilde{\Gamma}, \tilde{v}_0) \to$ $\to (\Gamma, v_0)$ such that $\operatorname{Im} w_* = H$. Recall that w_* is a monomorphism; thus $\pi_1(\tilde{\Gamma}, \tilde{v}_0)$ is isomorphic to H.



Observe that the total space of a covering of a graph is a graph. Hence the statement follows. □

This statement gives some taste of possible applications of coverings in group theory. For more on the subject, read [14] and check the references therein.

Chapter 20

Separation theorem

Let $J \subset \mathbb{R}^2$ be a subset homeomorphic to $\mathbb{S}^1$. The Jordan separation theorem states that the complement $\mathbb{R}^2 \setminus J$ is not connected. This theorem is known for its simple formulation and annoyingly tricky proofs. We will present the proof of Patrick Doyle [8] that is based on Klee's trick. This proof is among the shortest; it also uses several results in topology, which are worth knowing.

A Tietze–Urysohn extension theorem

The following statement is a special case of the so-called Tietze–Urysohn extension theorem, which provides an analogous statement for all normal spaces; see 9E.

20.1. Proposition. *Let Γ be a closed subset of $\mathbb{R}^2$. Then any continuous function $f\colon \Gamma \to \mathbb{R}$ admits a continuous extension to $\mathbb{R}^2$; that is, there is a continuous function $F\colon \mathbb{R}^2 \to \mathbb{R}$ such that $F(p) = f(p)$ for any point $p \in \Gamma$.*

Proof. We can assume that $\Gamma \neq \varnothing$; otherwise take $F(x) \equiv 0$.

Assume $f(\Gamma) \subset (1,2)$. Given $x \in \mathbb{R}^2$, let

$$a(x) = \inf_{y\in\Gamma}\{\,|x-y|\,\} \quad \text{and} \quad b(x) = \inf_{y\in\Gamma}\{\,f(y)\cdot|x-y|\,\}.$$

Observe that

$$a(x) < b(x) < 2\cdot a(x)$$

for any $x \notin \Gamma$. Therefore

$$F(x) = \begin{cases} f(x) & \text{if } x \in \Gamma, \\ \frac{b(x)}{a(x)} & \text{if } x \notin \Gamma \end{cases}$$

defines a function $F\colon \mathbb{R}^2 \to (1,2)$.

It remains to check that F is continuous; that is,

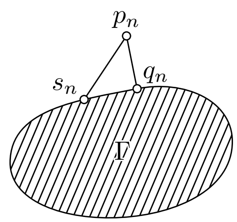
➊ $$F(p_\infty) = \lim F(p_n)$$

for any converging sequence of points $p_n \to p_\infty$ in $\mathbb{R}^2$.

We can assume that $\lim F(p_n)$ is defined; otherwise pass to a subsequence. Furthermore, by the triangle inequality, a and b are continuous functions. Since $a(x) > 0$ for any $x \notin \Gamma$, the function F is continuous in the complement $\mathbb{R}^2 \setminus \Gamma$.

It remains to check the case $p_\infty \in \Gamma$. Furthermore, if $p_n \in \Gamma$ for all large n, then the statement follows from the continuity of f. Therefore, passing to a subsequence, we can assume that $p_n \notin \Gamma$ for any n.

Let $q_n, s_n \in \Gamma$ be points that minimize $|p_n - q_n|$ and $f(s_n)\cdot|p_n - s_n|$, respectively; so



➋ $$\begin{aligned} a(p_n) &= |p_n - q_n|, \\ b(p_n) &= f(s_n)\cdot|p_n - s_n|, \\ F(p_n) &= f(s_n)\cdot\frac{|p_n - q_n|}{|p_n - s_n|}. \end{aligned}$$

The existence of q_n and s_n follows since dist_x and f are continuous on Γ and $\Gamma \cap \bar{\mathrm{B}}[x, R]$ is a nonempty compact set for large R.

Note that

$$\tfrac{1}{2}\cdot|p_n - s_n| \leqslant |p_n - q_n| \leqslant |p_n - p_\infty|;$$

in particular, $q_n \to p_\infty$, $s_n \to p_\infty$, and $f(s_n) \to f(p_\infty)$ as $n \to \infty$. Let

$$\varepsilon_n = \sup\left\{\, |f(p_\infty) - f(x)| \;:\; x \in \Gamma \cap \bar{\mathrm{B}}[\,p_\infty, 2\cdot|p_n - p_\infty|\,]\,\right\};$$

note that $\varepsilon_n \to 0$ as $n \to \infty$ and

$$1 - 2\cdot\varepsilon_n \leqslant \frac{|p_n - q_n|}{|p_n - s_n|} \leqslant 1.$$

It follows that $\frac{|p_n - q_n|}{|p_n - s_n|} \to 1$ as $n \to \infty$; together with ➋, this implies ➊.

In the general case, let $\hat f = h \circ f\colon \Gamma \to (1,2)$ for a homeomorphism $h\colon \mathbb{R} \to (1,2)$. If $\hat F\colon \mathbb{R}^2 \to (1,2)$ is an extension of $\hat f$, then $F = h^{-1} \circ \hat F$ is the required extension of f. □

20.2. Exercise. *Verify that the proof goes thru for any metric space in which every closed ball is compact. Try to modify the proof to make it work in arbitrary metric spaces.*

B Around van Kampen

20.3. Lemma. *Let us identify $\mathbb{R}^2$ with the (x,y)-plane in $\mathbb{R}^3$.*

Suppose $\Gamma \subset \mathbb{R}^2$ is a closed set. Then the complement $X = \mathbb{R}^2 \setminus \Gamma$ is connected if and only if the complement $Y = \mathbb{R}^3 \setminus \Gamma$ is simply-connected.

The proof is based on 14.11 and 19.12.

Proof of 20.3. Denote by Γ_- (respectively Γ_+) the sets that include Γ and the points below (respectively above) Γ; that is,

$$\Gamma_- = \{\, (x,y,z) \,:\, (x,y) \in \Gamma \text{ and } z \leqslant 0 \,\},$$
$$\Gamma_+ = \{\, (x,y,z) \,:\, (x,y) \in \Gamma \text{ and } z \geqslant 0 \,\}.$$

Consider the complements $V = \mathbb{R}^3 \setminus \Gamma_-$ and $W = \mathbb{R}^3 \setminus \Gamma_+$. Note that $X \times \mathbb{R} = V \cap W$ and $Y = V \cup W$.

The sets V and W are contractible; in particular, simply-connected. Indeed, the horizontal plane $z = 1$ is a deformation retract of V; a retraction can be defined by $(x,y,z) \mapsto (x,y,1)$ and the following homotopy shows that it is homotopic to the identity map:

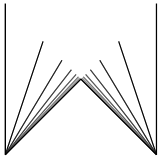
$$h_t(x,y,z) = (x,y,(1-t)\cdot z + t).$$

The plane $z = 1$ is contractible, hence so is V. Similarly, W is contractible since it has the plane $z = -1$ as a deformation retract.

If X is not connected, then so is $X \times \mathbb{R} = V \cap W$. In this case, by 19.12, $\mathbb{R}^3 \setminus \Gamma = V \cup W$ is not simply-connected.

On the other hand, if X is connected, then so is $X \times \mathbb{R} = V \cap W$. In this case, by 14.11, we have $\mathbb{R}^3 \setminus \Gamma = V \cup W$ is simply-connected. $\square$

20.4. Exercise. *Suppose $\Gamma \subset \mathbb{R}^2$, $X = \mathbb{R}^2 \setminus \Gamma$, and $Y = \mathbb{R}^3 \setminus \Gamma$ are as in 20.3. Show that if X has at least 3 connected components, then $\pi_1(Y) \not\simeq \mathbb{Z}$.*



Comments. If we do not assume that Γ is closed, then its complement $X = \mathbb{R}^2 \setminus \Gamma$ might be the flea and comb discussed in 12C. In this case, $Y = \mathbb{R}^3 \setminus \Gamma$ is the union of upper and lower open half-spaces with $X \times \{0\}$. The set X is connected (altho not path-connected), but $Y = \mathbb{R}^3 \setminus \Gamma$ is not simply-connected. This statement might look evident, but the actual proof is quite involved.

If the complement $X = \mathbb{R}^2 \setminus \Gamma$ is the union of two copies of the fan F (see 13.5) as shown in the picture then it is path-connected, but

$Y = \mathbb{R}^3 \setminus \Gamma$ is not simply-connected. The latter statement is closely related to 19.9; try to bootstrap a noncontractible loop in it and prove that it is not contractible.

C Klee trick

20.5. Proposition. *Suppose $\Gamma \subset \mathbb{R}^2$ is a closed subset homeomorphic to $\mathbb{R}$. Then there is a homeomorphism $\mathbb{R}^3 \circlearrowleft$ that maps Γ to the z-axis.*

The following proof uses the so-called Klee trick, which is quite useful in many topological problems. In fact, the construction gives a bit more; it implies the existence of a one-parameter family of homeomorphisms $h_t\colon \mathbb{R}^3 \circlearrowleft$ that starts at the identity map $\mathrm{id}_{\mathbb{R}^3}$ and ends at the homeomorphism we need.

Proof. Let $h\colon t \mapsto (a(t), b(t))$ be a homeomorphism $\mathbb{R} \to \Gamma$. Applying 20.1 to $f = h^{-1}$, we get a function $F\colon \mathbb{R}^2 \to \mathbb{R}$ such that

$$F(a(t), b(t)) = t$$

for any $t \in \mathbb{R}$.

Note that the map

$$p\colon (x, y, z) \mapsto (x, y, z + F(x, y))$$

is a homeomorphism. Indeed, this map is continuous and its inverse

$$p^{-1}\colon (x, y, z) \mapsto (x, y, z - F(x, y))$$

is continuous as well.

Similarly, the map

$$q\colon (x, y, z) \mapsto (x - a(z), y - b(z), z)$$

is a homeomorphism as well. Indeed, q is continuous and it has an inverse

$$q^{-1}\colon (x, y, z) \mapsto (x + a(z), y + b(z), z)$$

that is continuous as well.

It follows that the composition $q\circ p\colon \mathbb{R}^3 \to \mathbb{R}^3$ is a homeomorphism. Since $F(a(t), b(t)) = t$ for any t, we have

$$q \circ p(a(t), b(t), 0) = q(a(t), b(t), t) = (0, 0, t).$$

It follows that $q \circ p$ sends Γ to the z-axis as required. $\square$

20.6. Exercise. *Let $I \subset \mathbb{R}^2$ be a subset homeomorphic to $[0,1]$. Construct a homeomorphism $h\colon \mathbb{R}^3 \to \mathbb{R}^3$ that maps I to a line segment.*

20.7. Exercise. *Suppose $h\colon A \to B$ is a homeomorphism between closed subsets $A, B \subset \mathbb{R}$. Let us consider $\mathbb{R}$ as a subset of the plane $\mathbb{R}^2$. Show that h can be extended to a homeomorphism of the plane; that is, there is a homeomorphism $H\colon \mathbb{R}^2 \circlearrowleft$ such that $H(a) = h(a)$ for any $a \in A$.*

D Separation by a proper curve

20.8. Theorem. *Suppose $\Gamma \subset \mathbb{R}^2$ is a closed set homeomorphic to $\mathbb{R}$. Then $\mathbb{R}^2 \setminus \Gamma$ has at least two connected components.*

Note that the assumption that Γ is closed is necessary; indeed, a finite open interval I of a line in $\mathbb{R}^2$ is homeomorphic to $\mathbb{R}$, but its complement $\mathbb{R}^2 \setminus I$ is connected.

Proof. If $\mathbb{R}^2 \setminus \Gamma$ is connected, then by 20.3, $\mathbb{R}^3 \setminus \Gamma$ is simply-connected.

On the other hand, by 20.5, $\mathbb{R}^3 \setminus \Gamma$ is homeomorphic to the complement of the z-axis in $\mathbb{R}^3$, which is not simply-connected since it can be retracted to a circle[1] — a contradiction. □

20.9. Exercise. *Let A and B be disjoint closed subsets in $\mathbb{R}^2$ such that both complements $\mathbb{R}^2 \setminus A$ and $\mathbb{R}^2 \setminus B$ are connected. Show that $\mathbb{R}^2 \setminus (A \cup B)$ is connected.*

20.10. Exercise. *Suppose $I \subset \mathbb{R}^2$ is a subset homeomorphic to $[0,1]$. Show that $\mathbb{R}^2 \setminus I$ is connected.*

20.11. Exercise. *In the assumptions of 20.8, show that $\mathbb{R}^2 \setminus \Gamma$ has exactly two connected components.*

E Jordan's theorem

20.12. Theorem. *If $J \subset \mathbb{S}^2$ is a subset homeomorphic to $\mathbb{S}^1$, then $\mathbb{S}^2 \setminus J$ has at least two connected components.*

Proof. Choose a point $p \in J$. Note that $\Gamma = J \setminus \{p\} \simeq \mathbb{R}$, and it is a closed subset in $\mathbb{S}^2 \setminus \{p\} \simeq \mathbb{R}^2$. It remains to apply 20.8. □

[1] Alternatively one may apply 20.3 again.

Note that applying 20.11 in the proof of 20.12, one gets that $\mathbb{S}^2 \setminus J$ has exactly two connected components. Furthermore, this theorem has an even more precise version due to Arthur Schoenflies; it states that there is a homeomorphism $h\colon \mathbb{S}^2 \circlearrowleft$ that maps J to the equator. In particular, each connected component of $\mathbb{S}^2 \setminus J$ is homeomorphic to an open disc.

20.13. Exercise.

(a) Let $I \subset \mathbb{S}^2$ be a subset homeomorphic to $[0,1]$. Show that $\mathbb{S}^2 \setminus I$ is connected.

(b) Use part (a) to show that if $J \subset \mathbb{S}^2$ is as in 20.12, then J is the boundary of each connected component of $\mathbb{S}^2 \setminus J$.

20.14. Advanced exercise. *Let A and B be closed subsets in $\mathbb{S}^2$ such that $\mathbb{S}^2 \setminus A$ and $\mathbb{S}^2 \setminus B$ are connected and $\mathbb{S}^2 \setminus (A \cap B)$ is simply-connected. Show that $\mathbb{S}^2 \setminus (A \cup B)$ is connected.*

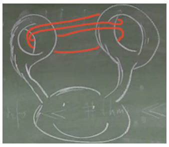
Appendix A

Semisolutions

0.1. Topologically speaking, a T-shirt is a disc with three holes, two for the hands and one for the head; pants have two holes, one for each leg. In an ideal universe socks should not have holes, but apparently our universe is not ideal.

0.2. He saw a hole in the bottom, so the topology of the mug is the same as a donut (topologically speaking, a solid torus).

0.3. Draw the hair tie on each picture, the last one is shown.



Source. Suggested by Rostislav Matveev.

0.4; *(a).* Push the loop up thru the other handle, bring it around over the points and back over the handles; the string will come off.

(b). Take the center of the string holding your wrists, push it up thru one of the loops on your friend's wrist and bring it down over his hand.

(c). Use the same trick as in *(b).*

Source. Part *(a)* and *(b)* appear in Harry Houdini's book [17]. A version of part *(a)* appears earlier in the family of Sam Loyd [21] under the name "The Gordian knot". Part *(b)* under the name "The prisoners' release puzzle" appears in Cassell's book [7] published in 1881. Both puzzles should be much older.

1.2. Check the triangle inequality for 0, $\frac{1}{2}$, and 1.

1.3. Check the conditions in 1.1.

1.4. Check all the conditions in Definition 1.1. Further we discuss the triangle inequality — the remaining conditions are nearly evident.

Let $a = (x_a, y_a)$, $b = (x_b, y_b)$, and $c = (x_c, y_c)$. Set

$$x_1 = x_b - x_a, \qquad y_1 = y_b - y_a,$$
$$x_2 = x_c - x_b, \qquad y_2 = y_c - y_b.$$

(a). The inequality

$$|a - c|_1 \leqslant |a - b|_1 + |b - c|_1$$

can be written as

$$|x_1 + x_2| + |y_1 + y_2| \leqslant |x_1| + |y_1| + |x_2| + |y_2|.$$

The latter follows since $|x_1 + x_2| \leqslant |x_1| + |x_2|$ and $|y_1 + y_2| \leqslant |y_1| + |y_2|$.

(b). The inequality

❶ $$|a - c|_2 \leqslant |a - b|_2 + |b - c|_2$$

can be written as

$$\sqrt{(x_1 + x_2)^2 + (y_1 + y_2)^2} \leqslant$$
$$\leqslant \sqrt{x_1^2 + y_1^2} + \sqrt{x_2^2 + y_2^2}.$$

Take the square of the left and the right-hand sides, simplify, take the square again and simplify again. You should get the following inequality:

$$0 \leqslant (x_1 \cdot y_2 - x_2 \cdot y_1)^2,$$

which is equivalent to ❶ and evidently true.

(c). The inequality

$$|a - c|_\infty \leqslant |a - b|_\infty + |b - c|_\infty$$

can be written as

$$❷\quad \max\{|x_1+x_2|,|y_1+y_2|\} \leqslant \\ \leqslant \max\{|x_1|,|y_1|\}+\max\{|x_2|,|y_2|\}.$$

Without loss of generality, we may assume that

$$\max\{|x_1+x_2|,|y_1+y_2|\} = |x_1+x_2|.$$

Further,

$$|x_1+x_2| \leqslant |x_1|+|x_2| \leqslant \\ \leqslant \max\{|x_1|,|y_1|\}+\max\{|x_2|,|y_2|\}.$$

Hence ❷ follows.

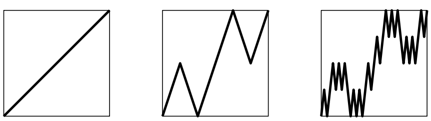
1.6. Show that the triangle inequality implies that $|f(x)-f(y)| < \varepsilon$ if $|x-y|_{\mathcal{X}} < \varepsilon$; draw a conclusion.

1.8. *(a)* Show that the triangle inequality implies that $|f(x)-f(y)|_{\mathcal{Y}} < \varepsilon$ if $|x-y|_{\mathcal{X}} < \varepsilon$; draw a conclusion.

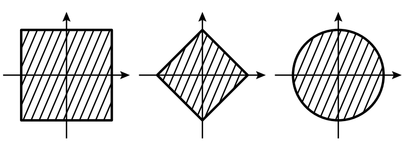
(b). Apply 1.1b.

1.9. Show and use that in 1.5 one can take $\delta = 1$ for any $\varepsilon > 0$.

1.10. Learn about space-filling curves and think. Alternatively, guess the pattern in the following graphs and pass to the limit.



1.11. Figure out which is which.



1.12. Apply the triangle inequality to x, y, and $z \in \mathrm{B}(y,R)\setminus \mathrm{B}(x,r)$. For the second part, consider the balls $\mathrm{B}(2,3)$ and $\mathrm{B}(0,4)$ in $[0,\infty)$.

Comment. Note that we used that $\mathrm{B}(y,R) \neq \neq \mathrm{B}(x,r)$, without this condition there are no general restrictions on r in terms of R. For example, the inclusion $\mathrm{B}(x,1000) \subset \mathrm{B}(y,1)$ holds in the discrete space.

1.14. Spell the definitions.

1.17; *(a)+(b).* Apply 1.15.

(c). The if part follows from 1.16 and part *(a).* It remains to prove the only-if part.

Let V be an open set. By 1.15, for any $x \in V$ there is $r_x > 0$ such that $\mathrm{B}(x,r_x) \subset V$. Observe that

$$V = \bigcup_{x\in V} \mathrm{B}(x,r_x).$$

1.18. Consider the open segments $(-\varepsilon,\varepsilon)$ for all $\varepsilon > 0$ in $\mathbb{R}$. Note that

$$\{0\} = \bigcap_{\varepsilon>0} (-\varepsilon,\varepsilon)$$

and the one-point set $\{0\}$ is not open.

1.19. Show and use that

$$\mathrm{B}(x,r)_1 \subset \mathrm{B}(x,r)_2 \subset \mathrm{B}(x,r)_\infty \subset \mathrm{B}(x,2\cdot r)_1;$$

here $\mathrm{B}(x,r)_1$, $\mathrm{B}(x,r)_2$, and $\mathrm{B}(x,r)_\infty$ denote the balls in the metrics $|*-*|_1$, $|*-*|_2$, and $|*-*|_\infty$ respectively.

1.21. Look at the image of $\mathbb{R}$ under the function $x \mapsto |x|$.

1.23. Assume the contrary; that is, a sequence $x_1, x_2, \ldots$ has two limits y and z. Set $r = |y-z|$. Note that $\mathrm{B}(y,\frac{r}{2})$ contains all but finitely many elements of the sequence $x_1, x_2, \ldots$; the same holds for $\mathrm{B}(z,\frac{r}{2})$. Observe that $\mathrm{B}(y,\frac{r}{2}) \cap \cap \mathrm{B}(z,\frac{r}{2}) = \varnothing$ and arrive at a contradiction.

1.24. Suppose f is not continuous. This means that there are x_∞ and $\varepsilon > 0$ such that for every n there is a point $x_n \in \mathrm{B}(x_\infty,\frac{1}{n})$ with $|f(x_n)-f(x_\infty)| > \varepsilon$. In particular, $y_n = f(x_n)$ does not converge to $y_\infty = f(x_\infty)$. It proves the if part of the exercise.

To prove the only-if part, suppose that there is a sequence $x_n \to x_\infty$ such that $y_n \not\to y_\infty$ as $n \to \infty$. Note that in this case we can pass to a subsequence so that $x_n \in \mathrm{B}(x_\infty,\frac{1}{n})$ and $|y_n - y_\infty| > \varepsilon$ for some fixed $\varepsilon > 0$. From above, f is not continuous.

1.25. Show that the semiopen interval $[0,1)$ is neither open nor closed in $\mathbb{R}$.

1.26. Choose a point $z \in \bar{\bar{A}}$. It means that there is a sequence $y_1, y_2, \ldots \in \bar{A}$ such that $y_n \to z$ as $n \to \infty$. The latter means that for each y_i there is a sequence $x_{i,1}, x_{i,2}, \ldots \in A$ such that $x_{i,n} \to y_i$ as $n \to \infty$. Try to choose a sequence of integers m_n such that $x_{n,m_n} \to z$ as $n \to \infty$. Draw a conclusion.

1.27. Show that Q is closed if and only if $x \in Q$ whenever $\mathrm{B}(x,\varepsilon) \cap Q \neq \varnothing$ for any $\varepsilon > 0$. Show that the latter is equivalent to $y \in V$ if and only if $\mathrm{B}(y,\varepsilon) \subset V$ for some $\varepsilon > 0$. Draw a conclusion.

2.2. Let $\mathcal{X}$ be an infinite set with cofinite topology. Show that any two nonempty open sets in $\mathcal{X}$ have nonempty intersection. Show that the latter does not hold for open balls in a metric space with at least two points.

2.3. Let $\mathcal{F}$ be a finite metric space. Observe that there is $\varepsilon > 0$ such that $|x-y| > \varepsilon$ for any two distinct points $x, y \in \mathcal{F}$. It follows that $\{x\} = \mathrm{B}(x,\varepsilon)$ for any $x \in \mathcal{F}$; in particular, each one-point set is open. By 2.1*b*, any set in $\mathcal{F}$ is open.

2.4. Choose $W \in \mathscr{W}$. By assumption, for any $w \in W$ there is $S_w \in \mathscr{S}$ such that $W \supset S_w \ni w$. Observe that

$$W = \bigcup_{w \in W} S_w.$$

It follows that $\mathscr{W} \subset \mathscr{S}$; in other words, any $\mathscr{W}$-open set is $\mathscr{S}$-open.

2.6; *(a)*. Consider the function defined by

$$f(x) = \begin{cases} a & \text{if } x < 0, \\ b & \text{if } x \geqslant 0. \end{cases}$$

(b). Suppose $f\colon \mathcal{X} \to \mathbb{R}$ is nonconstant; that is, $f(a) \neq f(b)$. Note that $W = \mathbb{R} \setminus \{f(a)\}$ is an open set containing b. Assume f is continuous. Then $\{b\} = f^{-1}(W)$ is an open set — a contradiction.

2.8. Note that $S \in \mathscr{T}$ if and only if for any $x_0 \leqslant x_1$, if $x_0 \in S$, then $x_1 \in S$.

(a). Check the conditions in 2.1 using the property above.

(b). Show that every two nonempty open sets in $(\mathbb{R}, \mathscr{T})$ intersect. Show that the latter statement does not hold in a metric space with at least two points. Draw a conclusion.

(c). To do the only-if part check the condition in 2.5.

To do the if part, suppose f is not nondecreasing; that is, we can find $x_0 < x_1$ such that $f(x_0) > f(x_1)$. Note that the inverse image $V = f^{-1}([f(x_0), \infty))$ contains x_0 but does not contain x_1. Conclude that V is not open in $(\mathbb{R}, \mathscr{T})$, so $f\colon (\mathbb{R}, \mathscr{T}) \to (\mathbb{R}, \mathscr{T})$ is not continuous.

2.9. Apply 2.7 together with the fact that $x \mapsto |x|$ is a continuous real-to-real function (the latter is assumed to be known from calculus).

2.11+2.12. Apply the definition of the convergence.

3.3; *(a)* Choose a metric that induces the topology on $\mathcal{X}$. Let W_n be the union of $\frac{1}{n}$-balls centered at points in Q. Show that W_n is open and that $Q = \bigcap W_n$.

(b). Try to find the needed closed set in the connected two-point space; see 2B.

3.4. Suppose V is an open subset and Q is its complement. Recall that Q is closed; see 3A. Show and use that $V \subset A$ if and only if $Q \supset B$.

3.5. Apply the definitions of neighborhood and dense set.

3.6; Show and use the following for any two subsets A and B:

- $\mathring{\mathring{A}} = \mathring{A} \subset A \subset \bar{A} = \bar{\bar{A}}$.
- if $A \subset B$, then $\bar{A} \subset \bar{B}$ and $\mathring{A} \subset \mathring{B}$.

For the second part, try to choose a subset A in $\mathbb{R}$ so that it meets the following conditions:

- A and $\mathbb{R} \setminus A$ contain isolated points,
- A and $\mathbb{R} \setminus A$ contain intervals,
- A and $\mathbb{R} \setminus A$ are dense in some interval.

3.7–3.9. Apply the definitions of boundary and closed set.

3.10. Let A be a subset of a topological space $\mathcal{X}$. Denote by B its complement; that is $B = \mathcal{X} \setminus A$. Show and use that the statement is equivalent to the following

$$\mathcal{X} \setminus \partial A = \mathring{A} \cup \mathring{B}.$$

3.12. Spell the definitions. The needed examples can be found for $\mathcal{A} = [0,1) \subset \mathbb{R} = \mathcal{X}$.

3.13. Set $a = f|_A$ and $b = f|_B$. Note that for any set $S \subset \mathcal{Y}$, we have

$$\begin{aligned} a^{-1}(S) &= f^{-1}(S) \cap A, \\ b^{-1}(S) &= f^{-1}(S) \cap B, \\ f^{-1}(S) &= a^{-1}(S) \cup b^{-1}(S). \end{aligned}$$

(a). Choose an open set $W \subset \mathcal{Y}$. Show that $a^{-1}(W)$ and $b^{-1}(W)$ are open in $\mathcal{X}$. Conclude that $f^{-1}(W)$ is open.

(b). Choose a closed set $Q \subset \mathcal{Y}$. Use 3.2 to show that $a^{-1}(Q)$ and $b^{-1}(Q)$ are closed. Apply 3.1*c* to show that $f^{-1}(Q)$ is closed. Apply 3.2 again.

(c). Consider a function f that is constant on disjoint sets A and B such that $\bar{A} \cap B \neq \varnothing$.

4.1. Use induction to show that $\mathcal{C}_n$ is the set of reals $x \in [0,1]$ that can be written in base 3 without 1 in the first n digits.

For the second part, show that every $x \in \mathcal{C}$ can be uniquely written in base 3 without using digit 1. (Note that $(0.0222\ldots)_3 = (0.1)_3$, so in general there is no uniqueness.) Use the last statement to construct a bijection from $\mathcal{C}$ to Power($\mathbb{N}$), which is the set of all subsets of the set of positive integers $\mathbb{N}$. By Cantor's theorem, Power($\mathbb{N}$) has greater cardinality than $\mathbb{N}$.

4.2. Denote by V_n the open set that is removed from $\mathcal{C}_{n-1}$ on the n-th step of construction of the Cantor set. So V_n is a union of 2^{n-1} open segments of length $\frac{1}{3^n}$ each. Consider three sets

$$\begin{aligned} W_1 &= V_1 \cup V_4 \cup V_7 \cup \ldots \\ W_2 &= V_2 \cup V_5 \cup V_8 \cup \ldots \\ W_3 &= V_3 \cup V_6 \cup V_9 \cup \ldots \end{aligned}$$

Show and use that $\partial W_i = \mathcal{C}$ for every i.

5.1. Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a homeomorphism. By the definition of inverse, we have $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$ for any $x \in \mathcal{X}$ and $y \in \mathcal{Y}$. Therefore, the existence of $f^{-1}\colon \mathcal{Y} \to \mathcal{X}$ implies that f is a bijection $\mathcal{X} \leftrightarrow \mathcal{Y}$.

For the second part, try to find a continuous bijection for the subspaces $A = [0,1) \cup \{2\}$ and $B = [0,1]$ of $\mathbb{R}$.

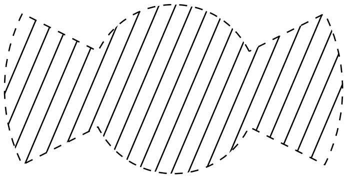
5.2. Observe that $y \mapsto \ln y$ is the inverse of $x \mapsto e^x$, and show that both functions are continuous. (You can use that differentiable functions are continuous.)

5.3. Try to build the needed function from $x \mapsto \arctan x$ or $x \mapsto e^{-e^x}$.

5.4. Apply the definitions of homeomorphism and 2.7.

5.5+5.6. Learn about inversion transformation and try to apply it.

5.7. Let V be an open star-shaped set with respect to the origin. Assume V can be described by the inequality $r < f(\theta)$ in the polar (r,θ)-coordinates, where $f\colon \mathbb{S}^1 \to \mathbb{R}$ is a continuous function. In this case it is not hard to prove the statement. But a general star-shaped set, for example the one on the diagram is problematic.



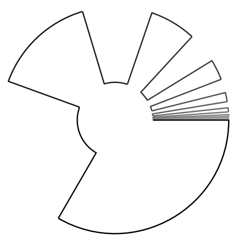
To do the general case, show that V can be presented as a union of a nested sequence of open sets $V_0 \subset V_1 \subset \ldots$ such that each V_n can be described by the inequality $r < f_n(\theta)$ in the polar (r,θ)-coordinates with continuous $f_n\colon \mathbb{S}^1 \to \mathbb{R}$. We can assume that V_0 is a round disc around the origin.

Further construct a sequence of homeomorphisms $\varphi_n\colon V_{n-1} \to V_n$ such that the compositions $\Phi_n = \varphi_n \circ \cdots \circ \varphi_1\colon V_0 \to V_n$ stabilize for each $x \in V_0$; that is, $\Phi_n(x)$ is a fixed point for all sufficiently large n. Set

$$\Phi(x) = \lim_{n\to\infty} \Phi_n(x),$$

and show that Φ defines the needed homeomorphism $V_0 \leftrightarrow V$.

Comment. Note that the boundary of V might not be homeomorphic to the circle; an example can be guessed from the picture.



5.8. Suppose that the sets are $P = \{p_1, p_2, \ldots\}$ and $Q = \{q_1, q_2, \ldots\}$. Try to construct a sequence of homeomorphisms $\Phi_n\colon \mathbb{R}^2 \leftrightarrow \mathbb{R}^2$ such that Φ_n converges to a homeomorphism $\Phi\colon \mathbb{R}^2 \leftrightarrow \mathbb{R}^2$ and for any n we have $\Phi_n(\{p_1,\ldots,p_n\}) \subset Q$ and $\Phi_n^{-1}(\{q_1,\ldots,q_n\}) \subset P$.

5.11. Try the maps between two-point spaces with appropriate topologies.

5.12. Construct a nonempty closed nowhere dense set K without isolated points. Map K to 0 and map each component of $\mathbb{R} \setminus K$ by a homeomorphism to $\mathbb{R}$.

Comment. If a function $f\colon \mathbb{R} \to \mathbb{R}$ is closed and open, then it has to be continuous; see the paper by Ivan Baggs [1].

6.3. Apply the observation together with the facts that $(x,y) \mapsto x+y$, $(x,y) \mapsto x{\cdot}y$, and $(x,y) \mapsto \max\{x,y\} = \frac{1}{2}{\cdot}|x+y| + \frac{1}{2}{\cdot}|x-y|$ are continuous functions on $\mathbb{R}^2$.

6.4. Show and use that $\overline{A \times B} = \bar{A} \times \bar{B}$ and $(A \times B)^\circ = \mathring{A} \times \mathring{B}$.

6.5. Check the following function

$$f(x,y) = \begin{cases} 0 & \text{if} \quad x = 0 \quad \text{or} \quad y = 0, \\ \frac{x}{y} & \text{if} \quad 0 < x \leqslant y, \\ \frac{y}{x} & \text{if} \quad 0 < y < x, \end{cases}$$

6.6. Spell the definitions.

6.10. Let $B_1 = \{a_1, a_1 + d_1, a_1 + 2{\cdot}d_1, \dots\}$ and $B_2 = \{a_2, a_2 + d_2, a_2 + 2{\cdot}d_2, \dots\}$. Show that if $a \in B_1 \cap B_2$, then

$$\{\, a, a+d, a+2{\cdot}d, \dots \,\} \subset B_1 \cap B_2,$$

where $d = d_1{\cdot}d_2$. Apply 6.9.

Show that the set $\{1\}$ is closed but not open.

Comment. This topology was introduced by Harry Furstenberg [11]; he used it to deduce from the last statement that the set of primes is infinite. The resulting proof is not truly new; it is just the classical proof from Euclid's Elements written in the topological language.

6.12; *(a)* Observe that $(V \times \mathcal{Y}) \cap (\mathcal{X} \times W) = {} = V \times W$. Further, show and use that all the sets $V \times W$ for open subsets $V \subset \mathcal{X}$ and $W \subset \mathcal{Y}$ form a base in $\mathcal{X} \times \mathcal{Y}$.

(b). To show that the map F is continuous, apply 6.11 to the prebase described before the exercise. Further, show and use that projection $G\colon (x, f(x)) \to x$ is a continuous left inverse; that is, $G(F(x)) = x$ for any x.

6.13. Show that *every two disjoint closed sets of a metric space have disjoint open neighborhoods*; that is, for any two closed sets A and B there are open sets $V \supset A$ and $W \supset B$ such that $V \cap W = \varnothing$. (Topological spaces that share this property are called n o r m a l; so you need to show that *any metrizable space is normal.*)

Observe that an arithmetic progression is a closed set in the initial topology. Construct two disjoint arithmetic progressions that do not admit disjoint open neighborhoods.

6.14. Check the conditions in 2.1 directly.

6.15+6.16. Spell out the definitions.

6.18. Check the conditions in the definitions of equivalence relation and equivalence class.

6.19. Choose a subset $A \subset [0,1]$ with nonempty complement and let $x \sim y$ if $x, y \in A$ or $x, y \notin A$. For part *(a)*, choose A to be open; for part *(b)* choose a set that is neither open nor closed in $[0,1]$.

6.20. Show that it has three points $a = [0]$, $b = [\frac{1}{2}]$, and $c = [1]$ and the open sets are

$$\varnothing,\ \{b\},\ \{a,b\},\ \{b,c\},\ \text{and}\ \{a,b,c\}.$$

6.21. Note that all positive numbers are in one $\mathbb{R}_+$-orbit. Similarly, all negative numbers are in one orbit and 0 forms another orbit. Thus, $\mathbb{R}/\mathbb{R}_+$ contains three points, say p, n, and z that correspond to positive numbers, negative numbers, and zero. It remains to describe all open subsets in $\{p, n, z\}$.

6.22. Recall that

$$f(x) = [x] = G{\cdot}x = \{\, g{\cdot}x \,:\, g \in G \,\}$$

Show that for any set $V \subset \mathcal{X}$ we have

$$f^{-1} \circ f(V) = \bigcup_{g \in G} g{\cdot}V.$$

(a). Apply this formula to show that if V is open, then so is $f^{-1} \circ f(V)$. Finally apply the definition of quotient topology.

(b). Apply this formula to show that if G is finite and V is closed, then so is $f^{-1} \circ f(V)$.

7.2. Apply 2.1b, 2.1c and 2.5.

7.4. Assume that the space is nonempty; otherwise there is nothing to prove. Choose a nonempty set V_0 from the cover. Its complement is a finite set, say $\{x_1, \dots, x_n\}$. For each x_i choose a set $V_i \ni x_i$ from the cover. Observe that $\{V_0, V_1, \dots, V_n\}$ is a subcover.

7.6; *(a).* Check subsets in a concrete space.

(b). Apply the definition of compact set.

(c). Choose a noncompact space $\mathcal{X}$ and the consider topology on the union $\mathcal{X} \cup \{a,b\}$ that includes $\mathcal{X} \cup \{a,b\}$, $\mathcal{X} \cup \{a\}$, $\mathcal{X} \cup \{b\}$ and all open sets in $\mathcal{X}$. Observe that the sets $\mathcal{X} \cup \{a\}$ and $\mathcal{X} \cup \{b\}$ are compact, but their intersection is not.

(d). If A is a subset of metric space $\mathcal{M}$ that is not closed, then we can choose a point $p \in \bar{A} \backslash A$. Check the definition of compactness for complements $V_n = \mathcal{M} \setminus \bar{\mathrm{B}}[p, \frac{1}{n}]$.

For the second part, try to use the argument in 7.13.

7.8. Apply the finite intersection property.

7.9. Choose a family of closed sets in $\mathcal{X}$, say $\{Q_\alpha\}$, with finite intersection property. Assume $\bigcap Q_\alpha = \varnothing$. Since $f^{-1}\{y\}$ is compact, for any $y \in \mathcal{Y}$ we can choose a finite family $Q_{\alpha_1}, \dots, Q_{\alpha_n}$ such that $f^{-1}\{y\} \cap Q_{\alpha_1} \cap \dots \cap Q_{\alpha_n} = \varnothing$. Let $C_y = Q_{\alpha_1} \cap \dots \cap Q_{\alpha_n}$. Show that (1) $y \notin f(C_y)$ and (2) $f(C_y)$ is a closed set for every $y \in \mathcal{Y}$.

Apply compactness of $\mathcal{Y}$ to choose a finite set of points $y_1, \dots, y_m \in \mathcal{Y}$ such that $f(C_{y_1}) \cap \dots \cap f(C_{y_m}) = \varnothing$. Conclude that $C_{y_1} \cap \dots \cap C_{y_m} = \varnothing$ and arrive at a contradiction.

7.12. Apply 7.11 and 7.5.

7.14. Apply 7.13 and 7.10.

7.16. Consider the map $h\colon \mathbb{R} \cup \{\infty\} \to \mathbb{S}^1$ defined by $x \mapsto (\frac{2\cdot x}{1+x^2}, \frac{1-x^2}{1+x^2})$ and $\infty \mapsto (0,-1)$. Show that h is a homeomorphism $\mathbb{R}^\infty \to \mathbb{S}^1$.

7.18. Apply 7.11 to the projections $\mathcal{X}\times\mathcal{Y} \to \mathcal{X}$ and $\mathcal{X}\times\mathcal{Y} \to \mathcal{Y}$ (which are continuous).

7.20. Spell the definitions.

7.22. The last statement is wrong. Show by example that in general, the products $V_{\beta_i} \times W_{\beta_i}$ do *not* cover $\mathcal{X}\times\mathcal{Y}$.

7.23; *(a)*. Write $\mathbb{S}^1$ is the image of closed interval under a continuous map, and apply 7.11.

(b). Use that $\mathbb{S}^1$ is a closed subset of $[-1,1]\times$ $\times[-1,1]$.

7.24. By 3.2, it is sufficient to show that any closed set $A \subset \mathcal{Y}$ has closed inverse image $B = f^{-1}(A) \subset \mathcal{X}$.

Observe that the set $C = \Gamma \cap (\mathcal{X} \times A)$ is closed, so its complement U can be presented as a union $\bigcup V_\alpha \times W_\alpha$.

Suppose B is not closed; choose a point $p \in$ $\in \bar{B}\backslash B$. Note that $\{p\}\times\mathcal{Y}$ is a compact set in U. Argue as in 7.17 to prove that there is an open set $N_p \ni p$ such that $N_p \times \mathcal{Y} \subset U$. Arrive at a contradiction.

Remark. This is a part of the closed graph theorem; compare to 9.6 and 6.12*b*. The following function $f\colon \mathbb{R}\to\mathbb{R}$ has closed graph, but is not continuous:

$$f(x) = \begin{cases} \frac{1}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

It shows that compactness of $\mathcal{Y}$ is a necessary assumption.

7.26. If there is a converging subsequence $\mathfrak{p}_{i_1}$, $\mathfrak{p}_{i_2}, \dots$, then we can choose a set $S \subset \mathbb{N}$ (which means $S \in \mathcal{A}$) such that $i_n \in S$ if n is even and otherwise $i_n \notin S$. Observe that $\mathfrak{p}_{i_n} \in \mathfrak{W}_S$ if n is even and otherwise $\mathfrak{p}_{i_n} \in \mathfrak{V}_S = \mathcal{B}\setminus\mathfrak{W}_S$. Draw a conclusion.

8.2. Consider an infinite set of points with discrete metric.

8.3. Let $r = \varepsilon - \varepsilon'$. Show the existence of a finite cover by r-balls. For each r-ball choose an element of the covering that contains it. Show that the obtained subcover has Lebesgue number ε'.

On the other hand, choose small $\varepsilon > 0$ and consider an open cover by *all* ε-balls of $[0,1]$. Show that this cover does not have a finite subcover with Lebesgue number ε.

8.4. Given a sequence $z_n = (x_n, y_n)$, pass to a subsequence such that the x-coordinate converges. Pass to a subsequence again to make the y-coordinate converge. Show and use that the obtained sequence converges in $\mathcal{X}\times\mathcal{Y}$.

8.10; *(a)*. Choose an open cover $\{V_\alpha\}$. Let W be the union of all intervals (a,b) such that $(a,b)\subset V_\alpha$ for some α and $a,b\in\mathbb{Q}$. Show and use that $\mathcal{X}\setminus W$ is countable.

(b). Consider a cover of $\mathcal{X}\times\mathcal{X}$ by the following open sets:

$$[x,\infty)\times[-x,\infty) \text{ and } (-\infty,x)\times(-\infty,-x)$$

for all $x\in\mathbb{R}$.

8.11. Consider two sequences of points $x_0, x_1, \dots$ and $y_0, y_1, \dots$ in $\mathcal{M}$ such that $x_i =$ $= f(x_{i-1})$ and $y_i = f(y_{i-1})$ for any i. Observe that

$$|x_0 - y_0| \leqslant |x_1 - y_1| \leqslant \dots$$

Show that one can pass to subsequences $x_{i_1}, x_{i_2}, \dots$ and $y_{i_1}, y_{i_2}, \dots$ such that $x_{i_n} \to$ $\to x_0$ and $y_{i_n} \to y_0$ as $n\to\infty$. Conclude that $|x_0 - y_0| = |x_1 - y_1|$; since the points x_0 and y_0 can be chosen arbitrarily, it implies that f is distance-preserving.

Observe that f is injective. Use the argument above to conclude that $f(\mathcal{M})$ is dense in $\mathcal{M}$. Note that $f(\mathcal{M})$ is compact and therefore closed (see 9.12). Conclude that f is surjective and therefore bijective.

Comment. This statement is used to prove key properties of so-called Gromov–Hausdorff metric — a metric on the space of all (isometry classes of) compact metric spaces.

8.13. We need to show that any Cauchy sequence $q_1, q_2, \dots$ in a closed subset Q of a complete space $\mathcal{M}$ converges to a point in Q. Convergence follows from completeness of $\mathcal{M}$. It remains to use that Q is closed.

8.14. If the space is not bounded, consider a sequence $x_1, x_2, \dots$ such that $|x_n - p| \to \infty$ for some (and therefore any) point p.

For the second part, check an infinite discrete space.

8.15. Choose a compact $\frac{1}{2\cdot n}$-net $K_n \subset \mathcal{M}$. By 8.5 and 8.6, K_n contains a finite $\frac{1}{2\cdot n}$-net, say F_n. Observe that F_n is a finite $\frac{1}{n}$-net of $\mathcal{M}$; that is,

$$\mathcal{M} = \bigcup_{z \in F_n} \mathrm{B}(z, \tfrac{1}{n}).$$

Choose a sequence of points $x_1, x_2, \dots \in \mathcal{M}$. Apply the diagonal procedure (should be known from calculus) to extract a subsequence $y_1, y_2, \dots$ such that for fixed choice $z_n \in F_n$ we have $y_m \in \mathrm{B}(z_n, \frac{1}{n})$ for any $m \geqslant n$. Observe that $y_1, y_2, \dots$ is Cauchy and use completeness of $\mathcal{M}$.

8.16. Consider a sequence $x_0, x_1, \dots \in \mathcal{M}$ such that $x_i = f(x_{i-1})$ for any i. Show that this sequence is Cauchy. Since $\mathcal{M}$ is complete, the sequence has a limit, say p. Observe that p is a fixed point. Furthermore, show that if $q \neq p$, then $|f(q) - p| < |q - p|$; in particular, $f(q) \neq q$ and so q is not fixed.

Comment. This is the Banach fixed-point theorem; it combines extreme simplicity and extreme usefulness.

8.17. Assume such a point does not exist. Then for any point p there is a point p' such that $|p - p'| < r(p)$ and $r(p') \leqslant (1-\varepsilon)\cdot r(p)$. Let $p_0, p_1, \dots \in \mathcal{M}$ be a sequence such that $p_i = p'_{i-1}$. Show and use that this sequence is Cauchy.

Comment. This statement can be used on complete spaces in place of existence of minimum point of a continuous function on a compact space; as far as I know it was discovered by Ivar Ekeland [9].

8.21; *(a).* Apply 7.10 and 7.12*b*. Check the conditions in 1.1.

(b). Choose a Cauchy sequence $\gamma_1, \gamma_2, \dots \in \Gamma$. Observe that $\gamma_n(t)$ is a Cauchy sequence in $\mathbb{R}^2$ for any fixed $t \in [0, L]$. Since $\mathbb{R}^2$ is complete, we can define

$$\gamma_\infty(t) = \lim_{n\to\infty} \gamma_n(t).$$

Show that $\gamma_\infty \in \Gamma$.

(c). Choose a sequence $\gamma_1, \gamma_2, \dots \in \Gamma$ and a countable dense set $\{t_1, t_2, \dots\}$ in $[0, L]$. Apply the diagonal procedure to extract a subsequence $\gamma_{i_1}, \gamma_{i_2}, \dots$ such that for any t_k the sequence $\gamma_{i_n}(t_k)$ converges as $n \to \infty$. Show that this subsequence converges in Γ.

8.23. Assume that the boundary curve γ is not a circle. Use convexity to show that all points of γ except for a countable set S are smooth; that is, their tangent lines are well-defined. Show that there is a circle σ that crosses γ at three distinct smooth points; furthermore, at one of these points, the tangent lines of γ and σ are distinct. Arrive at a contradiction.

9.3. Arguing by contradiction, assume a sequence has two limits x and y. Since the space is Hausdorff we can choose disjoint neighborhoods $V \ni x$ and $W \ni y$. Since the sequence converges to x, the set V contains all but finitely many elements of the sequence. The same holds for W — a contradiction.

9.4. Let V and W be a pair of open sets and $W' = \mathcal{X} \setminus \bar{V}$. Show and use that V and W meet 9.1 if and only if V and W' meet 9.1.

9.5. By the definition of a Hausdorff space, we can choose open subsets $W_{ij} \ni x_i$ such that $W_{ij} \cap W_{ji} = \varnothing$ if $i \neq j$. Show that the sets

$$V_i = \bigcap_{j \neq i} W_{ij}$$

meet the required condition.

9.6. The set Δ is closed if and only if its complement $U = (\mathcal{X} \times \mathcal{X}) \setminus \Delta$ is open. Show and use that the latter means that there is a family $\{(V_\alpha, W_\alpha)\}$ of disjoint pairs of open sets in $\mathcal{X}$ such that

$$U = \bigcup_\alpha V_\alpha \times W_\alpha.$$

9.7. Choose a point $(x, y_1) \notin \Gamma$. Let $y_0 = f(x)$. Let $V \ni y_0$ and $W \ni y_1$ be as in the neighborhoods provided by Hausdorffness of $\mathcal{Y}$. Show that $f^{-1}(V) \times W$ is a neighborhood of (x, y_1) that does not intersect Γ; draw a conclusion.

Remark. This is another part of the closed graph theorem; compare to 7.24 and 6.12*b*.

9.10 + 9.11. Spell the definitions.

9.14. By 9.13, for any $y \in L$ there is a pair of open sets V_y, W_y such that $V_y \supset K$ and $W_y \ni y$ such that $V_y \cap W_y = \varnothing$. Mimic the proof of 9.13 using these pairs.

9.17. Apply 9.15 for the following maps:

(a). The map $[0,1] \to \mathbb{S}^1$ defined by

$$t \mapsto (\cos(2 \cdot \pi \cdot t), \sin(2 \cdot \pi \cdot t)).$$

(b). The map $\mathbb{D}^2 \to \mathbb{R}^3$ that is written from polar to spherical coordinates as

$$(r, \theta) \mapsto (1, \theta, \pi \cdot r).$$

(c). Map $(s,t) \in \mathbb{S}^1 \times [0,1]$ to the point in $\mathbb{R}^2$ with the polar coordinates (s,t).

9.18. Let $K \subset \mathbb{R}^3$ be a convex body; we can assume that the origin lies in the interior of K.

Show that the boundary ∂K is compact. Show that any half-line that starts from the origin intersects ∂K at a single point. Conclude that $x \mapsto \frac{x}{|x|}$ defines a continuous bijection $\partial K \to \mathbb{S}^2$ and apply 9.16.

9.19. Use 9.13 to show that $\mathcal{X}/Q$ is Hausdorff. Denote the one-point compactification by V^∞; it is a compact space. Use 9.16 to show that the natural map $V^\infty \to \mathcal{X}/Q$ is a homeomorphism.

9.20. Compactness of $\mathcal{Y}$ follows from 7.11; it remains to show its Hausdorffness.

Since $\mathcal{X}$ is Hausdorff, one-point sets in $\mathcal{X}$ are closed. Since f is closed and onto, one-point sets in $\mathcal{Y}$ are closed as well.

Choose two points $y_1, y_2 \in \mathcal{Y}$. Observe that $K_1 = f^{-1}\{y_1\}$ and $K_2 = f^{-1}\{y_2\}$ are closed and therefore compact subsets of $\mathcal{X}$. Apply 9.14 to get disjoint open sets $V_1 \supset K_1$ and $V_2 \supset K_2$. Further, look at the sets $W_i = \mathcal{Y} \setminus f(\mathcal{X} \setminus V_i)$ and think.

9.22. Let N_1, $N_2, \dots$ be the sequence of neighborhoods of x provided by the definition of first-countable space. We can assume that $N_1 \supset \supset N_2 \supset \dots$; otherwise replace N_n by $N_1 \cap \dots \dots \cap N_n$.

If $x \in \bar{A}$, then we can choose $x_n \in N_n \cap A$ for every n. Show that $x_n \to x$. This proves the only-if part.

Show that the if part holds for any space, not necessarily first-countable.

9.23. Let V_1, $V_2, \dots$ be a sequence of neighborhoods of $x_0 = [0] \in \mathcal{X}$. Construct an open set W in $\mathcal{X}$ such that $V_n \not\subset W$ for any n. Conclude that $\mathcal{X}$ is not first-countable.

9.24. Argue as in 6.3 and do some algebra. Take $V = f^{-1}(-\infty, \frac{1}{2})$ and $W = f^{-1}(\frac{1}{2}, +\infty)$.

9.25. Consider sets

$$V = \bigcup_{a \in A} \mathrm{B}(a, \tfrac{1}{2} \cdot \mathrm{dist}_B a),$$
$$W = \bigcup_{b \in B} \mathrm{B}(b, \tfrac{1}{2} \cdot \mathrm{dist}_A b).$$

10.2; *(a).* Apply 9.13.

(b). Mimic the proof of 10.1 using 7.8.

10.3. Enumerate rationals, say as $q_1, q_2, \dots$ Choose a sequence of open dense sets $V_1, V_2, \dots$ such that $V_i \not\ni q_i$ and apply the Baire theorem.

10.4. Apply the Baire theorem to the complements $V_n = \mathcal{X} \setminus \bar{B}_n$.

Comment. This is the first step in the proof of the uniform boundedness principle.

10.5; *(a).* Let W be an open set in a Baire space $\mathcal{X}$, and let $V_1, V_2, \dots$ be a sequence of open dense sets in W. Apply the definition of Baire space to the sets $V_i' = V_i \cup (\mathcal{X} \setminus W)^\circ$.

(b). Spell the definitions.

10.6. Construct a strongly nested sequence of intervals $[0,1] = [a_0, b_0] \Supset [a_1, b_1] \Supset \dots$ such that $[a_n, b_n] \cap Q_m = \varnothing$ for any $m \leqslant n$. Apply the finite intersection property (7.7) to show that the intersection $X = [a_0, b_0] \cap [a_1, b_1] \cap \dots$ is nonempty. Note that $X \cap Q_m = \varnothing$ for any m.

Comment. By Sierpiński's theorem, the same conclusion holds for any nonempty compact connected metric spaces (instead of $[0,1]$); see for example [24, 5.16]. Without compactness, the statement fails to hold.

10.7. Argue as in 3.3*a*.

10.11; *(a)* Let $G = V_1 \cap V_2 \cap \dots$ be a G-delta set in a topological space $\mathcal{X}$; here each V_i is an open set. Observe that $D = G \cup (\mathcal{X} \setminus \bar{G})$ is a dense G-delta set, $\bar{G}$ is closed, and $G = \bar{G} \cap D$.

(b). Note that $S = [0, \infty)$ is a G-delta set in $\mathbb{R}$. Apply 10.10 to show that S is not an intersection of a dense G-delta set with an open set in $\mathbb{R}$.

10.12. Show and use that a complement of a countable set in $\mathbb{R}$ is G-delta.

10.13; *(a).* Denote by V_n the union of all open sets $W \subset \mathbb{R}$ such that $\operatorname{diam} f(W) < \frac{1}{n}$. Observe that V_n is open for any n. Further, show and use that $\operatorname{Cont} f = V_1 \cap V_2 \cap \dots$

(b). Apply 10.12.

(c). Show and use that for any integer n and any irrational number s there is $\varepsilon > 0$ such that ε-neighborhood of s contains none of the first n rational numbers $q_1, \dots, q_n$.

10.14. Let G be a G-delta subset of a complete metric space. Show and use that G admits another metric that induces the same topology and makes G a complete metric space.

For the second part, check the complement of $(\mathbb{R} \setminus \mathbb{Q}) \times \{0\}$ in $\mathbb{R}^2$, here $\mathbb{Q} \subset \mathbb{R}$ denotes the subset of rational numbers.

10.17. Let $\mathcal{X} = \mathbb{R}_d \times \mathbb{R}$ where $\mathbb{R}_d$ denotes the set $\mathbb{R}$ equipped with the discrete topology. Note that the graph Γ of any (not necessarily continuous) function $f \colon \mathbb{R}_d \to \mathbb{R}$ is a closed set in $\mathcal{X}$. Show and use that one can choose f so that the projection of Γ to $\mathbb{R}$ is any given set, say a Vitali set.

Comment. In fact there are Borel sets in $\mathbb{R} \times \mathbb{R}$, whose projections to $\mathbb{R}$ are not Borel, but this is out of scope of my notes.

10.18. Try to fix the following not very well-defined map:

$$(0.d_1 d_2 \dots)_2 \to ((0.d_1 d_3 \dots)_2, (0.d_2 d_4 \dots)_2),$$

where $(0.d_1 d_2 \dots)_2$ denotes a number in base 2.

11.3. Assume the contrary. Show and use that the sets $V = f^{-1}((0, \infty))$ and $W = f^{-1}((-\infty, 0))$ form an open separating pair of $\mathcal{X}$.

11.5. Apply 7.8 to show that Q_∞ is not empty. Further, assume V and W separate Q_∞. Apply 7.8 to the sequence $Q'_n = Q_n \setminus (V \cup W)$.

11.6. Let us denote by $\gcd(a, b)$ the greatest common divisor of positive integers a and b. Denote by $\mathbb{N}_0$ the set of all nonnegative integers, so the arithmetic progression $\{a, a+d, a+2\cdot d, \dots\}$ can be written as $a + \mathbb{N}_0 \cdot d$.

(a). Suppose $\gcd(a_1, d_1) = \gcd(a_2, d_2) = 1$. Show that the set $(a_1 + \mathbb{N}_0 \cdot d_1) \cap (a_2 + \mathbb{N}_0 \cdot d_2)$ is either empty or it can be written as $a_3 + \mathbb{N}_0 \cdot d_3$ for some positive integers a_3 and d_3 such that $\gcd(a_3, d_3) = 1$. Use it to check the conditions in 6.9.

(b). If a prime number d is larger than a_1 and a_2, then $(a_1 + \mathbb{N}_0 \cdot d) \cap (a_2 + \mathbb{N}_0 \cdot d) = \varnothing$. Use this to show that the space is Hausdorff.

Consider a finite sequence of elements of the base $a_1 + \mathbb{N}_0 \cdot d_1, \dots, a_n + \mathbb{N}_0 \cdot d_n$. Show that the product $d_1 \cdots d_n$ does not belong to any of them. Apply 7.20 and 7.21 to conclude that the space is noncompact.

Finally, assume V and W separate $\mathbb{N}$ with the described topology. Observe that $a_1 + \mathbb{N}_0 \cdot d_1 \subset V$ and $a_2 + \mathbb{N}_0 \cdot d_2 \subset W$ for some positive integers a_1, d_1, a_2 and d_2 such that $\gcd(a_1, d_1) = \gcd(a_2, d_2) = 1$. Let $a_3 = d_1 \cdot d_2$; without loss of generality we can assume that $a_3 \in V$. Observe that $a_3 + \mathbb{N}_0 \cdot d_3 \subset V$ for some d_3 such that $\gcd(a_3, d_3) = 1$. Show that $(a_3 + \mathbb{N}_0 \cdot d_3) \cap (a_2 + \mathbb{N}_0 \cdot d_2) \neq \varnothing$ and arrive at a contradiction.

Comment. This problem and solution with more details are discussed in the note by Solomon Golomb [12].

11.8. Apply 11.7.

11.10. Given a separating pair of $\mathcal{X}$ or $\mathcal{Y}$ construct a separating pair of $\mathcal{X} \times \mathcal{Y}$; this will prove the if part. For the only-if part, apply 11.9 to show that $(\{x\} \times \mathcal{Y}) \cup (\mathcal{X} \times \{y\})$ is connected for any $x \in \mathcal{X}$ and $y \in \mathcal{Y}$. And apply it again to show that $\mathcal{X} \times \mathcal{Y}$ is connected.

11.12+11.13. Apply 11.11, 11.7, and 11.9.

11.14; *(a)* Apply 11.4.

(b). Consider the following subspace of the real line $A = \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\}$. Show that the one-point set $\{0\}$ is a connected component in A and it is not open in A.

(c). Apply *(a)*.

Comment. The Cantor set $\mathcal{C}$ (see 4B) gives a more interesting example in *(c)*. Each connected component of $\mathcal{C}$ is a one-point set, but $\mathcal{C}$ does not have isolated points.

11.15. Denote by Q_x the intersection of all clopen sets containing x. Observe that Q_x is closed. Assume Q_x is not connected; subdivide it into two disjoint closed sets; apply 9.14 and try to reach a contradiction.

The closure of the following set in $\mathbb{R}^2$ gives a noncompact Hausdorff example:

$$\left\{ (x, n + \tfrac{1}{1-x^2}) \,:\, n \in \mathbb{Z} \text{ and } x \in (-1, 1) \right\},$$

here $\mathbb{Z}$ denotes the set of integers. For a compact, but not Hausdorff example, consider the set $\mathcal{X} = \mathbb{Z} \cup \{+\infty, -\infty\}$; declare any subset of $\mathbb{Z}$ and any complement of a finite set to be open.

Comment. The set Q_x is called the quasi-component of x; note that $y \in Q_x$ if and only if $f(x) = f(y)$ for any continuous map f from $\mathcal{X}$ to the discrete two-point space.

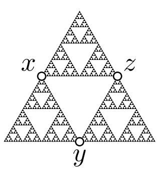
11.17. Show that $\mathbb{R}^2$ has no cut points, but $\mathbb{R}$ has.

11.18. Count points with given number of components in the complement for each space.

11.19. Let x be the cross point, let $c \neq x$ be a point on the circle, let e be one of the two endpoints of the segment, and let i be a point on the segment that is distinct from x and not an endpoint. Show that Q/H has just 4 points $[x]$, $[c]$, $[e]$, and $[i]$ (argue as in 11.18). List the open subsets of $\{[x], [c], [e], [i]\}$.

11.20; *(a)* Let T_n be the union of all sides of the 3^n triangles after the n-th iteration. Note that the sequence is nested; that is, $T_0 \subset T_1 \subset \dots$ Use induction to show that each T_n is connected. Conclude that the union $T = T_0 \cup T_1 \cup \dots$ is connected. Finally, show that the Sierpiński triangle is the closure of T and apply 11.4.

(b). Denote the Sierpiński triangle by $\triangle$.



Let $\{x, y, z\}$ be a 3-point set in $\triangle$ such that $\triangle \setminus \{x, y, z\}$ has 3 connected components. Show and use that there is a unique choice for the set $\{x, y, z\}$ and it is formed by the midpoints of the original triangle.

12.1. Passing to a subinterval, we can assume that $a(t) \neq x$ and $a(t) \neq y$ if $0 < t < 1$. Show that there is a maximal (with respect to inclusion) open set $V \subset [0,1]$ such that if (t_0, t_1) is a connected component of V, then $a(t_0) = a(t_1)$. Consider map $\hat{a}\colon [0,1] \to \mathcal{X}$ defined by $\hat{a}(t) = a(t)$ if $t \notin V$ and $\hat{a}(t) = a(t_0)$ if t lies in a connected component (t_0, t_1) of V. Show that $\hat{a}$ is a path. Read about Devil's staircase and apply similar construction to make $\hat{a}$ injective, or follow the argument in the note by Reinhard Börger [4]; see also [5].

Comment. From this statement we get that any path-connected Hausdorff space with at least two points must have at least continuum of points. Note that connected two-point space provides an example of a finite path-connected space. On the other hand, 11.6 provides an example of countable connected space.

12.2. Recall that connected two-point space has two points a and b, and its open sets are $\varnothing$, $\{a\}$, and $\{a, b\}$. Show and use that $f\colon [0,1] \to \mathcal{X}$ defined by

$$f(t) = \begin{cases} a & \text{if } t < 1, \\ b & \text{if } t = 1 \end{cases}$$

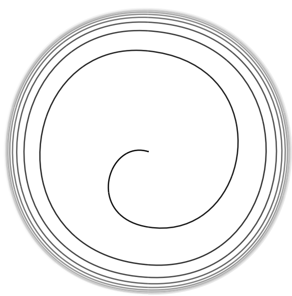
is a continuous map.

12.3. Show and use that for any continuous map φ and any path a, the composition $\varphi \circ a$ is a path.

12.4. Suppose that a and b are paths in $\mathcal{X}$ and $\mathcal{Y}$ respectively. Show and use that $t \mapsto (a(t), b(t))$ is a path in $\mathcal{X} \times \mathcal{Y}$.

12.6. The flea and the comb.

12.7. Look at the picture and think.



12.8. Choose a path-connected component V. Assume its complement, say W, is nonempty. Observe that W is a union of path-connected components. Therefore, W is open, and the pair V, W separates the space.

To prove the converse, apply 12.5

12.9. Read the note by Tatsuo Homma [16] and think.

Comments. This is a well-known puzzle; with the following interpretation, it appears in the book of Peter Winkler [28, p. 121]: *Two monks climb the mountain on the same day, starting at the same time and altitude, but on different* *paths. The paths go up and down on the way to the summit (but never dip below the starting altitude). Can they vary their speed (sometimes going backwards) so that at every moment of the day they are at the same altitude?*

12.10. Show and use that for any rational numbers $r \neq 0$ and q, the line $y = r\cdot x + q$ lies in $A \cup B$.

12.11. Read about pseudo-arc and think.

12.12. Describe paths with two-point images and use this description to construct a path between two given points in a connected finite space.

12.14; *(a)* Show that the local path-connectedness holds at all points of the comb, but does not hold at the flea.

(b). Consider the closure of the comb in $\mathbb{R}^2$.

Comment. Try to construct a path-connected space that is not locally path-connected at any point; can you make it compact?

12.15. Apply 11.16 and 12.13.

13.2. Consider linear homotopy in $\mathbb{R}^3$:

$$f_t(x) = (1-t)\cdot f_0(x) + t\cdot f_1(x).$$

Observe that $f_t(x) \neq 0$ for any t and x. Compose f_t with the radial projection $\mathbb{R}^3 \setminus \{0\} \to \mathbb{S}^2$ defined by $y \mapsto \frac{y}{|y|}$.

13.3. Argue as in 13.2 to construct a homotopy from f to the antipodal map $x \mapsto -x$. Further, use rotation of $\mathbb{S}^1$ to construct a homotopy from the antipodal map to the identity map. Consider concatenation of these two homotopies.

13.4. Choose $x \in \mathcal{X}$; let $y = r(x)$. Assume $y \neq x$. Let $V \ni x$ and $W \ni y$ be disjoint neighborhoods provided by Hausdorffness. Show and use that the set $V \cap r^{-1}(W)$ contains x and does not intersect $r(\mathcal{X})$.

13.5; *(a).* Consider homotopy h_t defined by

$$h_t(x) = \begin{cases} x & \text{if } x \in J_0 \\ t\cdot x & \text{if } x \notin J_0 \end{cases}$$

(b). Assume J_∞ is a deformation retract; let h_t be the corresponding homotopy relative to J_∞. Show that for any large n we have

$$|h_t(1, \tfrac{1}{2^n}) - (1,0)|_{\mathbb{R}^2} < \tfrac{1}{10}$$

for any t and arrive at a contradiction.

(c). Assume that F is a retract of the square. Show that in this case J_∞ is a deformation retract of F; the latter contradicts *(b).*

13.7. Let $h_t\colon \mathcal{X} \to \mathcal{X}$ be a homotopy from the identity map to a constant map; so, $h_1(x) = p$ for any $x \in \mathcal{X}$.

(a). Show and use that $t \mapsto h_t(x)$ is a path from x to p.

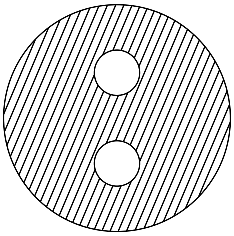
(b). Choose two continuous maps $f_0, f_1\colon \mathcal{Y} \to \mathcal{X}$. Note that $h_t \circ f_i$ is a homotopy from f_i to the constant map $y \mapsto p$. Concatenate the obtained homotopies.

(c). Choose two continuous maps $f_0, f_1\colon \mathcal{X} \to \mathcal{Y}$. Note that $f_i \circ h_t$ is a homotopy from f_i to the constant map $x \mapsto y_i = f_i(p)$. Choose a path from y_1 to y_2 and use it to construct a homotopy from $f_0 \circ h_1$ to $f_1 \circ h_1$. Concatenate the obtained homotopies.

13.8 + 13.9. Spell the definitions.

13.10. Show and use that $h \sim h \circ f \circ g \sim g$.

13.12. Observe and use that both sets form retracts of the plane figure in the picture.



13.14. Read about the mapping cylinder.

13.13. Show that each space has a circle as a deformation retract.

14.5; *(a).* Let $d = (a * b) * c = a * (b * c)$. Show that

$$d([0, \tfrac{t_0}{2}]) \supset d([0, t_0]) \text{ and}$$
$$d([0, 1 - t_0]) \supset d([0, 1 - \tfrac{t_0}{2}])$$

if $0 < t_0 \leqslant \frac{1}{2}$. Apply Hausdorffness.

(b). Use induction to show that given an integer $n \geqslant 1$ we have $a(t) = p$ for any $t \leqslant 1 - \frac{1}{2^n}$. Apply Hausdorffness.

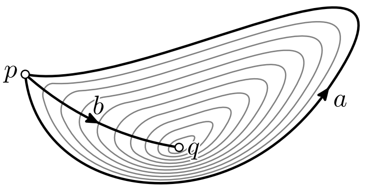
Comment. It is instructive to check that both statements fail in the connected two-point space.

14.7. Consider the map $f\colon [0,1]\times[0,1]\to\mathcal{G}$ defined by $f\colon (s,t)\mapsto a(t)\cdot b(s)$. Show that f is continuous. Write both $a*b$ and $b*a$ as compositions of f with a path from $(0,0)$ to $(1,1)$ in $[0,1]\times[0,1]$ and apply linear homotopy.

14.8. Let $r\colon \mathcal{X}\to A$ be a retraction. Choose points $p,q\in A$. Since $\mathcal{X}$ is path-connected, there is a path a in $\mathcal{X}$ from p to q. Observe that $r\circ a$ is a path in A from p to q. Further, let b_0 and b_1 be two paths in A from p to q. Since $\mathcal{X}$ is simply-connected, there is a homotopy of paths b_τ in $\mathcal{X}$ from b_0 to b_1. Show and use that $r\circ b_\tau$ is a homotopy of paths in A from b_0 to b_1.

14.10. The only-if part follows directly from 14.9. It remains to prove the if part.

Choose a loop a based at p. Let us identify $\mathbb{S}^1$ with $[0,1]/\{0,1\}$ (see 9.17*a*); so the loop a also defines a map $\mathbb{S}^1\to\mathcal{X}$.



Since any map $\mathbb{S}^1\to\mathcal{X}$ is null-homotopic, there is a free homotopy $a_\tau\colon [0,1]\to\mathcal{X}$ such that $a_0=a$, $a_1=e_q$ for some $q\in\mathcal{X}$, and $a_\tau(0)=a_\tau(1)$ for any τ. Consider the path $b(t):=a_t(0)$. Show and use that $a\sim b*e_q*\bar b\sim\sim e_p$.

For the last part, show and use that if $\mathcal{X}$ is a contractible space, it is path-connected and any map $\mathbb{S}^1\to\mathcal{X}$ is null-homotopic.

14.12; *(a).* Cover $\mathbb{S}^2$ by subsets V and W defined by $z>-\frac{1}{2}$ and $z<\frac{1}{2}$, respectively. Observe that V and W are homeomorphic to an open disc and the intersection $V\cap W$ is homeomorphic to the product $\mathbb{S}^1\times(0,1)$ and apply 14.11.

(b). Construct contractible (and therefore simply-connected) open sets $V\supset\mathbb{S}^2$ and $W\supset\supset\mathbb{D}^2$ and apply 14.11.

15.1. Show that the neutral element is unique and apply the definition of homomorphism and neutral element.

15.2. Check the condition in the definition of homomorphism.

For the second part observe that any inner homomorphism of a commutative group is trivial and find a commutative group with a nontrivial homomorphism.

15.3. Show and use that $(g\cdot h)^{-1}=h^{-1}\cdot g^{-1}$.

15.4. Let φ be a homomorphism. Show and use that $\varphi(g)=\varphi(h)$ if and only if $\varphi(g\cdot h^{-1})=1$.

15.5; *(a).* Do it by hand.

(b). Observe that [213] and [123] generate S_3; that is, any element in S_3 can be written as a finite product of these two elements. Check all variants for $\varphi[213]$ and $\varphi[123]$, not all of them lead to a homomorphism.

15.6. Note that a homomorphism $\varphi\colon\mathbb{Z}\to S_3$ is determined by the element $s=\varphi(1)\in S_3$; check all the cases.

15.7; *(a).* Show and use that $r\circ a=a$ for any loop a in A.

(b). Let a_τ be a homotopy in $\mathcal{X}$ between two loops a_0 and a_1 in A. Show and use that $r\circ a_\tau$ is a homotopy in A between the same loops.

(c). By *(a)* and *(b)*, it is sufficient to show that r_* is a monomorphism and i_* is an epimorphism.

The first statement means that if $r\circ a_0\sim\sim r\circ a_1$ in A for some loops a_0 and a_1 in $\mathcal{X}$, then $a_0\sim a_1$ in $\mathcal{X}$. Let r_τ be the homotopy from $r_0=\mathrm{id}_{\mathcal{X}}$ to $r_1=r$. Show and use that $r_\tau\circ a_i$ is a homotopy from a_i to $r\circ a_i$.

The same homotopies $r_\tau\circ a_i$ can be used to show that i_* is an epimorphism.

15.9; *(a)* Show and use that $f_1\circ a\sim\bar c*(f_0\circ a)*c$ for any loop a in $\mathcal{X}$ based at x_0.

(b). Let $f\colon\mathcal{X}\to\mathcal{Y}$ and $h\colon\mathcal{Y}\to\mathcal{X}$ be a homotopy equivalence and a homotopy inverse. Apply *(a)* to show that $(f\circ h)_*=f_*\circ h_*$ and $(h\circ f)_*=h_*\circ f_*$ are isomorphisms. Conclude that f_* is an isomorphism.

15.10. Spell the definitions.

15.11. Let $f\colon\mathcal{X}\times\mathcal{X}\to\mathcal{Y}$ be the quotient map. Choose a base point $x_0\in\mathcal{X}$ and let $y_0==f(x_0,x_0)$ be the base point in $\mathcal{Y}$. Given two loops a and b based at x_0 in $\mathcal{X}$, denote by $a\times b$ the loop in $\mathcal{X}\times\mathcal{X}$ defined by $t\mapsto(a(t),b(t))$. Observe that $f\circ(a\times e_{x_0})=f\circ(e_{x_0}\times a)$ for any loop a. Use it to show that

$$f\circ((a*b)\times e_{x_0})\sim f\circ(a\times b)\sim f\circ((b*a)\times e_{x_0})$$

and draw a conclusion.

16.3; *(a).* Suppose there is a homeomorphism $h\colon \mathbb{R}^3 \to \mathbb{R}^2$. Choose a point $p \in \mathbb{R}^3$ and let $q = h(p)$. Observe that $\mathbb{R}^3 \setminus \{p\}$ is homeomorphic to $\mathbb{R}^2 \setminus \{q\}$. Show that $\pi_1(\mathbb{R}^3 \setminus \{p\})$ is trivial, but $\pi_1(\mathbb{R}^2 \setminus \{q\})$ is not and arrive at a contradiction.

(b). Show and use that after removing one point the sphere $\mathbb{S}^2$ remains simply-connected, but $\mathbb{D}^2$ need not remain simply-connected.

(c). Apply 16.1, 15.7a, and the linear homotopy; see 13B.

(d). Look at the set of points p in each space that admits a neighborhood V such that $V \setminus \{p\}$ is simply-connected. Show and use that in the cylinder this set has two connected components and in the Möbius band just one.

(e). Construct a retract of $\mathbb{R}^2 \setminus F$ that is homeomorphic to the circle. Apply 14.10 and 16.1.

16.4; *(a).* Assume there is a homeomorphism $h\colon K_1 \to K_2$. Argue as in 11.18 and 16.3c to show that (1) h maps interior points of K_1 to interior points of K_2, (2) h maps boundary points of K_1 to boundary points of K_2, and (3) h maps the attached segments of K_1 to the attached segments of K_2. Try to arrive at a contradiction.

(b). Realize that both products $K_1 \times [0,1]$ and $K_2 \times [0,1]$ are solid tori with attached pairs of squares. To construct a homeomorphism, twist one solid torus.

16.7. Apply 16.6. For the second part, let b go around $\mathbb{S}^1$ and come back the same way.

16.9. Assume f exists. Consider a loop a in $\mathbb{S}^1$ based at 1 with nonzero degree. Then $\tilde{a} := f \circ a$ is its lift. Applying a shift to f we can assume that $\tilde{a}(0) = 0$, but then $\deg a = \tilde{a}(1) = f \circ a(1) = f \circ a(0) = 0$ — a contradiction.

16.11; *(a).* Apply 12.13 and 16.8.

(b). Observe that the winding number vanishes in the unbounded component, and when one crosses the curve the winding number increases or decreases by 1, depending on the direction of the curve at the point of crossing.

Comment. Observe that if the winding around a point is not zero, then the point lies in the image of any continuous map $\mathbb{D}^2 \to \mathbb{C}$, assuming that the boundary $\mathbb{S}^1 = \partial\mathbb{D}^2$ maps along the given curve.

17.4; *(a).* Let $V \ni p$ be an evenly covered open set with slices $\tilde{V}_\alpha$. Choose $q \in w^{-1}\{p\}$. Show and use that $\{q\} = w^{-1}\{p\} \cap \tilde{V}_\alpha$ for some α.

(b). Apply 17.3 to show that the image and its complement are open.

17.5. Spell the definitions of covering map and product topology.

17.6. Apply the definition of the covering map to show that the set of points in $\mathcal{Z}$ with exactly n points in their inverse image has to be open and closed.

17.7. Note that at each crossing, both types of arrows must point both in and out. (The solutions are not unique.)

17.8. Label Θ by three types of arrows that point, say from left to right. Now try to label the edges in the picture. At each crossing, the three arrows must be either all incoming or all outgoing. If such a labeling is impossible, then it is not a total space of a covering.

17.9+17.10. Spell the definitions.

17.11; *(a).* Spell the definitions, and be aware that composition of coverings might fail to be a covering; see 19.8.

(b). Apply the definition; start with the 4 crossing points, there should be two types of arrows coming in and two coming out; follow the arrows on the two pictures to figure out how they are connected.

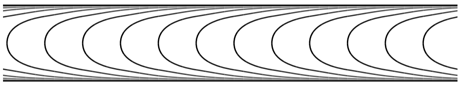
17.15; *(a)+(b)+(c).* Reformulate each statement using homotopy of paths and apply 17.12.

17.16. The inclusion $\operatorname{Im} f_* \subset \operatorname{Im} v_* \cap \operatorname{Im} w_*$ follows since for any loop a in $(\mathcal{P}, p_0)$, we have loops $\tilde{w} \circ a$ and $\tilde{v} \circ a$ in $(\mathcal{X}, x_0)$ and $(\mathcal{Y}, y_0)$, respectively.

To prove $\operatorname{Im} f_* \supset \operatorname{Im} v_* \cap \operatorname{Im} w_*$, apply 17.15$b$. Show and use that if a loop b in $(\mathcal{Z}, z_0)$ lifts to loops in $(\mathcal{X}, x_0)$ and $(\mathcal{Y}, y_0)$, then it lifts to a loop in $(\mathcal{P}, p_0)$.

18.2. By 7.11, the quotient space $\mathcal{X}/G$ is compact. Construct a finite cover of $\mathcal{X}/G$ by evenly covered open sets $V_1, \dots, V_n$. We can assume that the cover is minimal; that is, some point $G{\cdot}x \in \mathcal{X}/G$ is covered by one V_i. Show that the slices of V_i cover $\mathcal{X}$ and apply compactness of $\mathcal{X}$.

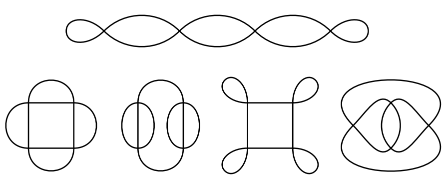
18.3. Let $\mathcal{X} = \mathbb{R} \times [0,1]$. Consider the action $\mathbb{Z} \circlearrowright \mathcal{X}$ generated by a homeomorphism that moves each point by length 1 along the Reeb foliation, which is shown on the picture; it is a subdivision of $\mathbb{R} \times [0,1]$ into smooth curves that include two horizontal lines $\mathbb{R} \times \{0\}$, $\mathbb{R} \times \{1\}$, and a one-parameter family of U-shaped curves with these two lines as asymptotes.



This action is freely discontinuous and the quotient space $\mathcal{Y} = (\mathbb{R} \times [0,1])/\mathbb{Z}$ is locally homeomorphic to $\mathcal{X}$, but $\mathcal{Y}$ is not Hausdorff; any two points $[(y_0,0)] \in \mathcal{Y}$ and $[(y_1,1)] \in \mathcal{Y}$ do not satisfy the Hausdorff condition.

Comment. Note that both quotient spaces $(\mathbb{R} \times [0,1))/\mathbb{Z}$ and $(\mathbb{R} \times (0,1])/\mathbb{Z}$ are homeomorphic to $\mathbb{S}^1 \times [0,\infty)$; We can assume that each curve in the Reeb foliation is mapped to horizontal circles $\mathbb{S}^1 \times \{t\}$. Therefore $\mathcal{Y}$ is homeomorphic to the space obtained gluing two copies of $\mathbb{S}^1 \times [0,\infty)$ by a homeomorphism $h\colon \mathbb{S}^1 \times (0,\infty) \to \mathbb{S}^1 \times (0,\infty)$ that preserves the second coordinate. This homeomorphism twists $\mathbb{S}^1 \times \{t\}$ more and more as $t \to 0$.

18.5. The solutions are not unique. Choose spaces from the following picture (or draw your own). Label arcs by arrows, depending on the choice of labeling the same space might solve different parts of the exercise.



18.7. Let $w\colon \tilde{\mathcal{X}} \to \mathcal{X}$ be a 2-fold cover. Consider the involution i that swaps the points in each inverse image $w^{-1}\{x\}$ for $x \in \mathcal{X}$. Prove that i generates a $\mathbb{Z}_2$-action on $\tilde{\mathcal{X}}$ and identify $\mathcal{X}$ with the quotient space $\tilde{\mathcal{X}}/\mathbb{Z}_2$.

18.8. Consider the set of all 6 vertices, 12 edges and 8 faces of the octahedron. Equip it with a topology, by stating that each face is open, the set formed by an edge with two adjacent faces is open, and the set formed by a vertex with 4 adjacent edges and 4 adjacent faces is open. Take a quotient of this space by an involution.

18.10. Apply 9.15 and 18.4.

Comment. Note that the covering in the exercise can be written as $z \mapsto z^2$.

18.12; *(a)* Assume the contrary and apply the Borsuk–Ulam theorem for the map

$$\tilde{h}(x) = \frac{f(x) - f(-x)}{|f(x) - f(-x)|}.$$

(b). Given a unit vector $u \in \mathbb{S}^2$ consider a perpendicular plane $\Pi(u)$ that divides C evenly. Note that $\Pi(u) = \Pi(-u)$. Denote by $a(u)$, and $b(u)$, the volumes of A and B on the side of Π in the direction of u. Note that $a(u) + a(-u)$ is the total volume of A and $b(u) + b(-u)$ is the total volume of B for any u. Apply *(a)* to the map $\mathbb{S}^2 \to \mathbb{R}^2$ defined by $u \mapsto (a(u), b(u))$.

(c). Show and use that if $f(x) = f(-x)$, then x and $-x$ lie in one of the sets A, B, or C. For the last part, try to cover the sphere by four spherical triangles.

Comments. Part *(a)* of the exercise can be interpreted in the following way: *at any moment of time there exists a pair of antipodal points on the surface of the earth that have the same temperature and pressure.*

18.14; *(a).* Note that $\tilde{z}_1$ can be connected by a path to any point in $w^{-1}\{z_0\}$ and apply 18.13.

(b). By *(a)* the monodromy homomorphism is surjective; it remains to show that it is injective. Choose two loops a and b based at z_0; let $\tilde{a}$ and $\tilde{b}$ be their lifts such that $\tilde{a}(0) = \tilde{b}(0) = \tilde{z}_1$. If $m_{[a]} = m_{[b]}$, then $\tilde{a}(1) = \tilde{b}(1)$. Since $\tilde{\mathcal{Z}}$ is simply-connected, $\tilde{a} \sim \tilde{b}$. Therefore $a = w \circ \tilde{a} \sim w \circ \tilde{b} = b$; in other words, $[a] = [b]$.

(c). Show and use that the lift of any loop in $\mathcal{Z}$ is a loop in $\tilde{\mathcal{Z}}$.

18.15; *(a).* Show and use that $|g{\cdot}p - p| \geqslant 1$ for any point $p \in \mathbb{R}^2$ and any $g \neq 1$ in G.

(b). Just apply.

(c). Follow the same arrows in different order and see that the result is different.

(d). Consider the subaction of $G \circlearrowright \mathbb{R}^2$ of the motions

$$(x,y) \mapsto (m + x, 2{\cdot}n + y)$$

for all integers m and n.

19.1. Apply 17.2*b*.

19.6. Show that $\mathcal{F}_\infty$ is reasonable and apply 19.5.

19.7. A sufficiently small circle $S \subset \mathcal{H}$ must lie in an open set evenly covered by w. Construct a retraction $r\colon \mathcal{H} \to S$; consider a nontrivial connected covering v with base space S, and pass to its pullback along r (apply 17.10).

19.8. Taking the pullback of the wrapping map along a retraction $r\colon \mathcal{H} \to S$ to some circle $S \subset \mathcal{H}$ produces a normal $\mathbb{Z}$-covering. The total space will look like a cactus train. Now for the double cover, unwrap different circles in cacti of this train, so that the circle sizes approach zero when you move to infinity.

19.9. Read about the Griffiths space.

19.11. Apply 17.2*b* in each case, and 19.10 once.

Comment. Each statement fails if $\mathcal{Z}$ is not reasonable. Exercise 19.8 provides an example for part *(a)*. The other parts are simpler, but require patience.

Recall that $\mathbb{N}^\infty$ denotes one-point compactification of the positive integers $\mathbb{N}$; this space can be identified with $\{0, 1, \frac{1}{2}, \frac{1}{3}, \ldots\} \subset \mathbb{R}$; this space is not locally connected and therefore not reasonable.

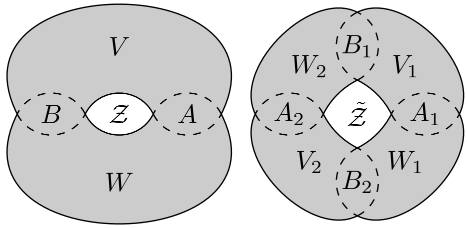
For *(b)*, take $\mathcal{X} = \mathbb{N}^\infty \times \mathbb{N}$, $\mathcal{Y} = \mathbb{N}^\infty \times \{0,1\}$, and $\mathcal{Z} = \mathbb{N}^\infty$; furthermore, define u and w as projections and let

$$v(n,k) = \begin{cases} (n,0) & \text{if } n \leqslant k \\ (n,1) & \text{if } n > k \end{cases}$$

Observe that $(\infty, 0) \in \mathcal{Y}$ does not admit a neighborhood evenly covered by v.

For *(c)*, take $\mathcal{X} = \mathbb{N}^\infty \times \mathbb{N}$, $\mathcal{Y} = \mathcal{X}/\sim$ where $(k,m) \sim (k,n)$ if $m > k$ and $n > k$, and $\mathcal{Z} = \mathbb{N}^\infty$. Furthermore, let u be the projection and let v be the quotient map; these uniquely determine w. Observe that $\infty \in \mathcal{Z}$ does not admit a neighborhood evenly covered by w.

19.12. Suppose $V \cap W = A \cup B$, where A and B are disjoint open sets; see the first picture. Take 2 copies of each set: $A_1, A_2, B_1, B_2, V_1, V_2, W_1, W_2$ and glue them to form a space $\tilde{\mathcal{Z}}$ as shown in the second picture.



More precisely, the inclusions of A and B into V and W induce 8 embeddings: $A_i \hookrightarrow V_j$, $A_i \hookrightarrow W_j$, $B_i \hookrightarrow V_j$, $B_i \hookrightarrow W_j$ for all pairs i and j. Glue the copies along all the embeddings in the diagram and show that the obtained space $\tilde{\mathcal{Z}}$ covers $\mathcal{Z}$ twice.

$$\begin{tikzcd}
A_1 \arrow[r, hook] \arrow[d, hook] & V_1 & B_1 \arrow[l, hook'] \arrow[d, hook] \\
W_1 & & W_2 \\
B_2 \arrow[r, hook] \arrow[u, hook] & V_2 & A_2 \arrow[l, hook'] \arrow[u, hook]
\end{tikzcd}$$

Comment. This is another partial case of the van Kampen theorem, see also 14.11.

One can also construct a normal $\mathbb{Z}$-covering, by preparing $\mathbb{Z}$ copies of each set and gluing them along the following maps:

$$\begin{aligned}
\cdots \hookleftarrow A_{n-1} \hookrightarrow V_{n-1} \hookleftarrow B_{n-1} \hookrightarrow \\
\hookrightarrow W_n \hookleftarrow A_n \hookrightarrow V_n \hookleftarrow B_n \hookrightarrow \\
\hookrightarrow W_{n+1} \hookleftarrow A_{n+1} \hookrightarrow V_{n+1} \hookleftarrow \cdots
\end{aligned}$$

19.15. Let $a\cdot b(t) := a(t)\cdot b(t)$ and consider the product $[a]\cdot[b] := [a\cdot b]$ on $\tilde{\mathcal{G}}$.

20.2. Follow the proof and see that compactness of $\Gamma \cap \bar{\mathrm{B}}[x, R]$ is the only place we use special properties of the Euclidean plane. This was used to prove the existence of points q_n and s_n. Use instead their approximate versions $\tilde{q}_n$ and $\tilde{s}_n$, so the equalities in ❷ hold with an error that approaches zero as $n \to \infty$, and check that the remaining part of the proof works with these points.

20.4. Show and use that Y retracts onto a figure eight; see 17D.

20.6. Follow the proof of 20.5.

20.7. By 20.1, $h\colon A \to B$ and $h^{-1}\colon B \to A$ can be extended to continuous maps $p\colon \mathbb{R} \to \mathbb{R}$ and

$q\colon \mathbb{R} \to \mathbb{R}$. Show and use that the maps

$$(x,y) \mapsto (x, y + p(x)),$$
$$(x,y) \mapsto (x - q(y), y),$$
$$(x,y) \mapsto (y, x)$$

are homeomorphisms $\mathbb{R}^2 \to \mathbb{R}^2$.

20.9. Assume the contrary. Identify $\mathbb{R}^2$ with the (x,y)-plane in $\mathbb{R}^3$. Show the following:

- $\mathbb{R}^3 \setminus (A \cup B)$ has nontrivial fundamental group, but
- $\mathbb{R}^3 \setminus A$ and $\mathbb{R}^3 \setminus B$ have trivial fundamental groups.

Construct a continuous function $f\colon \mathbb{R}^2 \to \to [0,1]$ such that $A = f^{-1}\{0\}$ and $B = f^{-1}\{1\}$. Consider the homeomorphism $h\colon \mathbb{R}^3 \circlearrowleft$ defined by $(x,y,z) \mapsto (x, y, z + f(x,y))$. Observe that h moves A and B to different horizontal planes. Use the triviality of fundamental groups of $\mathbb{R}^3 \setminus A$ and $\mathbb{R}^3 \setminus B$ to show that $\mathbb{R}^3 \setminus h(A \cup B)$ has trivial fundamental group. Arrive at a contradiction.

Comment. A more standard solution is done by applying Alexander duality to Čech cohomology.

20.10. Apply 20.6.

Comment. In fact any compact contractible set in $\mathbb{R}^2$ has connected complement; a way more general statement is provided by Alexander's duality theorem.

20.11. Apply 20.4.

20.13; *(a)* Apply 20.10.

(b). Assume the contrary. Then there is a neighborhood V of a point $p \in J$ that fails to meet one of the components of $\mathbb{S}^2 \setminus J$. Choose a subset $I \subset J$ that is homeomorphic to $[0,1]$ and such that $I \supset J \setminus V$ and apply *(a)*.

Comment. Recall that by 20.11, $\mathbb{S}^2 \setminus J$ has exactly two connected components, but the solution above does not use this.

20.14. Removing a point $p \in A \cap B$ from $\mathbb{S}^2$ we get $\mathbb{R}^2 \simeq \mathbb{S}^2 \setminus \{p\}$; let us keep the notation A and B for these sets without p. Note that $\mathbb{R}^2 \setminus A$ and $\mathbb{R}^2 \setminus B$ are connected and $\mathbb{R}^2 \setminus (A \cap B)$ is simply-connected.

As before, let us think that $\mathbb{R}^2$ is the (x,y)-plane in $\mathbb{R}^3$. Consider the set

$$C = A \times \{0\} \cup B \times \{1\} \cup (A \cap B) \times [0,1] \subset \mathbb{R}^3.$$

Show that $\mathbb{R}^2 \setminus (A \cup B)$ is connected if and only if $\mathbb{R}^3 \setminus C$ is simply-connected. Argue as above to prove the latter statement.

Comment. This statement also follows from Alexander's duality theorem.

Index

Bibliography

[1] I. Baggs. "Conditions implying continuity of open and closed functions". *Proc. Amer. Math. Soc.* 33 (1972), pp. 185–190.

[2] R. Baire. "Sur les fonctions de variables réelles". *Annali di Matematica Pura ed Applicata (1898-1922)* 3.1 (1899), pp. 1–123.

[3] K. Baker. *Bing's House*. Sketches of Topology. Mar. 25, 2010. URL: `https://sketchesoftopology.wordpress.com/2010/03/25/bings-house/` (visited on 03/18/2026).

[4] R. Börger. "How to make a path injective". *Recent developments of general topology and its applications*. Vol. 67. Math. Res. 1992, pp. 57–59.

[5] J. Brazas. "Constructing arcs from paths by collapsing subloops". *Rocky Mountain J. Math.* 54.4 (2024), pp. 965–974.

[6] G. Cantor. "De la puissance des ensembles parfaits de points". *Acta Math.* 4.1 (1884), pp. 381–392.

[7] *Cassell's Book of In-door Amusements, Card Games and Fireside Fun*. 1881.

[8] P. H. Doyle. "Plane separation". *Proc. Cambridge Philos. Soc.* 64 (1968), p. 291.

[9] I. Ekeland. "On the variational principle". *J. Math. Anal. Appl.* 47 (1974), pp. 324–353.

[10] M. Fréchet. "Les ensembles abstraits et le calcul fonctionnel." *Rend. Circ. Mat. Palermo* 30 (1910), pp. 1–26.

[11] H. Furstenberg. "On the infinitude of primes". *Amer. Math. Monthly* 62 (1955), p. 353.

[12] S. Golomb. "A connected topology for the integers". *Amer. Math. Monthly* 66 (1959), pp. 663–665.

[13] A. Hatcher. *Notes on introductory point-set topology*. URL: `https://pi.math.cornell.edu/~hatcher/Top/TopNotes.pdf`.

[14] A. Hatcher. *Algebraic topology*. 2002.

[15] F. Hausdorff. *Grundzüge der Mengenlehre*. 1914.

[16] T. Homma. "A Theorem on Continuous Functions". *Kodai Mathematical Seminar Reports* 4.1 (1952), pp. 13–16.

[17] H. Houdini. *Houdini's Book of Magic and Party Pastimes: Fascinating Puzzles, Tricks and Mysterious Stunts*. 1927.

[18] С. Иванов. *Геометрия и топология*. URL: `http://www.pdmi.ras.ru/~svivanov/uni/uni.html`.

[19] C. Kosniowski. *A first course in algebraic topology*. 1980.

[20] K. Kuratowski. "Sur l'opération $\bar{A}$ de l'Analysis Situs". *Fundamenta Mathematicae* 3.1 (1922), pp. 182–199.

[21] S. Loyd. *Cyclopedia of 5000 Puzzles, Tricks and Conundrums*. 1914.

[22] E. J. McShane. "Partial orderings and Moore–Smith limits". *Amer. Math. Monthly* 59 (1952), pp. 1–11.

[23] J. R. Munkres. *Topology*. 2000.

[24] S. Nadler Jr. *Continuum theory*. Vol. 158. Monographs and Textbooks in Pure and Applied Mathematics. An introduction. Marcel Dekker, Inc., New York, 1992, pp. xiv+328.

[25] F. Petrov. *17 camels trick*. MathOverflow. eprint: `https://mathoverflow.net/q/271608`.

[26] L. A. Steen and J. A. Seebach Jr. *Counterexamples in topology*. 1995.

[27] O. Ya. Viro, O. A. Ivanov, N. Yu. Netsvetaev, and V. M. Kharlamov. *Elementary topology*. 2008.

[28] P. Winkler. *Mathematical Puzzles: A Connoisseur's Collection*. Natick, MA, 2004.